

YMNOTES MEDICAL

Digestive System

Hepatogastroenterology
Hepatogastrointestinal Surgery

Version 3 STUDY EDITION

Clear structure. Connected understanding.

Interactive Contents

Tap a chapter to jump directly to it.

01	GI Symptoms and Investigations	1
	Internal Medicine · Radiology / Investigation	
02	The Esophagus	10
	Internal Medicine · Surgery	
03	The Stomach and Duodenum	33
	Internal Medicine · Pharmacology · Surgery	
04	Pathology of GIT Tumours	70
	Pathology · Surgery	
05	Diarrhea and Malabsorption	76
	Internal Medicine	
06	Constipation	95
	Internal Medicine	
07	Foodborn Infection	106
	Microbiology · Internal Medicine	
08	Inflammatory Bowel Disease	115
	Internal Medicine · Pharmacology	
09	Parasitic GI Diseases	122
	Parasitology	
10	Small Intestine and Colon Surgery	140
	Surgery	
11	Diverticulosis of the Colon	145
	Surgery · Internal Medicine	
12	Appendix	151
	Surgery	
13	Intestinal Obstruction	162
	Surgery · Emergency / Acute Care	
14	Polyposis of Colon	189
	Surgery · Pathology	
15	Colorectal Cancer	195
	Surgery · Pathology	
16	Anorectal Diseases	210
	Surgery	
17	Hepatology and Hepatic Surgery	231
	Surgery	
18	Liver Disease Assessment	236
	Radiology / Investigation · Internal Medicine	
19	Acute Hepatitis	251
	Internal Medicine	
20	Chronic Hepatitis B	264
	Internal Medicine · Pharmacology	

Interactive Contents

Tap a chapter to jump directly to it.

21	Autoimmune Hepatitis Internal Medicine	274
22	Liver Cirrhosis Internal Medicine · Pathology	280
23	Portal Hypertension Internal Medicine · Surgery	308
24	Hydatid Cyst Parasitology · Surgery	318
25	Pyogenic Liver Abscess Surgery · Internal Medicine · Parasitology	324
26	Hepatocellular Carcinoma Surgery · Pathology	332
27	Ascites Internal Medicine	337
28	Renal Impairment in Cirrhosis Internal Medicine	345
29	Jaundice Internal Medicine	349
30	Hepatic Encephalopathy Internal Medicine	355
31	Acute Hepatic Failure Internal Medicine · Emergency / Acute Care	361
32	Biliary System Anatomy Surgery	367
33	Gallbladder Diseases Surgery	373
34	Gallstones and Cholecystitis Surgery	376
35	Obstructive Jaundice Surgery · Radiology / Investigation	389
36	Acute Pancreatitis Surgery · Emergency / Acute Care	399
37	Cancer Pancreas Surgery · Pathology	407
38	Acute Abdomen Emergency / Acute Care · Surgery	413
39	Abdominal Trauma Surgery · Emergency / Acute Care	418
40	Splenic Trauma Surgery · Emergency / Acute Care	425

Chapters 21–40 · Page numbers refer to the book body.

Introduction



01



Learning objectives

The student will

- 1) Know the various symptoms of gastrointestinal diseases
- 2) Understand the basic mechanism underlying each of the symptoms of gastrointestinal tract
- 3) Understand the basic diagnostic tools for gastrointestinal diseases

Symptoms and signs of gastrointestinal diseases

Alarm symptoms- Red flag symptoms:

- Dysphagia.
- Weight loss.
- Persistent vomiting.
- Anorexia.
- Haematemesis or melaena.

NB: Patients with these features have a higher possibility of significant gastrointestinal pathology and should be investigated properly.

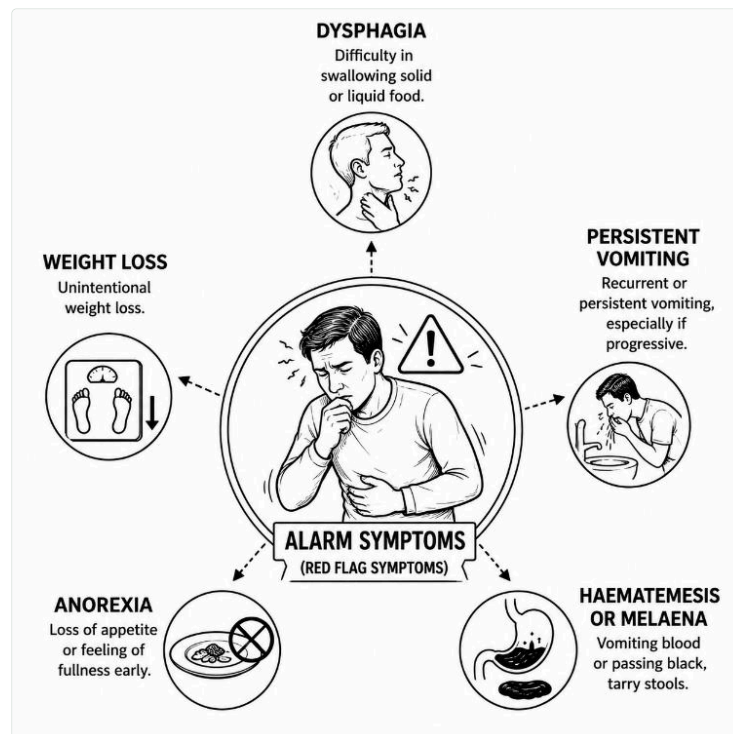


Figure 1.1 — Alarm symptoms (red flag symptoms)

- Stomatitis : It is an inflammation in the mouth from any cause.



- Angular stomatitis : is inflammation of the corners of the mouth .
- Dysphagia : Difficulty of swallowing.
- Odynophagia : Painful swallowing.
- Dyspepsia : Describe a number of upper abdominal symptoms such as heartburn, acidity, pain or discomfort, nausea, wind, fullness or belching.
- Vomiting : causes of vomiting
- Any Gastrointestinal disease.
- Infections: viral and bacterial.
- CNS diseases: Increase Intracranial pressure , migraine.
- Metabolic causes : Chronic renal failure , diabetic Ketoacidosis.
- Drugs : Digoxin.
- Reflex : Myocardial infarction.
- Psychogenic.
- Pregnancy.
- Alcohol.

Diarrhea & Constipation

Diarrhea

- Common complaint not always due to organic disease.
 - Defined as a change in bowel habit with:
 - Increase Frequency.
 - Increase Amount.
 - Decrease Consistency (loose/watery stool).
 - More than 200 g stool/day with low dietary fiber considered diarrhea.
 - Clinical:
 - Large-volume watery stool → usually organic cause
 - Bloody diarrhea → usually colonic pathology
 - Fever + acute diarrhea for several days → usually infective
 - Single episode → dietary change or psychogenic

Constipation

- Diagnosis is difficult because daily bowel opening is NOT required for normal bowel habit.
 - Constipation = ≥ 2 of the following:
 - < 3 stools/week.
 - Hard stools in $> \frac{1}{4}$ of defecations.



- Straining in $>1/4$.
- Incomplete evacuation in $>1/4$.
- Manual maneuvers (digital evacuation/pelvic floor support) in $>1/4$.

Abdominal Pain:

Mainly due to stretching of smooth muscle or organ capsules. Severe acute pain consider acute abdomen ; may also be referred (pneumonia) or metabolic (DKA).

Assess:

intensity, character, duration, frequency, aggravating/relieving factors & associated symptoms.

- Common Causes:
- Epigastric: functional dyspepsia, Peptic ulcer ; heartburn ,GERD.
- Right hypochondrium: biliary disease; hepatitis/cardiac failure.
- Biliary pain: severe, intermittent, lasts hours, recurrent.
- Left iliac fossa: diverticulitis; chronic pain often functional.
- Right iliac fossa: appendicitis/ileocaecal disease.
- Female : consider gynaecological causes.
- Abdominal wall pain: localized tenderness, persistent pain not relieved by tensing muscles.

Anorexia & Weight Loss:

- Anorexia: reduced appetite; common in systemic /psychiatric disease. May occur with cancer but is usually late and nonspecific.
- Weight loss usually due to reduced intake; in malabsorption mainly due to anorexia.
- Weight loss despite normal or increased intake $\rightarrow$ hyperthyroidism / catabolic states.

GIT Bleeding

- Hematemesis: vomiting blood.
- Melena: black, tarry stool $\rightarrow$ usually bleeding proximal to right colon; requires >50 mL blood.
- Massive upper GI bleed: fresh blood per rectum + shock.
- Hematochezia: bright-red/maroon rectal bleeding $\rightarrow$ usually lower GI source.





Investigations of gastrointestinal diseases

Investigations are selected according to the clinical presentation and include laboratory tests, stool studies, endoscopy, and imaging.

1. Laboratory & Stool Investigations:

- Routine blood tests: CBC, renal function, liver function.
- Stool examination: useful for detecting blood, fat (steatorrhoea), parasites and infection.
- Fecal occult blood test (FOBT): detects non-visible GI bleeding.
- Fecal Immunochemical test (FIT) : more accurate less false-positive results.
- Fecal calprotectin elevated in inflammatory bowel disease (IBD).
- Tumor markers:
 - CEA → colorectal cancer
 - CA 19-9 → pancreaticobiliary cancer
 - AFP → hepatocellular carcinoma

Tumor markers are mainly supportive and are not diagnostic alone.

2. Endoscopy:

Upper GI Endoscopy (EGD)

Investigation of choice for most upper GI disorders.

- Allows direct visualization, biopsy and treatment.
- Diagnoses: reflux esophagitis, gastritis, ulcers and malignancy.
- Therapeutic : treatment of upper GI bleeding and obstruction.
- Relative contraindications :recent myocardial infarction and severe COPD.

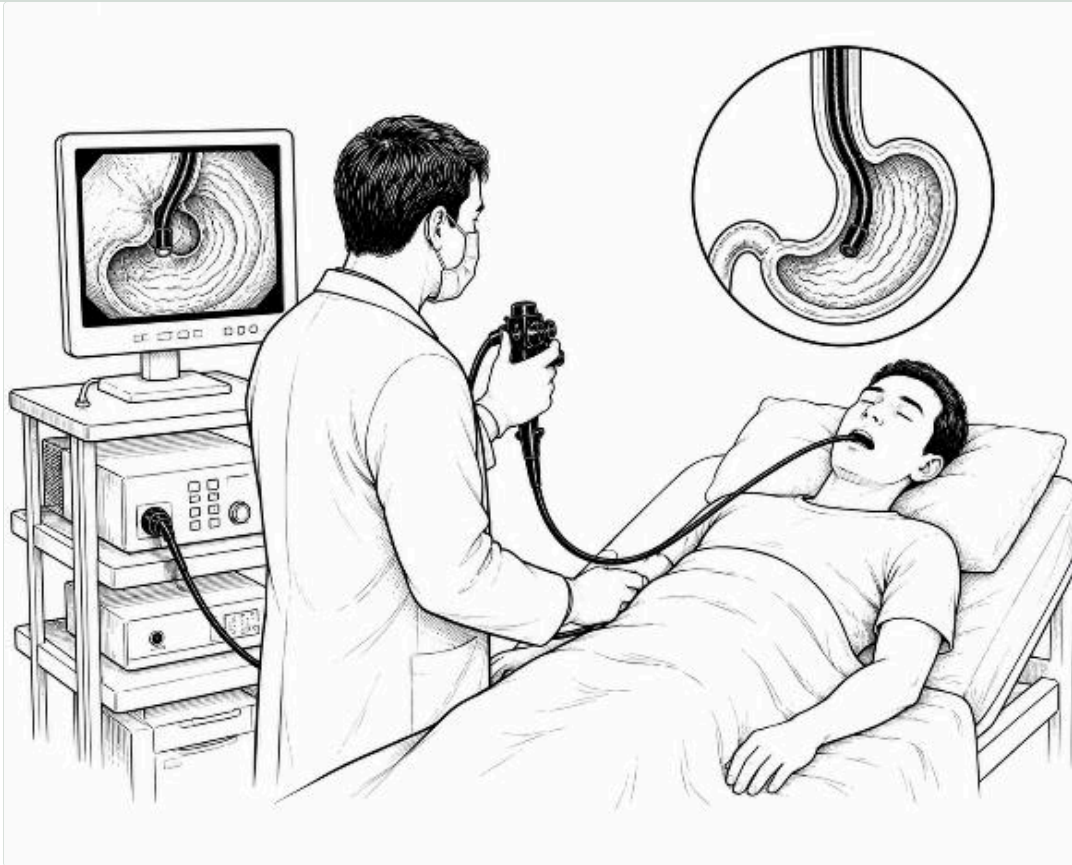


Figure 1.2 — Upper gastrointestinal endoscopy

Colonoscopy

- Visualizes the entire colon and terminal ileum.
 - Allows biopsy, polypectomy, stricture dilatation and stenting.
 - Important findings include cancer, polyps and diverticular disease.
 - Sigmoidoscopy (Rigid / Flexible) : can be use in rectal bleeding.

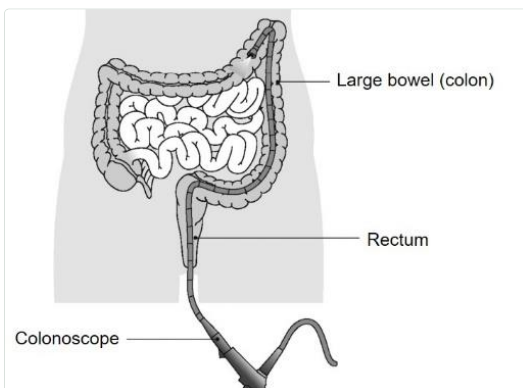


Figure 1.3 — Colonoscopy

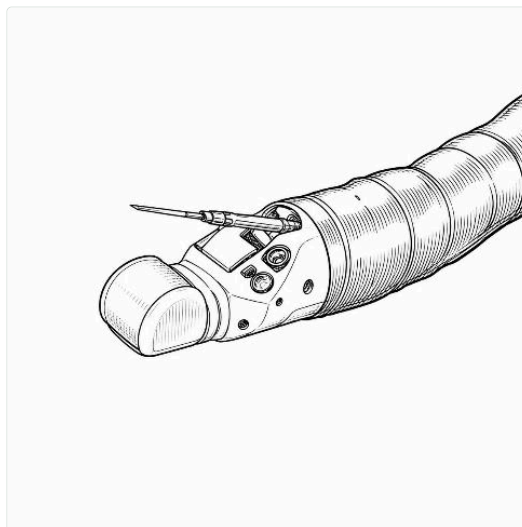


Figure 1.4 — Colonoscope



Small-Bowel Endoscopy

- Balloon enteroscopy: allows examination and treatment of the small bowel.
 - Capsule endoscopy: useful for obscure GI bleeding and small-bowel lesions/IBD after negative gastroscopy and colonoscopy.

Endoscopic Ultrasound (EUS)

- Combines endoscopy with ultrasound.
 - Useful for GI wall lesions, staging of esophageal/gastric cancer, and detection/biopsy of pancreatic tumors and cysts.

3. Imaging

Plain X-ray

- Mainly useful in acute abdomen:
 - Bowel obstruction
 - Perforation (free subdiaphragmatic air)
 - Toxic megacolon
 - May show calcification from gallstones or chronic pancreatitis.



Figure 1.5 — Plain abdominal X-ray





Ultrasound

- No radiation; first-line for many abdominal conditions.
 - Useful for ascites, masses, biliary disease, cholecystitis, appendicitis and inflammatory conditions.
 - Can demonstrate bowel dilatation and wall thickening.
 - Allows image-guided biopsy and drainage.

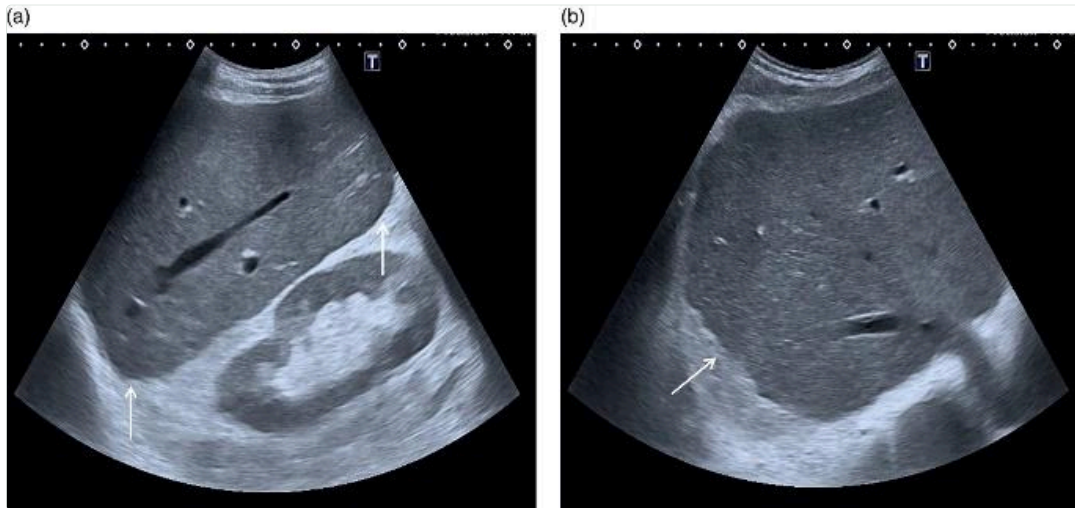


Figure 1.6 — Abdominal ultrasound

CT

- A major first-line investigation in acute abdomen.
 - Detects perforation, obstruction and its cause.
 - Demonstrates appendicitis, diverticulitis, abscesses and Crohn's disease/complications.
 - Important for cancer staging and image-guided biopsy.
 - Assess renal function before IV contrast.

MRI

- Advantages: no ionizing radiation and excellent soft-tissue contrast.
 - Particularly useful for rectal cancer, perianal fistulae/abscesses, small-bowel disease, and hepatobiliary/pancreatic disorders.
- MR Enterography: small-bowel disease, particularly useful in IBD.

PET

- Uses FDG metabolic activity to assess:
 - Cancer staging
 - Metastatic disease



- Useful in oesophageal, gastric and colorectal cancers.

4. Contrast Studies

- Barium swallow: mainly for dysphagia and esophageal disorders.
- Double-contrast barium meal: evaluates stomach and duodenum; largely replaced by gastroscopy.
- Barium enema: evaluates the colon but has largely been replaced by colonoscopy and CT colonography.
- If perforation is suspected → use water-soluble contrast (e.g. Gastrografin), not barium.

THE ESOPHAGUS



02



STRUCTURE AND PHYSIOLOGY:

- The esophagus is **20 cm** long that connects the pharynx to the stomach.
- The esophagus is lined by **stratified squamous epithelium**.
- The esophagus is separated from the pharynx by the upper esophageal sphincter (UES).

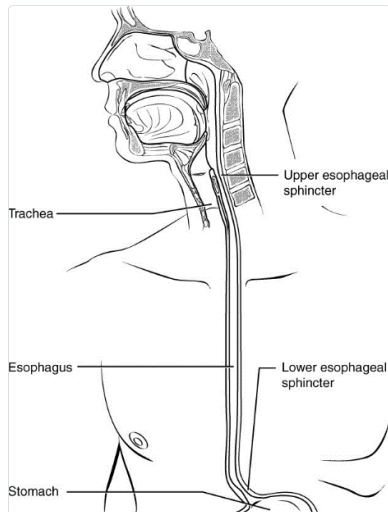


Figure 2.1 — Structure of the esophagus

The lower esophageal sphincter (LES) consists of a 2-4 cm in the distal end of the esophagus that responsible for the prevention of gastric reflux.

SYMPTOMS OF ESOPHAGEAL DISORDERS

1.Dysphagia:

- Difficulty of swallowing.
- The characteristics of the progression of dysphagia can be helpful Progression over a few weeks suggests a **malignant stricture**. The slow onset of dysphagia for solids and liquids at same time suggest **motility disorder** (e.g. Achalasia).

CAUSES Causes of dysphagia

- **Disease of mouth and tongue e.g.** tonsillitis .
- **Neuromuscular disorders e.g.** Bulbar palsy , myasthenia gravis .
- **Esophageal motility disorders e.g.** Achalasia ,Scleroderma ,diffuse esophageal spasm, Diabetes mellitus , Chagas' disease.
- **Extrinsic pressure e.g.** Mediastinal glands , goiter ,enlarged left atrium
- **Intrinsic lesion e.g.** Foreign body, Stricture: benign or malignant - carcinoma, esophageal web ,Pharyngeal pouch.



2. Odynophagia:

is pain during swallowing and is suggestive of esophagitis.

3. Substernal discomfort, heartburn:

This is a common symptom of reflux of gastric contents into the esophagus.

4. Regurgitation:

is the effortless reflux of esophageal contents into the mouth and pharynx.

SIGNS OF ESOPHAGEAL DISORDERS

The main sign of esophageal disease is **weight loss** due to reduced food intake.

INVESTIGATIONS FOR ESOPHAGEAL DISORDERS

1. Esophagoscopy.

2. Barium swallow and meal.

3. Manometry

It is used to assess esophageal motor activity.



Figure 2.2 — Esophageal radiographic investigation



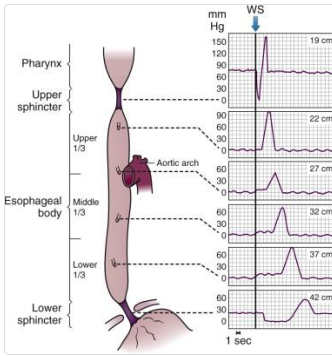


Figure 2.3 — Esophageal manometry

4. pH monitoring

- 24-hour ambulatory monitoring uses a pH-sensitive probe positioned in the lower esophagus

• GASTROESOPHOGEAL REFLUX DISEASE-

(GERD)

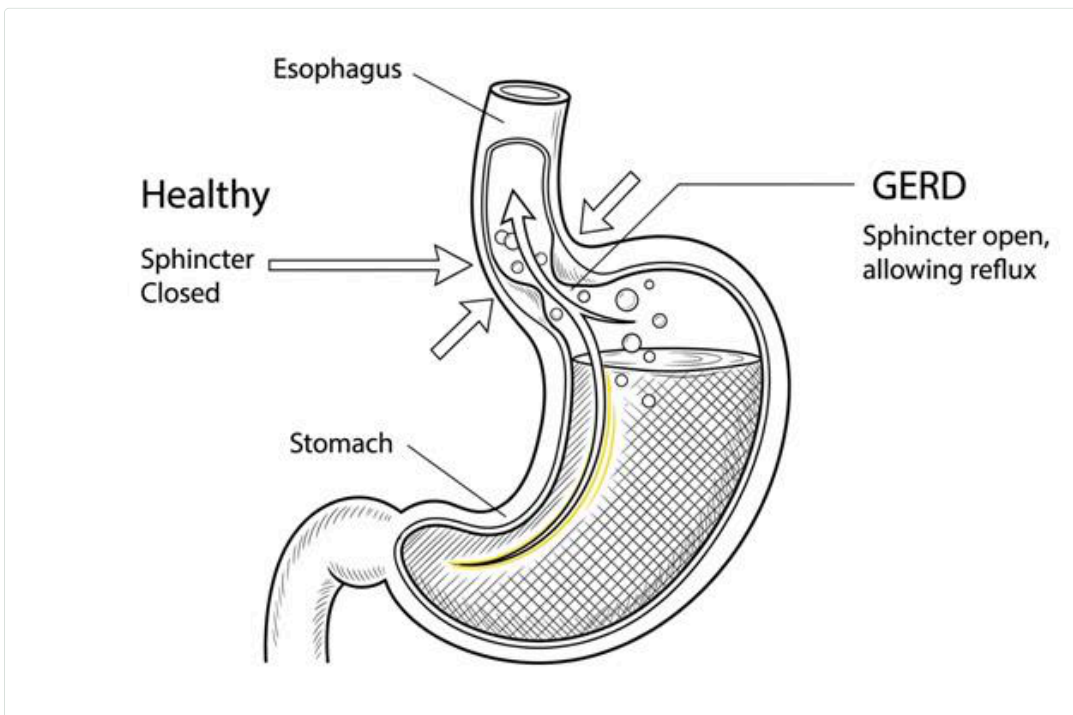


Figure 2.4 — Gastroesophageal reflux disease

PATHOPHYSIOLOGY

- The reflux occur when the anti-reflux mechanisms fail, allowing acidic gastric contents to make prolonged contact with the lower esophageal mucosa.



- Decrease tone of lower esophageal sphincter (LES).

RISK FACTORS:

- 1) **Increase intra-abdominal pressure** : Pregnancy or obesity
- 2) Diet:
Fat, chocolate, coffee, alcohol and large meals.
- 3) Drugs:
antimuscarinic, calcium-channel blockers, nitrates
- 4) Cigarette smoking
- 5) Hiatus hernia (described below)

CLINICAL FEATURES

- **Heartburn is the major feature.**
The burning is aggravated by lying down and relieved by oral antacids.

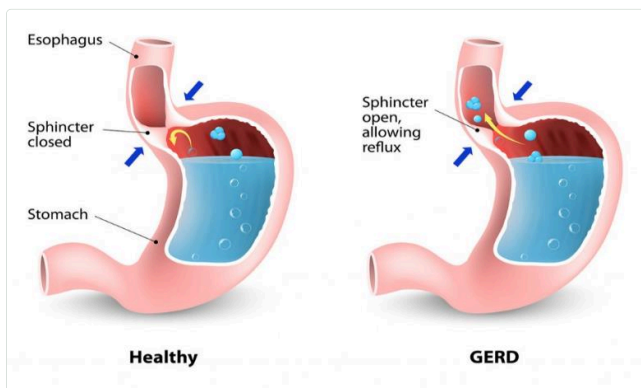


Figure 2.5 — Gastroesophageal reflux

- **Regurgitation** of food and acid into the mouth.
- **Aspiration pneumonia**

DIAGNOSIS AND INVESTIGATIONS

- The clinical diagnosis can usually be made without investigation.
 - If there is **alarm signs e.g.** weight loss, no response to PPI the following investigations:
- 1) 24-hour pH monitoring.
 - 2) Upper endoscopy.
 - 3) Manometry.

COMPARE Esophageal reflux and cardiac ischemia

- | | |
|--|--|
| <ul style="list-style-type: none"> • Reflux pain: burning, worse on lying down , don't radiates to the arms ,worse with hot drinks or alcohol and is relieved by antacids | <ul style="list-style-type: none"> • Cardiac ischemic pain: Gripping or crushing radiates to neck or left arm ,worse with exercise accompanied by dyspnea |
|--|--|



TREATMENT

- **Simple measures :**
 - Simple antacids.
 - Loss of weight and raising the head of the bed at night.
 - Reduction in caffeine consumption
 - Stop smoking and alcohol intake.
- **Drugs:**
 - **Proton pump inhibitors** e.g. omeprazole, reduce gastric acid secretion and are the drugs of choice for mild cases.
 - **H2 Blockers** e.g. ranitidine.
 - **Antacids.**
 - **Dopamine antagonist** as metoclopramide and domperidone are increase peristalsis and speed emptying .
- **Endoluminal gastroplication:**
 - This recent technique could be performed by endoscopy:
- **Endoscopic full-thickness plication (GERDX):**
 - A endoscopic device has been designed to inhibit gastroesophageal reflux by placing a transmural plication near the gastroesophageal junction under direct endoscopic visualization.
 - Anti-reflux mucosal ablation

(ARMA):

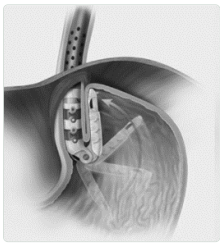


Figure 2.6 — Anti-reflux mucosal ablation

- Mucosal ablation around the gastroesophageal junction to cause a healing process to re-form the cardiac opening
- **Surgery:** management of sliding hiatus hernia (described below)

COMPLICATIONS

- **Peptic stricture:**
- Since the introduction of PPIs peptic strictures have become less common.
- **Barrett's esophagus :**
- The normal esophageal squamous epithelium is replaced by metaplastic columnar mucosa.



- Premalignant: esophageal adenocarcinoma.
- Barrett's is commonest in middle-aged obese men.
- Diagnosis: endoscopy, Chromo-endoscopy.
- Treatment : Radiofrequency ablation (RFA).

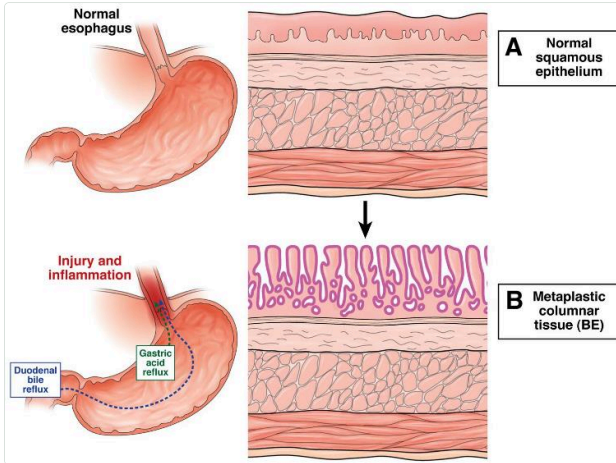


Figure 2.7 — Esophageal epithelial changes

BENIGN ESOPHAGEAL STRICTURE

- Peptic stricture secondary to reflux is the most common cause of benign strictures
- They also occur after the ingestion of corrosives, after radiotherapy, after sclerosis of varices.
- Dysphagia is usually managed by endoscopic dilatation.
- Surgery is sometimes required.

ESOPHAGEAL INFECTIONS

- Infection is a cause of painful swallowing and is seen in immunosuppressed (e.g. on chemotherapy) and AIDS.
- Infection can occur with:
 - Candida - Herpes simplex - Cytomegalovirus - tuberculosis.
- In candidiasis, the characterize by white plaques.
- Most patients on large doses of immunosuppressive agents are treated prophylactically for candida with nystatin or amphotericin.



HIATUS HERNIA-

- A hiatus hernia describes the **protrusion of an organ** from the abdominal cavity into the chest through the esophageal hiatus.
- This is **typically the stomach** that herniates.
- Extremely common disease.

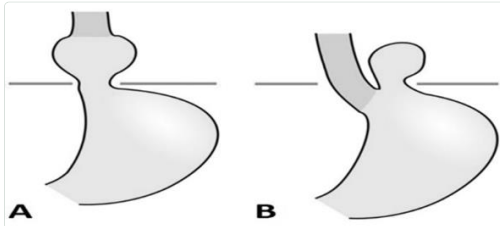


Figure 2.8 — Hiatus hernia

Classification:

3 subtypes:

Sliding hiatus hernia (85 %)	Rolling or Para - esophageal hernia (10%):	A mixed type hiatus hernia
The gastro-oesophageal junction (G O J), the abdominal part of the oesophagus, and the cardia of the stomach 'slide' upwards through the diaphragmatic hiatus into the thorax. (fig. A)	An upward movement of the gastric fundus through esophageal hiatus occurs to lie alongside a normally placed GOJ. This is a true hernia with a peritoneal sac. (fig. B)	which has both a rolling and sliding Components

- RISK FACTORS:
 - **Age:**
 1. Is an important risk factor, due to:
 - A combination of age- related loss of diaphragmatic tone, increasing intra-abdominal pressures (e.g. repetitive coughing), and widening of diaphragmatic hiatus.
 - **Increased intra-abdominal pressure:**
 2. Pregnancy, obesity, and ascites due to upwards displacement of the abdominal viscera.
- CLINICAL FEATURES OF SLIDING H.H.:
 - The majority of hiatus hernias are **asymptomatic**.



- Gastro-esophageal reflux symptoms:

1) Marked burning

epigastric pain:

- Made worse by lying flat and on leaning forwards.
 - Patient experiences relief on raising head of bed.
- 2) **Vomiting** – regurgitation.
- 3) **Bleeding and anemia** secondary to esophageal ulceration leading either melena, occult blood in stool or hematemesis in severe cases.
- 4) **Dysphagia** (esophageal stricture on long-standing cases)
- 5) Sometimes **Weight loss** in case of long duration vomiting and dyspepsia.
 - Epigastric tenderness in case of associated gastritis
 - INVESTIGATIONS:

1. Upper endoscopy :

- Is the **standard investigation** for esophageal disorders showing upward displacement of the Gastro-Esophageal Junction also (Z-line).
 - During inspiration the GOJ is opened and closed in expiration
 - It shows evidence of reflux as hyperemia, fissuring, ulceration or stricture of the lower esophagus
 - Biopsy may be taken from suspicious areas (areas of Barrett's)

2. Plain x-ray chest:

fundic air may be evident in the chest – detects chest complications as aspiration pneumonia.

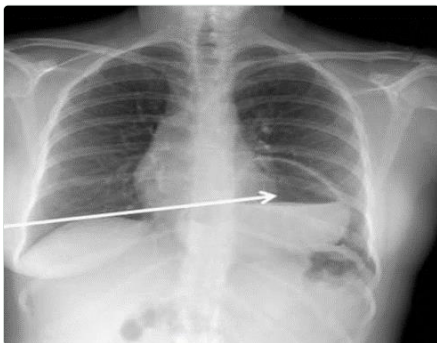


Figure 2.9 — CXR: stomach in the chest

3. Barium swallow:

- Evidence of complication as stricture esophagus.



4. Barium meal in Trendelenberg's position:

- Reflux of the barium to the esophagus in GERD
 - Stomach above the diaphragm in big Hiatus Hernia.

5. CT abdomen and chest

6. pH test:

shows increased acid level in the esophagus.(diagnostic for reflux)

Barium meal: sliding H.H

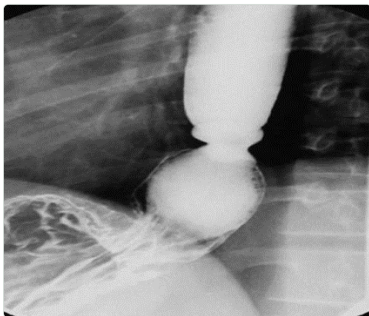


Figure 2.10 — Barium meal: sliding hiatus hernia

- **TREATMENT:**
 - **Life-style changes as:**
 - Losing weight.
 - Decreasing the size of meals.
 - Avoiding certain acidic foods, such as tomato sauce and citrus
 - Limiting fried and fatty foods, foods containing caffeine (including chocolate), • Eating meals at least three to four hours before lying down, and avoiding bedtime snacks • Keeping head higher than the rest of body when lying on the back.
 - Stop smoking
 - Avoid a tight belt or tight clothing.
 - **Medications:**
- **Antacids** that neutralize stomach acid.
- **H2 blockers.**
- **Proton pump inhibitors.**

PPI stronger acid blockers than H₂ blockers and allow for damaged esophageal tissue to heal.



Surgical management of GERD and hiatal hernia

Aim to restore the intra-abdominal length of the esophagus, repair the diaphragmatic defect, and recreate a functional lower esophageal sphincter (LES) barrier.

Indications for Surgery

Operative repair is recommended for all symptomatic patients

1. Failure of medical treatment for type I
2. due to the risk of volvulus, strangulation, and ischemia. In type II

Principles of Surgical Management

Laparoscopic repair is the gold standard approach:

1. Hernia Sac Reduction & Dissection

- Complete excision of the mediastinal sac
- Reduce the herniated stomach into the abdomen and
- completely mobilize and excise the hernia sac from the posterior mediastinum.

2. Hiatal Closure (Cruroplasty)

- Reapproximating diaphragmatic crura
- Close the enlarged diaphragmatic hiatus posteriorly using non-absorbable sutures. Synthetic or biological mesh reinforcement may be used in large defects to reduce recurrence.

3. Anti-Reflux Procedure (Fundoplication)

Wrap the gastric fundus around the distal esophagus to recreate the high-pressure zone:



Figure 2.11 — Fundoplication

- Nissen Fundoplication: 360° total wrap (gold standard)
- Toupet Fundoplication: 270° posterior partial wrap
- (preferred if underlying esophageal dysmotility exists to minimize post-operative dysphagia).



Postoperative Complications :

1) Acute Dysphagia: tight wrap, or tight crural closure

Treatment: Soft diet for 2-4 weeks. If persistent beyond 6 weeks, perform endoscopic balloon dilation.

2) Gas-Bloat Syndrome : Inability to belch/vomit due to hypercompetent 360° wrap treatment: Dietary modification, simethicone, and reassurance; resolves spontaneously in most patients.

3) Early Hernia Recurrence / Wrap Slippage

Treatment: revision surgery

4) Recurrent GERD : Breakdown or migration of the wrap into the chest

Treatment: revision surgery.

ACHALASIA OF ESOPHAGUS-

- It is a primary esophageal motility disorder characterized by failure of LES relaxation and loss of peristalsis in the distal esophagus.

PATHOPHYSIOLOGY:

- Vagal degeneration
- LES contraction and relaxation are regulated through vagus nerve
- Degeneration of ganglion cells in myenteric plexus concerned with the inhibitory neurons → increased tone in LES → severe spasm.

CAUSES OF ACHALASIA:

I- Primary Achalasia:

Etiology unknown, Hereditary, degenerative, autoimmune

II- Secondary Achalasia:

(e.g. secondary to Chaga's disease)

CLINICAL PRESENTATIONS:

- Dysphagia for liquids more than solids most significant symptom. (effect of weight). • Difficulty Belching.
- Regurgitation (especially when lying flat at night)
- Heartburn and Chest Pain in case of esophagitis
- Weight loss is minimal and late
- Hiccups

COMPLICATIONS ASSOCIATED WITH ACHALASIA:

Pneumonic Aspiration

Esophageal cancer



INVESTIGATIONS:

I- Radiographic Evaluation

II- Manometric Evaluation

III- Endoscopic Evaluation

I- Radiographic Evaluation

1- Chest x-ray:

- Markedly dilated Esophagus with widening of mediastinum
- Absent gastric air bubble: swallowed air does not reach the fundus

2- Barium swallows: diagnostic tool



Figure 2.12 — Smooth narrowing of the distal esophagus at the level of the lower esophageal sphincter with the characteristic “bird beak” appearance.

- The proximal esophagus is markedly dilated



Absent gastric air bubble



Figure 2.13 — Radiographic findings in achalasia

II- Esophageal manometry: confirm the diagnosis

- Elevated resting LES pressure (spasm).
 - Incomplete LES relaxation.
 - Weak peristalsis of esophageal wall (aperistalsis).



III- Endoscopic Evaluation:

- To exclude cancer of the gastro-esophageal junction.
 - Detects spasm of the LES.

MANAGEMENT:

Pharmacological agents:

Patients who are unable to tolerate more invasive treatment modalities.

1- Calcium channel blockers and nitrates decrease LES pressure.

2- Endoscopic Intra-sphincteric Injection of Botulinum Toxin:

- blocks release of acetylcholine at the level of the LES → relaxation of the sphincter

Endoscopic techniques

A- Pneumatic dilatation:

- Is the most effective non-surgical treatment option for patients with achalasia.
- The balloon is inflated at the level of the gastro-esophageal junction to dilate the muscle fibers while leaving the mucosa intact.

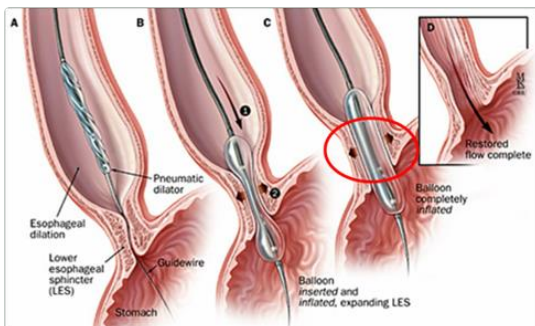


Figure 2.14 — Pneumatic dilatation

B- The Per-Oral Endoscopic Myotomy, or

POEM, :

- is a minimally invasive endoscopic procedure for the treatment of achalasia wherein the inner circular muscle layer of the lower esophageal sphincter is divided through a submucosal tunnel.

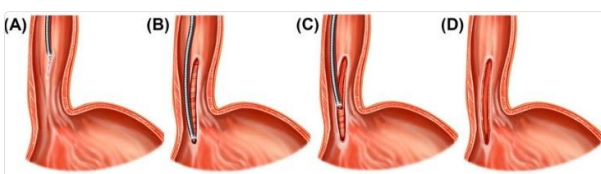


Figure 2.15 — Per-oral endoscopic myotomy



Surgical: Laparoscopic Heller's Cardio-Myotomy

- Weakens LES by cutting muscle fibers through performing an anterior myotomy across the LES.

Combined with Partial Fundoplication: Always paired with a partial fundoplication to prevent postoperative reflux.

c) Refractory / End-Stage (Megaesophagus): Esophagectomy with gastric conduit reconstruction.

Esophageal dysphagia

Primarily caused by

1. Mechanical obstructions
2. Neuromuscular motility disorders.

I- Mechanical Causes

- These conditions physically narrow the esophageal lumen, leading predominantly to dysphagia for solids before progressing to liquids:
 - 1) Esophageal Carcinoma: Progressive dysphagia starting with solids and rapidly advancing to liquids, accompanied by weight loss (Squamous Cell Carcinoma or Adenocarcinoma).
 - 2) Peptic Stricture: Benign narrowing of the distal oesophagus resulting from chronic gastroesophageal reflux disease (GERD) and scarring.
 - 3) Esophageal Webs: constrict the lumen and are often associated with Plummer-Vinson syndrome.
 - 4) Extrinsic Compression: Pressure on the esophagus from outside structures, such as a mediastinal tumor, aortic aneurysm, or retrosternal goiter.
 - 5) Foreign Body Impaction: Sudden onset of complete dysphagia
 - 6) Esophageal Diverticula: Outpouchings like Zenker's (pharyngoesophageal), traction, or epiphrenic diverticula that trap food and cause regurgitation.
 - 7) Corrosive Stricture: Severe scarring following ingestion of caustic agents or prior mediastinal radiotherapy.

II- Neuromuscular / Motility Causes

These functional disorders impair peristalsis or lower esophageal sphincter (LES) relaxation, often causing equal dysphagia for both solids and liquids from the onset:

- 1) Achalasia: Failure of the LES to relax alongside absent peristalsis due to degeneration of the myenteric plexus.



2) Distal Oesophageal Spasm (DES): Uncoordinated, high-amplitude contractions causing intermittent dysphagia and chest pain.

3) Hypercontractile Oesophagus: Severe, high-pressure muscular contractions during peristalsis.

4) Systemic Sclerosis (Scleroderma): Atrophy of smooth muscle leading to an incompetent LES and severe aperistalsis.

A definitive diagnosis for esophageal dysphagia relies on a clinical history to endoscopy, radiology, and functional/motility testing.

KEY HIGHLIGHTS diagnostic strategy to identify the most common mechanical and neuromuscular causes:

Step 1: Detailed Clinical History

Step 2: Upper Endoscopy (EGD) – The Initial Diagnostic Procedure

Step 3: Barium swallow

Step 4: Oesophageal Manometry: Gold Standard for Motility

Step 5: Advanced Imaging (CT / Endoscopic Ultrasound EUS)

Used when extrinsic compression or advanced local pathology is suspected

Surgical management of esophageal disorders

Surgical management of esophageal dysphagia depends entirely on whether the primary etiology is a mechanical obstruction, a motility disorder, or an external pathology.

1. Mechanical Obstructive Lesions: Esophageal Carcinoma

2. Benign Strictures (Peptic / Corrosive / Radiation)

• **First-line / Endoscopic** : Serial endoscopic balloon dilation or wire-guided bougie dilation.

• **Refractory Strictures:** Endoscopic steroid injection into the stricture, temporary stent placement, or formal esophageal resection and reconstruction (colon interposition or gastric pullup) for complete caustic destruction.

3. Esophageal Webs & Rings

• **Primary Treatment:** Endoscopic rupture using balloon dilation or targeted electrosurgical incision (using specialized knife).

4. Distal Esophageal Spasm & Hypercontractile Esophagus

• **Endoscopic (POEM):** Extended Peroral Endoscopic Myotomy targeting the long spastic segment in the mid-to distal esophagus when medical therapy fails.



5. Zenker's Diverticulum (Pharyngoesophageal)

- Endoscopic Approach (Transoral): Diverticulectomy using stapling, laser, or electrocautery (flexible or rigid endoscopic cricopharyngeal myotomy).
- open Surgical Approach: Cricopharyngeal myotomy combined with diverticulectomy (excision of the pouch)

6. Extrinsic Compression & Emergency Impactions

- Extrinsic Tumors / Goiters: Surgical resection of the compressive mediastinal mass or thyroidectomy for retrosternal goiter.
- Foreign Body: Emergent endoscopic removal using snare, basket, or grasping forceps; surgical exploration is reserved solely for cases complicated by perforation.

Cancer Esophagus

Pathology:

Age: old age > 50 years

Sex: males > females

Predisposing factors:

- Dietetic: spicy foods
- Excessive alcohol intake – smoking
- Achalasia
- Long standing reflux esophagitis
- Barret's esophagus (most important P.F)
- Corrosive stricture of the esophagus

Macroscopic picture:

- **Site:** 1- Middle 1/3 → 50 % (COMMONEST)
- 2- Lower 1/3 → 33 %
- 3- Upper 1/3 → 10: 15 %

Gross appearance:

1. Malignant ulcer: raised evereted edge, hard base and necrotic floor
2. Stricture: hard, long and irregular
3. Fungating mass: cauliflower, bleeding and necrotic surface





Microscopic picture:

1. Squamous cell carcinoma:

- The most common
- Grade: may be well, moderate or undifferentiated carcinoma

2. Adenocarcinoma:

- On top of Barrett's esophagus
- Local spread from gastric carcinoma at the cardia to lower esophagus.

Spread:

A. local :

- the tumor spread in vertical and horizontal (circumferentially)
- Vertical spread from the mucosa to submucosa to muscle layer to adventitia then penetrate to the surrounding structures as trachea, bronchus, pleura, diaphragm, stomach,....

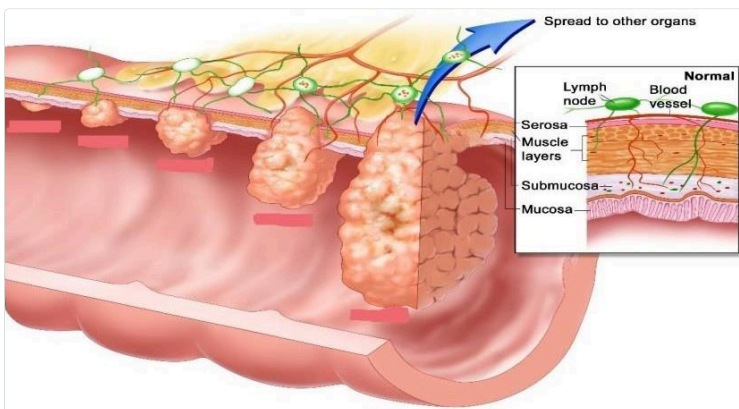


Figure 2.16 — Local spread

A. Lymphatics:

to regional L.N. according to the level:

- Cervical L.N. from upper 1/3
- Hilar and Mediastinal L.N. from the middle 1/3
- Left gastric and celiac L.N. from the lower 1/3

B. Blood :

late: to lung - liver - bone - brain.

Lower third may give early liver metastasis (portal)



Clinical picture:

General symptoms and signs:

- Marked progressive weight loss up to cachexia of rapid onset
 - Enlarged cervical L.N. in upper third cancer
 - Anemia
 - Symptoms of spread as recurrent aspiration in tracheal infiltration, pleural effusion

Local Symptoms:

1) Dysphagia:

- Old aged male – history of P.F. as GERD
- Presented when about 2/3 of the circumference is involved by the tumor
- Progressive and continuous
- First to solids than progresses to fluids as well then event to saliva
- Dribbling of saliva (severe form of dysphagia).

2) Regurgitation → leads to recurrent aspirations and chest infections

3) Hematemesis:

bleeding malignant tissues.

Investigations:

I. Upper endoscopy:

the most valuable and diagnostic

May show malignant ulcer, hard irregular stricture or cauliflower mass

Biopsy is taken for histo-pathology: Confirms the diagnosis.

A stent may be introduced to relieve dysphagia (palliative).

• Endo-luminal ultrasound:

- To detect infiltration of the tumor within and outside the wall of esophagus to surrounding structure.
- Detects resectability and operability

III- Radiological:

1. Barium swallow:

- It may show irregular filling defect.
- It may show signs of malignant stricture: elongated, irregular narrowing, characteristic shouldering and dilatation of the proximal esophagus (Rat-tail appearance)



Figure 2.17 — Barium swallow shows irregular stricture and filling defect

2. C.T. scan abdomen and chest:

for staging of the tumor

- Detect local spread and infiltration of surrounding structures
- Detect metastases in the liver, lungs
- Detect enlarged celiac L.N.
- Detects mediastinal L.N.
- Detects pleural effusion
- Evaluates the stage, operability and resectability

IV- LAB.:

- CBC – Liver, kidney functions to evaluate the general condition and fitness for surgery.

Treatment:

Usually the prognosis is bad as the diagnosis is usually late

■ Treatment options:

1- Palliative 2- Radical Surgery

Palliative treatments:

to relieve dysphagia (temporary effects)

Indications: advanced cases – bad general conditions

a)Intubation:

Insertion of metal stent inside the strictured segment to create a path for feeding

Introduced by endoscopy

Examples: Celestin and Souttar tubes

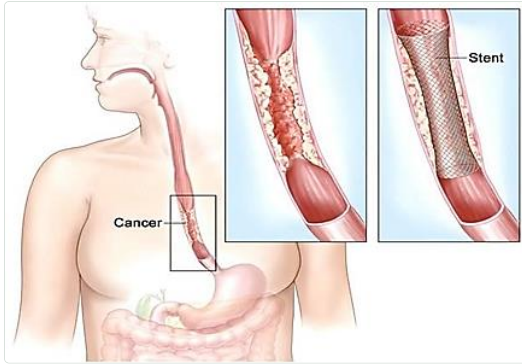


Figure 2.18 — Esophageal stent placement

b)YAG LASER:

Photocoagulation induces coagulation inside the strictured segment to create a path

c) Palliative surgery:

Gastrostomy or jejunostomy used for feeding:



Figure 2.19 — Feeding gastrostomy

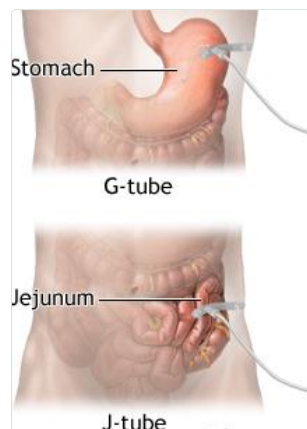


Figure 2.20 — Gastrostomy and jejunostomy



d) Radiotherapy

Updated multimodal management

-A. Early Stage Disease (T1):

- Endoscopic Resection: Endoscopic Mucosal Resection (EMR) or Endoscopic Submucosal Dissection (ESD) is curative for intramucosal (T1) disease without lymphovascular invasion, followed by radiofrequency ablation (RFA) of surrounding Barrett's mucosa.

B. Locally Advanced but Resectable Disease:

Management differs significantly based on pathology:

1. Esophageal Adenocarcinoma (EAC):

- Preferred Standard: Perioperative chem & immunotherapy + surgery.

2. Esophageal Squamous Cell Carcinoma (ESCC):

- Preferred Standard: Neoadjuvant Chemoradiation followed by surgery.

Surgical Approaches (Esophagectomy):

Techniques:

Radical surgery:

Curative surgery – used in early cancer and good general condition. **Aim:** resection of the tumor with safety margin and restore continuity .

- **Ivor Lewis Esophagectomy** : Transthoracic approach (right thoracotomy + laparotomy) with gastric conduit reconstruction.
- **McKeown Esophagectomy**: Three-incision approach (neck, chest, abdomen) for upper or mid-thoracic lesions
- **Transhiatal Esophagectomy**: Abdominal and cervical incisions without formal thoracotomy.

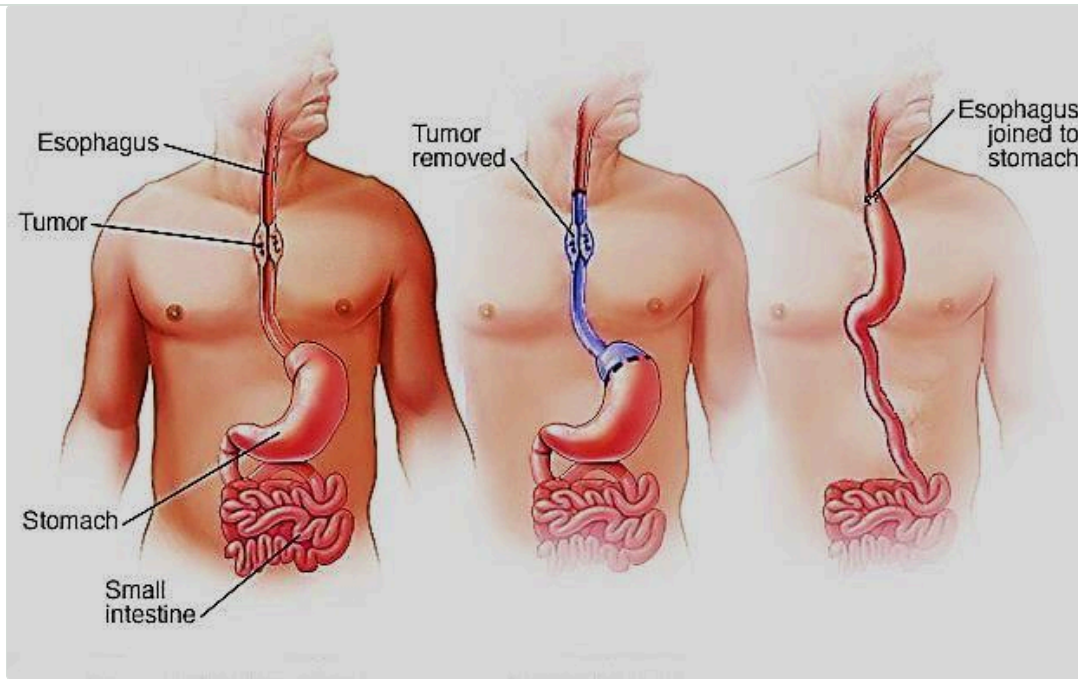


Figure 2.21 — Esophagectomy and gastric reconstruction



THE STOMACH AND DUODENUM



03



Learning Objectives

By the end of this chapter, the medical student will be able to:

1. Analyze the clinical manifestations, diagnostic approach, and management of acid-related gastric and duodenal diseases.
2. Evaluate the clinical presentation, diagnosis, and treatment of peptic ulcer disease (PUD).
3. Formulate modern diagnostic pathways and eradication regimens for *Helicobacter pylori*.
4. Identify clinical manifestations, diagnostic approaches, and surgical indicators for PUD complications.
5. Manage acute and chronic gastrointestinal bleeding.
6. Recognize the clinical presentation, diagnosis, and staging of gastric carcinoma.

Anatomy and Physiology

Gastric Anatomy

The stomach consists of five distinct anatomical regions:

- **Cardia:** Adjacent to the gastroesophageal junction.
- **Fundus:** Dome-shaped region extending above and to the left of the cardia, toward the left hemidiaphragm.
- **Body (Corpus):** Extends from the fundus to the lower end of the lesser curve, known as the incisura angularis.
- **Antrum:** Extends from the incisura angularis to the pyloric canal.
- **Pylorus:** The muscular region between the antrum and the duodenum.

The stomach is bordered by two main functional sphincters: the gastroesophageal sphincter and the pyloric sphincter.

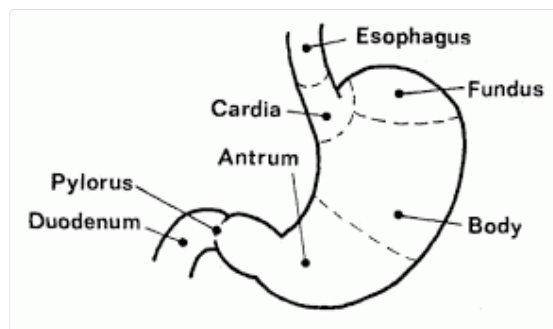


Figure 3.1 — Structure of the stomach



Histological Structure and Secretions

- Upper Two-Thirds (Fundus and Body): The mucosa contains parietal cells that secrete hydrochloric acid and intrinsic factor, and chief cells that secrete pepsinogen to initiate proteolysis.
- Lower One-Third (Antrum): The antral mucosa contains mucus-secreting cells, G cells (which secrete gastrin to stimulate acid production), and D cells (which secrete somatostatin, a potent suppressant of acid secretion).
- Duodenal Mucosa: Features villi and contains Brunner's glands that secrete alkaline mucus. This secretion, alongside pancreatic and biliary inputs, neutralizes acidic gastric chyme entering the duodenum.

CLINICAL PEARL: Gastric Secretory Regulation

- Acid secretion is under tight neural and hormonal control.
- Acetylcholine (vagal stimulation) and Gastrin stimulate acid production directly, and indirectly by releasing histamine from enterochromaffin-like (ECL) cells.
- Somatostatin acts as a primary negative feedback brake, inhibiting both histamine and gastrin release.

The stomach also serves as a protective mucosal barrier composed of cell plasma membranes and a heavy mucus-bicarbonate layer. This barrier shields the epithelium against auto-digestion by acid, pepsin, and ingested irritants (aspirin, NSAIDs, alcohol, and bile salts). Local prostaglandins (PGE₂) are critical to maintaining this barrier by stimulating mucus and bicarbonate synthesis.

Gastritis

Definition

Gastritis: Gastritis is inflammation of the gastric mucosa, usually diagnosed histologically, and may result from infectious, immune, chemical, or drug-related causes.

- **Gastropathy:** A gastric mucosal disorder featuring epithelial injury and subsequent regeneration without associated histological inflammation.

Note: Gastritis and gastropathy are not mutually exclusive and frequently coexist in clinical practice.

Acute Gastritis

Acute gastritis refers to a sudden onset of mucosal inflammation. A disruption in the mucosal barrier triggers an immune response that recruits neutrophils (polymorphonuclear leukocytes) to the site of injury. Severe mucosal damage can progress to acute erosive gastritis, featuring shallow mucosal lesions (erosions), ulcerations, and focal intraluminal bleeding.



Etiology and Pathogenesis

- NSAIDs and Aspirin: Inhibit cyclooxygenase (COX-1), blocking the synthesis of prostaglandin E2. This depletes protective gastric mucus and increases baseline acid output.
- Critical Illness (Stress Ulcers): Extreme physiological stress—such as major surgery, severe trauma, massive burns (Curling ulcers), or central nervous system injury (Cushing ulcers)—induces mucosal ischemia.
- Exogenous Irritants: Elevated intake of alcohol, heavy caffeine consumption, and exposure to cigarette smoke directly irritate and inflame the gastric mucosa.
- Infectious: The early, acute phase of an *Helicobacter pylori* infection.

Signs and Symptoms

Most individuals are asymptomatic or experience mild upper abdominal discomfort, loss of appetite, belching, nausea, and vomiting. Severe erosive cases can present acutely with upper gastrointestinal bleeding (hematemesis or melena).

Diagnosis and Management

Diagnosis begins with a thorough medical history to pinpoint specific triggers (such as chronic NSAID use or alcohol intake). Because acute gastritis is typically short-lived and self-resolving, removing the offending trigger is often sufficient. If bleeding occurs or the diagnosis remains uncertain, an Upper Endoscopy (EGD) is indicated to visualize the mucosa and obtain biopsy samples.

Acid suppression via Proton Pump Inhibitors (PPIs) or Potassium-Competitive Acid Blockers (PCABs) effectively aids symptom relief and accelerates mucosal healing.

Gastritis and NSAIDs

- Aspirin and other NSAIDs deplete mucosal prostaglandins by inhibiting the cyclo-oxygenase (COX) pathway, which leads to mucosal damage.
- Cyclo-oxygenase occurs in two main forms: COX-1, the constitutive enzyme, and COX-2, inducible by cytokine stimulation in areas of inflammation.
- COX-2 specific inhibitors have less effect on the COX-1 enzyme in the gastric mucosa. It still produce gastric mucosal damage but less than with other conventional NSAIDs. -Because of the large number of patients on NSAIDs including low-dose aspirin for vascular prophylaxis, this is a significant problem, particularly in the elderly.

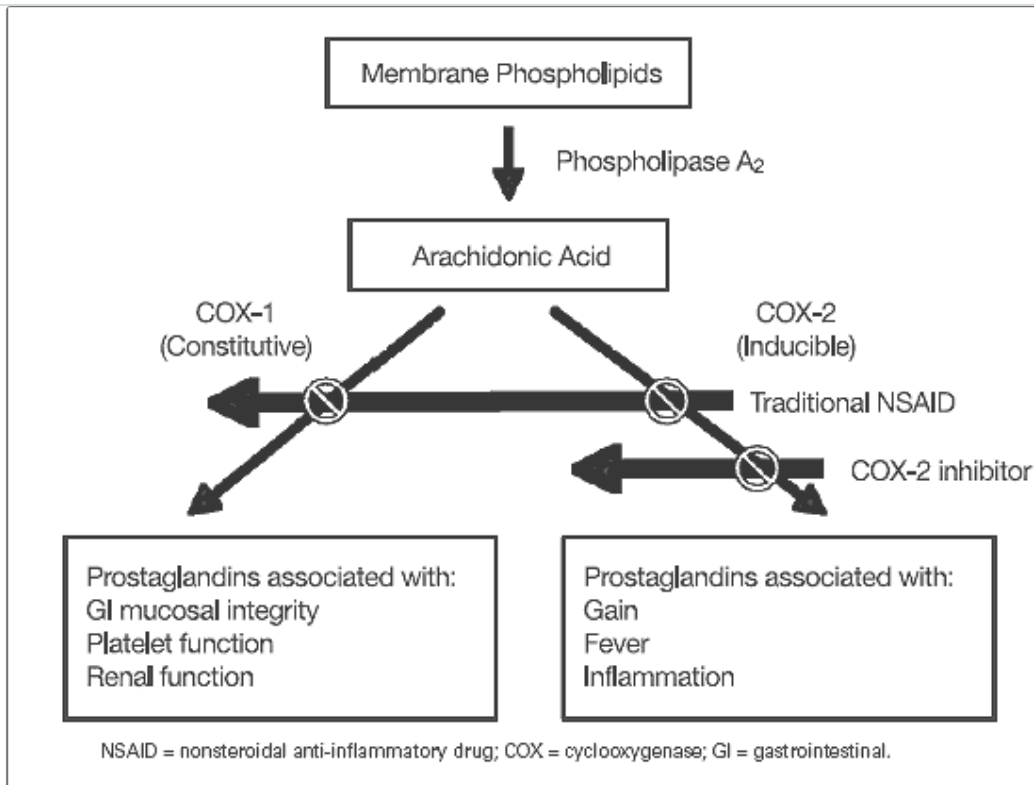


Figure 3.2 — Mechanism of action of traditional NSAID and anti COX-2 inhibitors

Treatment

- Stop the ingestion of NSAIDs.
- A PPI should be given. -H. pylori eradication therapy if H. pylori positive.
- In many people with severe arthritis, stopping NSAIDs may not be possible, therefore use: -A NSAID with low GI side-effects at lowest dose possible or if there is no cardiovascular risk, a COX-2 NSAID can be used

Prophylactic cytoprotective therapy, e.g. PPI or misoprostol (a synthetic analogue of prostaglandin for all high-risk patients, i.e. over 65 years; those with a peptic ulcer history, particularly with complications, and patients on therapy with corticosteroids or anticoagulants. PPIs reduce the risk of endoscopic duodenal and gastric ulcers and are better tolerated than misoprostol, which causes diarrhoea.

Management of NSAID-Induced Ulcers

1. Discontinue the offending NSAID immediately whenever clinically feasible.
2. Initiate daily healing doses of a PPI.
3. If chronic NSAID therapy cannot be stopped due to severe comorbidities (e.g., severe arthritis), transition the patient to the lowest effective dose, consider a selective COX-2 inhibitor, and co-prescribe daily cytoprotective prophylaxis via a PPI or Misoprostol (a synthetic prostaglandin E₂ analogue).



Chronic Gastritis

Chronic gastritis involves long-standing, persistent inflammation of the gastric mucosa. It is characterized histologically by a lymphocytic and plasma cell infiltrate, and is broadly divided into two major distinct pathological types:

- Chronic autoimmune Gastritis
- Helicobacter Pylori Gastritis

Chronic autoimmune gastritis,

It affects the body and fundus of the stomach; and it represents an example of a type IV hypersensitivity. In chronic autoimmune gastritis, there are autoantibodies against H⁺/K⁺ ATPase on parietal cells and intrinsic factor. Autoantibodies against H⁺/K⁺ ATPase decrease gastric acid secretion, thereby causing hypochlorhydria, which is associated with impaired iron absorption and hyperplasia of gastrin cells. As a result, there's an increased secretion of gastrin, hypergastrinemia, which is a hormone that has a trophic effect on enterochromaffin-like cells, or ECL cells; therefore, these individuals are at risk for developing neuroendocrine tumors.

Autoantibodies against intrinsic factor lead to decreased absorption of vitamin B12, resulting in pernicious anemia.

Finally, in chronic autoimmune gastritis chief cells are also lost and this results in decreased levels of serum pepsinogen. Diagnosis of chronic autoimmune gastritis includes endoscopic biopsy, but also detection of anti-parietal cell antibodies, anti-IF antibodies; and levels of serum gastrin and pepsinogen

Helicobacter pylori Gastritis

H. pylori is a slow-growing spiral Gram-negative flagellate urease-producing bacterium which plays a major role in gastritis and peptic ulcer disease.

- It colonizes the mucous layer in the gastric antrum, but is also found in the duodenum in areas of gastric metaplasia.
- Antral gastritis is the usual effect of H. pylori infection. It is usually asymptomatic, although patients without ulcers do sometimes experience relief of dyspeptic symptoms after Helicobacter eradication. Chronic antral gastritis causes hypergastrinaemia due to gastrin release from antral G cells. The subsequent increase in acid output is usually asymptomatic, but can lead to duodenal ulceration

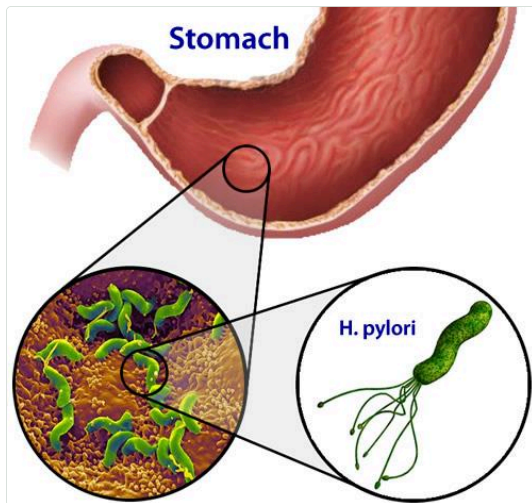


Figure 3.3 — H Pylori gastritis

Epidemiology

The prevalence of *H. pylori* is high in developing countries (80-90% of the population), and much lower (20-50%) in developed countries.

- Infection rates are highest in lower income groups. Infection is usually acquired in childhood; although the exact route is uncertain, it may be fecal-oral or oral-oral. The incidence increases with age.

Pathogenesis

- The pathogenetic mechanisms are not fully understood, with the majority of the colonized population remaining asymptomatic throughout their life.
H. pylori is highly adapted to the stomach environment, exclusively colonizing gastric epithelium and inhabiting the mucous layer, or just beneath.
- The production of urease enables the conversion of urea to ammonium and chloride, which are directly cytotoxic.
- Ulcers are commonest when the infecting strain expresses CagA (cytotoxic-associated protein) and VacA (vacuolating toxin) genes secondary to a more pronounced inflammatory and immune response
- The precise mechanism of duodenal ulceration is unclear, as only 15% of patients infected with *H. pylori* (50-60% of the adult population worldwide) develop duodenal ulcers.
- Factors that have been implicated include: increased gastrin secretion, smoking, bacterial virulence and genetic susceptibility.
- Gastric ulcer (GU) : Gastric ulcers are associated with a gastritis affecting the body as well as the antrum of the stomach (pangastritis) causing parietal cell loss and reduced acid production. The ulcers are thought to occur because of reduction of gastric mucosal resistance due to cytokine production by the *H. pylori* infection or perhaps to alterations in gastric mucus.



■ Other *H. pylori* associated diseases:

- Gastric adenocarcinoma. The incidence of distal (but not proximal) gastric cancer parallels that of *H. pylori* infection in countries with a high incidence of gastric cancer. Serological studies show that people infected with *H. pylori* have a higher incidence of distal gastric carcinoma (Gastric B cell lymphoma). Over 70% of patients with gastric B cell lymphomas (mucosal-associated lymphoid tissue - MALT) have *H. pylori*.
H. pylori gastritis has been shown to contain the clonal B cell that eventually gives rise to the MALT lymphoma.

Diagnosis of *Helicobacter pylori* infection

Diagnosis of *H. pylori* is necessary if the clinician plans to treat a positive result. This is usually in the context of active peptic ulcer disease, previous peptic ulcer disease or

MALT lymphoma, or to 'test and treat' patients with dyspepsia under the age of 55 with no alarm symptoms (i.e. weight loss, anemia, dysphagia, vomiting, or family history of gastrointestinal cancer).

Non-invasive methods

■ Stool antigen test. This is beginning to be the method of choice to determine the *H. pylori* status. A specific immunoassay using monoclonal antibodies for the qualitative detection of *H. pylori* antigen is now widely available. The overall sensitivity and specificity is over 95%. It is useful in the diagnosis of *H. pylori* infection and for monitoring efficacy of eradication therapy. Patients should be off PPIs for 2 weeks but can continue with H2 blockers.

■ ¹³C Urea breath test. This is a quick and reliable test for *H. pylori* and can be used as a screening test. The measurement of ¹³CO₂ in the breath after ingestion of ¹³C urea. The test requires a mass spectrometer. The test is sensitive (90%) and specific (96%), but the sensitivity can be improved by insuring the patient has not taken antibiotics in the 4 weeks prior and proton pump inhibitors in the 2 weeks before the test.

■ Serological tests detect IgG antibodies are reasonably sensitive (90%) and specific (83%). They have been used in diagnosis and in epidemiological studies. IgG titres may take up to 1 year to fall by 50% after eradication therapy and therefore are not useful for confirming eradication or the presence of a current infection.

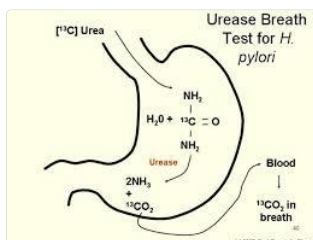


Figure 3.4 — Urea breath test for detection of *H. pylori*



■ Invasive (endoscopy)

■ **Biopsy urease test.** Gastric biopsies, usually antral are added to a substrate containing urea and phenol red. If *H. pylori* are present, the urease enzyme that they produce splits the urea to release ammonia which raises the pH of the solution and causes a rapid colour change (yellow to red). This enables a patient's *H. pylori* status to be determined before they leave the endoscopy unit. The test may be falsely negative if patients are taking PPIs or antibiotics.

■ **Histology.** *H. pylori* can be detected histologically on routine (Giemsa) stained sections of gastric mucosa obtained at endoscopy.

■ **Culture.** Biopsies obtained can be cultured on a special medium, and in vitro sensitivities to antibiotics can be tested. This technique is typically used for patients with refractory *H. pylori* infection to identify the appropriate antibiotic regimen and routine culture is rare.

Eradication therapy

Proton pump inhibitor or PCAB twice / day + Amoxicillin 1g twice / day + clarithromycin 500 mg twice /day . Treatment was given for 14 days.

However due to global clarithromycin resistance, classic standard triple therapy is no longer recommended as an empirical first-line choice unless local resistance is confirmed to be low.

1. Bismuth Quadruple Therapy (Preferred First-Line Choice)

- **Acid Suppressant** : Standard-dose PPI twice daily or Vonoprazan (PCAB) 20 mg twice daily.
- **Bismuth Subcitrate** : 120–300 mg four times daily.
- **Tetracycline** : 500 mg four times daily.
- **Metronidazole** : 250–500 mg four times daily.
- *Duration* : 10–14 days

2. PCAB Dual Therapy

- **Vonoprazan** : 20 mg twice daily.
- **Amoxicillin** : 1g three times daily (total 3g/day).
- *Duration* : 14 days.

3. Rescue Therapy

- Implements quinolones (e.g., Levofloxacin 500 mg once daily + PPI + Amoxicillin) when standard first-line options fail. Avoid reusing antibiotics previously taken by the patient.

In developing countries reinfection is common, compliance with treatment may be poor and metronidazole resistance is high (>50%) as it is frequently used for parasitic infections), so failure of eradication is common





SUBTYPES OF CHRONIC GASTRITIS

Parameter	Chronic Autoimmune Gastritis	Helicobacter pylori Gastritis
Anatomical Site	Fundus and Body (spares the gastric antrum).	Antrum initially; can progress to the body over time(pangastritis).
Mechanism	Type IV Hypersensitivity ; autoantibodies target H ⁺ /K ⁺ ATPase on parietal cells and Intrinsic Factor(IF).	Chronic active inflammation induced by bacterial virulence factors (CagA, VacA).
Serum Gastrin	Markedly elevated (Severe hypergastrinemia due to lack of acid negative feedback on G cells).	Variable; can cause hypergastrinemia locally if limited to the antrum.
Gastric pH	Elevated (Achlorhydria / Hypochlorhydria) due to parietal cell loss.	Normal to low (antral disease); elevated if it progresses to pangastritis.
Hematologic Signs	Pernicious Anemia (macrocytic anemia due to Vit B ₁₂ malabsorption) and impaired iron absorption.	Iron Deficiency Anemia (secondary to occult bleeding from associated peptic ulcers).
Neoplastic Risk	Trophic effect of gastrin causes ECL cell hyperplasia → Neuroendocrine (Carcinoid) Tumors ;Adenocarcinoma.	Distal Gastric Adenocarcinoma and MALT Lymphoma (B-cell lymphoma).
Diagnosis	Serology (anti-parietal/anti-IF antibodies), low serum pepsinogen, elevated gastrin, fundic biopsy.	Non-invasive testing (Stool Antigen, Breath Test) or endoscopic biopsy.

ACUTE AND CHRONIC GASTRITIS

Diagnostic Feature	Acute Gastritis	Chronic Gastritis
Onset & Duration	Sudden onset; typically self- limiting within days to weeks.	Insidious onset; long- standing, persistent mucosal inflammation over years.
Primary Cell Type	Neutrophils (Polymorphonuclear leukocytes).	Lymphocytes and Plasma cells .
Dominant Etiology	NSAIDs/Aspirin, alcohol, critical illness (ischemic stress), or early acute H.pylori phase.	2 Types : Autoimmune and . : Chronic active H. pylori colonization.
Pathophysiology	Direct mucosal irritation, toxic injury, or acute ischemia disrupting the surface barrier.	Progressive immune-mediated destruction or persistent bacterial-induced cytokine inflammation.
Endoscopic Findings	Hyperemic mucosa, punctate petechiae, mucosal erosions, and shallow lesions.	Atrophic, thinned mucosa with visible submucosal vessels, or nodular architecture.
Malignancy Risk	No increased risk of malignancy.	Increased risk of gastric adenocarcinoma and MALT lymphoma.

There are two types of chronic gastritis: chronic autoimmune gastritis and Helicobacter pylori-associated gastritis.

PEPTIC ULCER DISEASE

- A peptic ulcer consists of a break in the superficial epithelial cells penetrating down to the muscularis mucosa of either the stomach or the duodenum; there is a fibrous base and an increase in inflammatory cells.



- Erosions, by contrast, are superficial breaks in the mucosa alone. Most DUs are found in the duodenal cap; the surrounding mucosa appears inflamed, hemorrhagic or friable (duodenitis). GUs are most commonly seen on the lesser curve near the incisura, but can be found in any part of the stomach.

Epidemiology of peptic ulcer disease :

- Duodenal ulcers affect approximately 10% of the adult population and are two to three times more common than gastric ulcers.
- Ulcer rates are declining rapidly for younger men and increasing for older individuals, particularly women.
- Both DUs and GUs are common in the elderly.
- There is considerable geographical variation, with peptic ulcer disease being more prevalent in developing countries related to the high H. pylori infection.
- In the developed world the percentage of NSAID-induced peptic ulcers is increasing, as the prevalence of H. pylori declines.

Clinical features of peptic ulcer disease

- The characteristic feature of peptic ulcer is recurrent, burning epigastric pain. It has been shown that if a patient points with a single finger to the epigastrium as the site of the pain this is strongly suggestive of peptic ulcer disease.
- The relation-ship of the pain to food is variable and on the whole not helpful in diagnosis. The pain of a DU classically occurs at night (as well as during the day) and is worse when the patient is hungry, but this is not reliable. The pain of both gastric and duodenal ulcers may be relieved by antacids. Nausea may accompany the pain; vomiting is infrequent but can relieve the pain. Anorexia and weight loss may occur, particularly with GUs. Persistent and severe pain suggests complications such as penetration into other organs.
- Back pain suggests a penetrating posterior ulcer. Severe ulceration can occasionally be symptomless, as many who present with acute ulcer bleeding or perforation have no preceding ulcer symptoms. -Untreated, the symptoms of a DU relapse and remit spontaneously. The natural history is for the disease to remit over many years due to the onset of atrophic gastritis and a decrease in acid secretion.
- Examination is usually unhelpful ; epigastric tenderness is common in non-ulcer dyspepsia.

Investigation of suspected peptic ulcer disease

- Patients under 55 years of age with typical symptoms of peptic ulcer disease who test positive for H. pylori can start eradication therapy without further investigation.
- Endoscopic diagnosis and exclusion of cancer is required in older patients .All gastric ulcers must be biopsied to exclude an underlying malignancy and should be followed up endoscopically until healing has taken place.
- Endoscopy is required in all patients with alarm symptoms' .



- Patients with a risk of bleeding or those with complications, i.e. haemorrhage or perforation, should always have a stool test for *H. pylori* or a ¹³C urea breath test 6 weeks after the end of treatment to be sure eradication is successful.



Figure 3.5 — Endoscopic view of a duodenal ulcer

Zollinger-Ellison Syndrome (ZES)

Zollinger-Ellison syndrome is a rare, severe cause of aggressive peptic ulcer disease driven by a gastrinoma. A gastrinoma is a neuroendocrine tumor typically located within the duodenal wall or the pancreas that autonomously secretes abnormal, excessive amounts of gastrin.

- **Pathophysiology:** Unregulated gastrin hypersecretion continuously overstimulates parietal cells, causing massive parietal cell hyperplasia and extreme hydrochloric acid output. This overwhelming acid load completely destroys the native mucosal defense mechanisms.
- **Clinical Presentation:** Multiple, recurrent ulcers extending into atypical structural locations (such as the distal duodenum or proximal jejunum) that remain completely refractory to standard medical management. This is frequently accompanied by chronic watery diarrhea due to structural mucosal injury and acid-induced inactivation of pancreatic digestive enzymes.
- **Diagnosis:** Confirmed by documenting markedly elevated fasting serum gastrin levels paired with gastric fluid pH verification (≤ 2).
- **Genetic Profile:** May occur sporadically or as an autosomal dominant manifestation of Multiple Endocrine Neoplasia Type 1 (MEN1).

Pharmacology of treatment of Gastritis and Peptic ulcer

A- ANTI - SECRETORY DRUGS

1. Proton Pump Inhibitors (PPIs)

- Omeprazole - Esomeprazole - Lansoprazole - Dexalansoprazole - Pantoprazole - Rabeprazole (oral - IV - 20-40 mg/d)
- **Mechanism of Action**
- Act on the final step of HCl secretion → Most effective acid suppressants.



- They are prodrugs → after absorption → parietal cells → secretory canaliculi → converted by acid into active form → inhibits H^+/K^+ ATPase (proton pump) → Inhibit basal & meal stimulated acid secretion (98%, 1-2 h after 1st dose).
- They have anti- *H. pylori* effect (lansoprazole most potent)

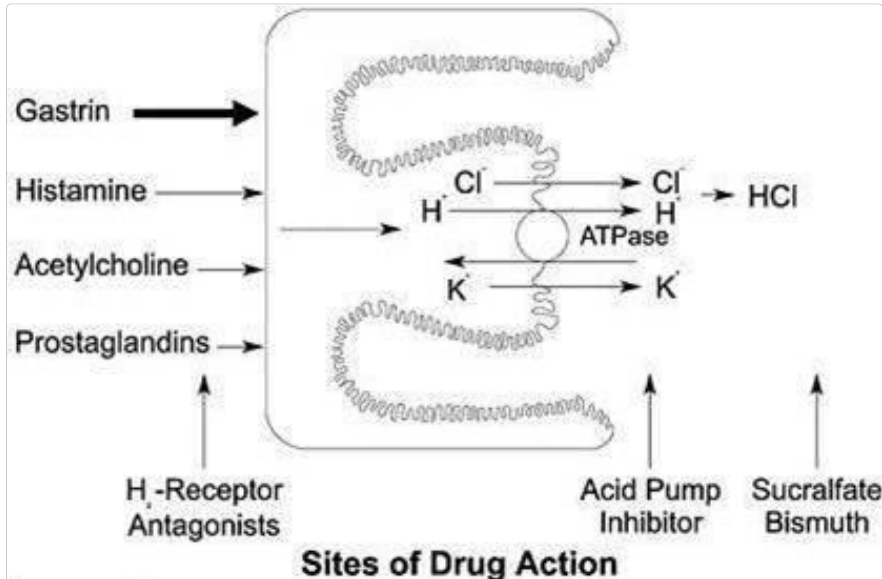


Figure 3.6 — Sites of drug action in gastric acid secretion

■ Pharmacokinetics

1. Weak bases → destroyed by gastric acidity & ionized in acid stomach (not readily absorbed) → given in enteric-coated form
2. Should be taken 30 – 60 minutes before meals (on empty stomach): only inhibit active pumps - Food → ↓ bioavailability by 50%
3. Repeated administration → ↓ gastric acidity → ↑ bioavailability.
4. Long duration (onset 1 hour & duration 24 hours ($t_{1/2} = 1$ h but PPIs covalently bind to PP and synthesis of new pump requires 24 hours)).
5. Metabolized in the liver.

■ Adverse effects

1. GIT: abdominal pain, nausea, vomiting, diarrhea or constipation.
2. CNS: headache, dizziness, somnolence.
3. Hypochlorhydria → ↑ Risk of infection (e.g. hospital acquired pneumonia & *H. pylori*) and Gastric tumor (in animals): bacterial colonization → ↑ carcinogenic nitroso compounds.
4. Vitamin B12 deficiency (on long term use)
5. Enzyme inhibition (clinically insignificant but inhibit activation of clopidogrel).



1. Potassium Competitive Acid Blockers (P-CAB)

- Vonoprazan, revaprazan, tegoprazan (newer version of PPIs): differs from PPIs in:

Omeprazole	Vonoprazan
Irreversible PP inhibition (↓ active pumps)	Competitive reversible PP inhibition (↓ both active and inactive PP)
Takes 3-5 days for maximal acid suppression	Take 1 day for maximal acid suppression
T1/2 = 1-2 hrs (but longer duration of action being irreversible)	T1/2 = 6-9 hrs
Prodrug	Not prodrug
Not acid stable (Taken 30-60 minutes before meals + need acid for activation)	Acid stable (Taken before or after meals + Not need acid for activation)

2. H2-Receptor Antagonists

- Cimetidine (prototype) – ranitidine (discontinued due to risk of cancer) - famotidine – nizatidine (oral – IV)
- **Mechanism & Pharmacological Actions**
- Reversible competitive antagonists of H2 receptors on parietal cells → Complete inhibition of histamine mediated but partial inhibition of ACh or gastrin mediated HCl secretion.
- Potent inhibition of nocturnal or fasting but less inhibition of daytime (meal-stimulated) acid secretion.
- **Individual H2 Antagonists**

1. Cimetidine: limited use due to its enzyme inhibiting, antiandrogenic & CNS side effects. 2. Ranitidine, Famotidine, Nizatidine developed to minimize adverse effects of cimetidine (More potent than cimetidine):

Ranitidine (discontinued)	Nizatidine 150 mg	Famotidine 20 mg
Less enzyme inhibiting & antiandrogenic effects.	No enzyme-inhibiting or antiandrogenic effects.	No enzyme-inhibiting or antiandrogenic effects.
Anti H. pylori effect (most efficient).	The most potent.	Cytoprotective effect.
50% bioavailability (↑hepatic first pass effect)	100% bioavailability (no 1st pass effect)	50% bioavailability. 50% decomposed by gastric acidity.

Eliminated by hepatic metabolism → renal excretion of the drugs or their metabolites → dose reduction if ↓ Creatinine clearance or severe liver disease

- **Adverse Effects of H2 Blockers (Mostly with cimetidine)**

1. Headache, diarrhea or constipation.
2. Tolerance, rebound hyperacidity & recurrence (upregulation of H2 receptors).
3. CNS: sedation, confusion, hallucination (mostly with cimetidine especially in elderly, renal or hepatic dysfunction).



4. Antiandrogenic, ↑ prolactin, gynecomastia, galactorrhea, impotence (mostly with cimetidine).
5. Enzyme inhibition → drug interactions (mostly with cimetidine).
6. Hypotension & bradycardia if given rapidly IV in intensive care (Block cardiac H₂ receptors)

Advantages of PPIs and P-CABs over H₂ antagonists:

1. Higher efficacy → better symptomatic relief & higher response rates.
2. More prolonged effect (= 24hs → given in single daily dose; versus twice daily dosing with H₂ antagonists).
3. More effective in anti H. pylori regimen.

■ **Indications of Antisecretory Drugs**

- 1- Peptic ulcer: PPIs preferred especially in severe cases. H₂ antagonists in mild or moderate cases (healing is delayed for 8 weeks & recurrence is common).
- 2- Anti- H. Pylori regimens: PPIs preferred.
- 3- GERD: PPIs preferred. H₂ antagonists are used to cover the nocturnal HCl secretion.
- 4- Zollinger Ellison syndrome: PPI are preferred. High dose of parenteral preparations are given early to cover vomiting period.
- 5- Stress ulcer: to ↓ risk of bleeding.
- 6- Acute gastritis in children & gastric erosions (Parenteral)
- 7- Bleeding oesophageal varices: to prevent recurrence (Parenteral)
- 8- To avoid aspiration pneumonia during surgery e.g. obstetric anesthesia before caesarian section (Parenteral)



MUCOSAL PROTECTIVES (CYTOPROTECTIVES)

	1. Sucralfate	2. Misoprostol	3. Colloidal Bismuth
Mech.	It is an aluminium salt of sulfated sucrose. In acid medium → -ve sulfate groups bind +ve proteins in the ulcer base → protective barrier (also inactivates bile and pepsin) ↑ secretion of bicarbonate, mucus & PGs.	Synthetic PGE1 analog cytoprotective through: ↓ HCl secretion by ↓ histamine-stimulated c-AMP production. ↑ mucosal blood flow ↑ secretion of mucus & bicarbonate.	Anti- H pylori activity (main effect). Mucosal protective: Chelates proteins in ulcer base → protective coat against acid & pepsin (inactivates pepsin) ↑ secretion of bicarbonate, mucus & PGs.
Uses:	Not used - Stress ulcer prophylaxis (mainly). Antacids or H2 blockers and some drugs e.g. tetracycline are given 1-2 hour apart.	Selective for NSAID-induced gastric ulcer if PPIs fail. Prophylactically with NSAIDs or corticosteroids.	Anti- H. pylori therapy
ADRs	Nausea, vomiting, dry mouth. Flatulence, constipation. Al+3 toxicity: chelates phosphates → osteomalacia & encephalopathy in renal diseases	Diarrhea & colic (most common). Vaginal bleeding & uterine contractions (CI in pregnancy → abortion).	Teeth & tongue discoloration - black stools (confused with bleeding). Encephalopathy and neurotoxicity especially in renal disease → not give >2 months & not restarted within 1 year.

ANTI H PYLORI THERAPY

- Eradication of H **pylori** infection results in rapid healing & ↓ recurrence.
- It is achieved by combination therapy with **High efficacy anti- H. pylori Triple therapy**.
- Triple therapy: 10-14 days: Lansoprazole + Clarithromycin + Amoxicillin (or metronidazole in patients allergic to amoxicillin).
- Quadruple therapy: 10-14 days: PPI + metronidazole 250 mg + bismuth subsalicylate 300 mg + tetracycline 500 mg four times daily
- Sequential therapy: PPI + amoxicillin 1 g twice a day for 5 days followed by PPI + clarithromycin 500 mg and tinidazole/ metronidazole 500 mg twice a day for 5 days
- Or: PPI + amoxicillin 1 g twice a day + levofloxacin 250 or 500 mg twice a day for 10 days
- Followed by 4-6 weeks: PPI therapy to allow ulcer heal.

N.B.: PPIs may be substituted by P-CABs in Peptic acid diseases and eradication of H. Pylori

ANTACIDS

- Antacids are weak bases that reduce gastric acidity by neutralizing HCL
- **Pharmacological Actions**
 1. Neutralize gastric acid (duration 1-2 hrs).
 2. Decrease peptic activity (pepsin is inactive at pH > 4.5).



3. Reduce *H. pylori* colonization.
4. ↑ mucosal PGs production (promote defense mechanism).
5. Adsorbent (to pepsin, bile & toxins), astringent (Precipitate surface proteins) and demulcent (soothing) effects (with Al+3 & Mg+2 salts only).

◦ **Therapeutic Uses (taken 1 and 3 hours after meals & at bedtime)**

1. Rapidly relieve pain & help healing of peptic ulcer.
2. Relieve dyspeptic symptoms.
3. GERD (rapid pain relief).

Drug Interactions:

Antacids can affect rates of dissolution, absorption, bioavailability and renal elimination of many drugs through:

1. Changing gastric or urinary pH.
2. Delaying gastric emptying.
3. Chelation: Al+3 salts can form insoluble complexes with fluoroquinolones, tetracyclines, digoxin and iron salts → ↓ absorption (do not give antacids within 2 hrs from administration of these drugs).

Types of Antacids

Local Antacids	Systemic Antacids
Al+3 Hydroxide Disadvantage: constipation. Mg+2 Salts Disadvantage: diarrhea. Ca+2 carbonate Disadvantages Constipation. Rebound hyperacidity (Ca+2 → ↑ acid secretion) → ↓ pain relief & healing. Milk-alkali syndrome (systemic alkalosis - hypercalcemia). ↑ CO₂ → flatulence & eructation.	Na HCO₃ Advantages Rapid onset. Potent (↑ pH > 7). Disadvantages Systemic alkalosis on long term use. Rebound hyperacidity. Na⁺ content is dangerous in HF, hypertension or renal disease. ↑ CO₂ → flatulence, eructation. Alkaline urine → renal stones. N.B.: Sodium bicarbonate is added for effervescent formulations or in immediate release PPIs preparations (e.g. omeprazole).

Choice of antacids

- Al+3 hydroxide & Mg+2 salts do not result in CO₂ release or systemic alkalosis thus they are the most commonly used antacids, alone or combined (e.g. Magaldarate - to neutralize the effects of each other on bowel habit).
- Antacids should be cautiously used in elderly & renal impairment since:
- Al+3 antacids → chelate phosphates → osteomalacia, hypophosphatemia, encephalopathy & ↑ risk of Alzheimer dementia.
- Mg+2 antacids → CNS depression.



Precautions to Avoid NSAIDs-induced Ulcer

1. Use low doses and avoid combinations of NSAIDs (↑ toxicity without added benefit).
2. Use topical preparations whenever possible.
3. Co-administer PPIs or misoprostol.
4. Avoid combination with corticosteroids except in selected cases.
5. Use cautiously in the elderly.

Gastrointestinal Bleeding

Acute Upper GI Bleeding

- Hematemesis: vomiting blood.
- Melena: black, tarry stool due to altered blood.
- Fresh blood per rectum may occur with massive upper GI bleeding, usually with shock.

Etiology:

- Peptic ulcer disease : one of the commonest causes.
 - NSAIDs/aspirin → gastric and duodenal ulcers/erosions and bleeding.
 - Anticoagulants increase bleeding severity rather than directly causing bleeding.
 - Other important causes: oesophageal varices, Mallory-Weiss tear, gastric cancer and Dieulafoy lesion.

Clinical approach

a) Initial Management: ABCDE

A – Airway: ensure airway patency. B – Breathing: assess and treat life-threatening respiratory problems. C – Circulation: assume hypovolaemia until proven otherwise; 2 large-bore IV lines and restore circulating volume. D – Disability: assess consciousness and neurological status. E – Exposure: complete examination while maintaining patient dignity and preventing heat loss.

b) Unstable Patient

- Monitor pulse and BP frequently.
- Blood tests: Hemoglobin, urea/electrolytes, liver function, coagulation profile, group & cross-match.
- IV fluids and blood transfusion when indicated.
- Oxygen therapy and keep NPO until endoscopy.
- Stop NSAIDs/aspirin where appropriate; interruption of antiplatelet therapy requires specialist assessment because of thrombosis risk.



- Urgent endoscopy after resuscitation.
- Persistent bleeding → repeat endoscopy → angiographic embolization → surgery if uncontrolled.
- Management is ideally multidisciplinary.

NB : Hemoglobin levels are generally a poor indicator of the need to transfuse because anemia does not develop immediately .

Endoscopic Treatment

Endoscopy provides diagnosis and hemostasis.

High-risk bleeding ulcers may require combination therapy, such as:

- Adrenaline injection + thermal therapy
- Endoscopic clipping

Variceal bleeding → endoscopic band ligation.

For actively bleeding ulcers or a visible vessel, IV PPI therapy reduces rebleeding.

Initial Endoscopy

- Bleeding ulcer / recent bleeding stigmata: Endoscopic hemostasis with ≥2 methods:
 - Diluted epinephrine injection + thermal therapy or clips.
 - H. pylori: Take antral biopsies.
 - Positive urease test → reliable.
 - Negative urease test → confirm with gastric histology.

Drug Therapy

- IV Omeprazole 80 mg bolus → 8 mg/h infusion for 72 h
 - Indicated for active bleeding or visible vessel.
 - Reduces rebleeding and need for surgery.

Persistent / Recurrent Bleeding

- Repeat endoscopy to identify and treat bleeding source.
 - If unsuccessful → angiography + embolization by interventional radiology.
 - Surgery if bleeding remains persistent/uncontrolled → primarily to control hemorrhage.

Causes of Upper GI Bleeding:

- oesophageal varices.
 - Mallory-Weiss Tear.
 - Mucosal tear at the gastro-oesophageal junction.



- Usually follows retching or coughing.
- Most bleeding stops spontaneously.
- Early endoscopy confirms the diagnosis and allows treatment.
- Dieulafoy Lesion
- Large submucosal vessel that bleeds without an associated ulcer or erosion.
- May cause intermittent or massive haemorrhage.
- Requires prompt endoscopic diagnosis and haemostasis.
- Peptic Ulcer Bleeding
- Treat bleeding endoscopically.
- Eradicate *H. pylori* when present.
- Continue PPI therapy to promote ulcer healing.
- Persistent/recurrent bleeding → angiographic embolization or surgery.

Acute Lower GI Bleeding

- Major bleeding: diverticular disease, ischemic colitis.
- Small-volume bleeding: hemorrhoids, anal fissures.

Assessment

- Start with ABCDE.
 - Unstable patients require resuscitation as for upper GI bleeding.
 - Focused history and examination, including digital rectal examination.
 - Investigations include CBC, coagulation profile, blood grouping/cross-match and appropriate imaging/endoscopy.

Painless hematochezia	Painful hematochezia
• Diverticular bleeding • Colorectal cancer • Angiodysplasia • Dieulafoy lesion	• Infectious /inflammatory colitis • Inflammatory bowel disease. • Haemorrhoids /anal fissure • Radiation proctitis/telangiectasia

Figure 3.7 — Painless and painful hematochezia

Diagnostic Modalities:

- Proctoscopy: anorectal causes.
 - Flexible sigmoidoscopy/colonoscopy: IBD, cancer, diverticular disease, ischemic colitis and angiodysplasia.
 - Angiography: selected cases of ongoing bleeding, particularly vascular lesions.



NB: Most acute lower GI bleeding stops spontaneously; surgery is rarely required.

Isolated episodes of rectal bleeding in the young (<45years) usually only require rectal examination and flexible sigmoidoscopy because the probability of a significant proximal lesion is very low unless there is a strong family history of colorectal cancer at a young age, Individual lesions are treated as appropriate.

Chronic GI Bleeding

Usually presents as iron-deficiency anaemia.

- In men and postmenopausal women, unexplained iron-deficiency anaemia should prompt investigation for GI blood loss, particularly malignancy and coeliac disease.
- Common causes include lesions capable of causing acute bleeding; hookworm is an important worldwide cause.
- If the bleeding site is unclear, investigate both upper and lower GI tract.

Investigation

1. Upper GI endoscopy + duodenal biopsy → exclude coeliac disease.
2. Colonoscopy → identify and biopsy/remove lesions.
3. If negative → small-bowel evaluation, with capsule endoscopy as the preferred diagnostic test.
4. Positive capsule findings may be treated using balloon-assisted enteroscopy.
5. CT colonography may be used when colonoscopy is unsuitable.

Treatment

- Treat the underlying cause.
 - Iron replacement for iron-deficiency anemia.
 - IV iron or maintenance transfusion may be required in selected patients.

Complications of peptic ulcer

A. Acute complications: (surgical emergency)

1. Bleeding peptic ulcer
2. Perforated peptic ulcer

B. Chronic complications: (surgical indications)

1- Stenosis:

- Pyloric stenosis
- Duodenal obstruction



- Hour glass stomach

2- Malignant change:

malignant gastric ulcer

I-Perforated peptic ulcer

Pathology:

- Perforation of peptic ulcer occurs due to deepening of the ulcer through the wall
- Perforation occurs due to exposure to ulcerogenic agent
- Chronic DU > Chronic GU
- Anterior wall ulcer > posterior wall
- According to the rate of perforation and the size of perforation the degree of peritonitis develops

Stages:

a.Chemical irritation:

irritation of the peritoneal membrane by the escaped acid contents first at the upper abdomen then escape of the contents to the supra-hepatic, hepato-renal and right para-colic gutter to the right iliac fossa

b.Stage of illusion:

in which the peritoneal membrane secretes a large amount of alkaline fluids to neutralizes the acidic fluid → temporary relieve

c.Stage of septic peritonitis:

infection occurs by the gut bacteria (gram -ve bacilli) → worsen the condition and may lead to septic shock

Clinical presentation:

1. history of peptic ulcer dyspepsia

2. Chemical irritation :

- Sudden severe pain in the epigastrium then becomes generalized
- General examination: tachycardia, normal temperature, rapid respiration
- Local exam: severe tender epigastrium then become generalized + rebound tenderness and guarding rigidity extends to right iliac fossa

3. stage of illusion:

- Partial relief of symptoms due to alkaline peritoneal effusion
- **Generally** : Tachycardia, low grade fever and looks better but still ill.



Local Examination :

Inspection: Abdominal distension. (ileus + effusion)

Palpation: Tenderness, rebound tenderness and guarding decreased but still present
Percussion: hyperresonance + shifting dullness

Auscultation: silent abdomen (due to ileus)

4. Stage of septic peritonitis:

- Pain is more severe agonizing and generalized
- **Generally** : Tachycardia, high **fever**, tachypnea and toxic look
- **Local Examination** :
 - Inspection: Abdominal distension, Palpation: tenderness, rebound and board like rigidity
 - Percussion: hyper-resonance + shifting dullness
 - Auscultation: silent abdomen (due to ileus)

How to diagnose perforated peptic ulcer:

1. Clinical suspicion

2. Investigations

How to investigate?

A. Radiological:

1. Plain x-ray abdomen in erect position (very important):

Air under diaphragm (= perforated viscus)

2. Abdominal ultra-sound or C.T. abdomen (basic investigation)

- Detect the presence of peritoneal fluid collection
- Exclude other causes of acute abdominal pain as acute cholecystitis, pancreatitis or leaking abdominal aortic aneurysm

3. Gastro-graffin meal:

may be used to detect escape of the dye into the peritoneal cavity in doubtful cases.(not commonly used)

B. Lab.:

- CBC: shooting Leucocytosis.
 - Serum amylase and lipase to exclude pancreatitis.
 - Cardiac enzymes: to exclude MI (in doubtful cases)



Figure 3.8 — Air under diaphragm

D.D. from all causes of acute abdomen

- 1- Acute cholecystitis
- 2- Acute pancreatitis
- 3- Acute peptic ulceration and gastritis
- 4- Acute liver abscess
- 5- Acute pyelonephritis
- 6- MVO
- 7- Primary peritonitis
- 8- Strangulation
- 9- Others as MI

You need both clinical diagnosis + investigations to reach proper diagnosis.

Management:

General treatment (preparation for surgery):

- Insertion of nasogastric tube
 - Intravenous fluids (ringer and saline)
 - Broad spectrum antibiotics for infection especially against gram negative bacilli (third generation cephalosporins)
 - Intravenous proton pump inhibitor
 - Sedative for pain.

Surgical exploration of the abdominal cavity:

- Mid line incision
 - Thorough examination of the abdominal cavity to detect other pathologies
 - Peritoneal toilet and irrigation with normal saline and remove all septic collections in different peritoneal spaces.
 - Determine the site of perforation (gastric or duodenal)
 - **Closure of the perforation with omental patch fixed with 3 or 4 overlying**



stitches at the site of perforation

- In case gastric ulcer in suspicious case for malignancy gastrectomy can be done.
- Leave drains inside the peritoneal cavity

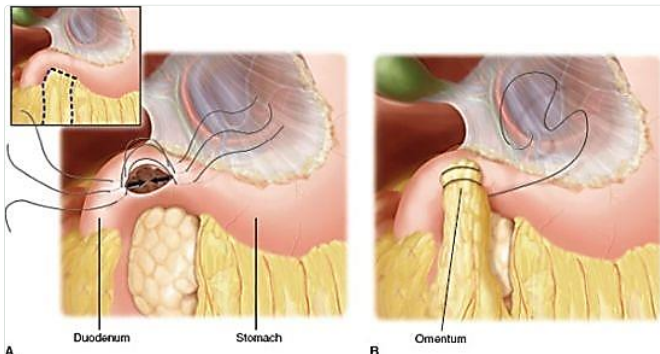


Figure 3.9 — Closure + Omental patch closure

II- Surgery for bleeding peptic ulcer:

Indications:

- 1- Failure the above measures to control bleeding with persistent shock state
- 2- Recurrence of bleedings with failed conservative measures.
- 3- Old atherosclerotic patient (prone for severe and recurrent bleeding).
- 4- Endoscopic evident of large vessel at the floor of the ulcer (prone for sudden severe bleeding).

Type of surgery:

- exploration of the **bleeding duodenal ulcer** and **under-running** of the ulcer base with sutures + **vagotomy (commonly used procedure)**
- In bleeding suspicious **gastric ulcer**, partial gastrectomy might be required.

III- Pyloric obstruction (stenosis)

Problem solving (history of ulcer dyspepsia + picture of Gastric Outlet Obstruction)

Pathogenesis:

long standing pre-pyloric or duodenal ulcer healed by excessive fibrosis leads to narrowing, stenosis and gastric outlet obstruction.



Clinical presentations:

Symptoms:

- History of long duration with ulcer dyspepsia
 - Pain: dull aching, in the central part of the upper abdomen. Pain was more in empty stomach and relieved with intake of food. Pain is periodic in nature.
 - Vomiting: is projectile in nature, contains food materials taken more than 12 hours ago, more in the evening, sour taste (acidic) and is colorless (does not contain bile).
 - Abdominal distension (dilated stomach)
 - Patient complains of sensation of a rolling mass in the abdomen moving from left to right more after taking meals.

General examination:

- In severe cases signs of dehydration (dry tongue, sunken eyes,...)

Local examination:

- Inspection: upper abdominal distension – visible peristalsis running from left to right • Palpation: tender epigastrium – succussion splash (retained fluids)
 - Percussion: hyper-resonance
 - Decreased intestinal sounds

Complications:

- Dehydration
- Persistent and recurrent vomiting lead to hypo-natremia, hypokalemia, hypocalcemia, hypochloremia and metabolic alkalosis + paradoxical aciduria
- Repeated aspiration with chest infection

Differential diagnosis: (clinical + investigational)

1. Gastric outlet obstruction due to chronic duodenal ulcer
2. Malignant tumor at the pyloric area
3. Extrinsic compression near the pylorus due to enlarged lymph node mass (due to lymphoma for example)
4. Lymphoma of distal stomach
5. Gastric outlet obstruction due to a bezoar ingestion (foreign body).
6. Cancer head of the pancreas causing gastric outlet obstruction.
7. Adult hypertrophic pyloric stenosis.



Investigations:

1) Upper endoscopy:

- Markedly dilated stomach
 - Excess old food residue in the stomach
 - Signs of mucosal irritation, hyperemia, due to stasis
 - Evidence of scarring at the first part of duodenum or pyloric canal
 - Biopsy from suspicious lesions
 - Exclude other luminal causes of gastric outlet obstruction.

2) Barium meal:

- hugely dilated stomach (may reach up to the pelvis)
 - delayed passage of dye to the duodenum
 - Soap-dish appearance (fluid level in the stomach)

3) Abdominal ultra-sound:

detect other causes of Gastric Outlet Obstruction

4) Other radiological investigation as C.T. when suspecting other causes of G.O.O.

5) Lab.:

- Renal functions
- Blood for serum electrolyte Na, K, Cl, Ca, pH

Treatment of G.O.O. due to Chronic Peptic Ulcer:

I. General pre-operative treatment:

- Insert naso-gastric tube: for suction of the gastric contents and relieve distension • Correction of dehydration and electrolytes disturbances by intravenous fluid and crystalloids administration.
 - Treatment of associated chest infections.

II. Surgical management:

- 1) Truncal vagotomy and gastro-jejunostomy.
- 2) Highly selective vagotomy with gastro-jejunostomy.
- 3) Post-operative proton pump inhibitors.
- 4) Partial gastrectomy + vagotomy + drainage in hugely dilated stomach

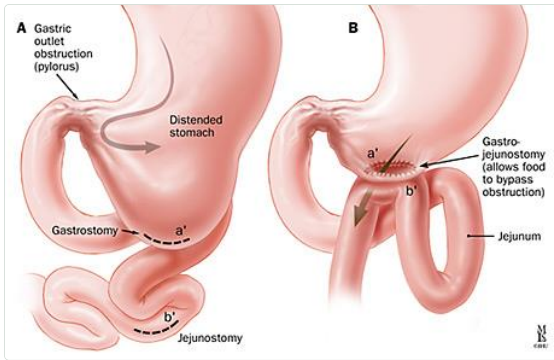


Figure 3.10 — Surgical management of gastric outlet obstruction

Congenital hypertrophic pyloric stenosis (CHPS)

It is a case of congenital hypertrophy of the pyloric sphincter detected in newborn or during infancy.

Incidence: 2 to 5 per 1,000 births /year. The male: female ratio is 4:1.

Etiology: familial inheritance plays a role a) Deficient vagal innervation of the muscular layer leads to failure of relaxation of the pyloric muscle.

b) Increased synthesis of growth factors, and subsequent pyloric muscle hypertrophy, hyperplasia leads to obstruction of the pylorus.

c) The hyper-gastrinemia: increases acid production and repeated pyloric contractions leads to muscle hypertrophy and subsequent obstruction.

Patho-physiology:

- Pyloric obstruction leads to increased gastric contractility to overcome the obstruction → gastric dilatation → projectile vomiting → loss of HCL → Metabolic alkalosis and dehydration and loss of sodium leads to hyponatremia.
- A secondary hyper-aldosteronism develops due to the decreased blood volume. The high aldosterone level causes the kidneys to retain Na⁺ (to correct the intravascular volume depletion), and excrete increased amounts of K⁺ into the urine resulting in a low blood level of potassium → hypokalemia and increased H⁺ loss in urine → paradoxical aciduria.
- Net results are hypochloremia, hypokalemia, hyponatremia, metabolic alkalosis and paradoxical aciduria.

Signs and symptoms

- Babies with this condition usually present any time in the first weeks to months of life with progressively vomiting.



- The vomiting is non-bile stained and projectile.
- Constant hunger, belching, colic and the baby is unable to eat properly.
- Dehydration

Abdominal examination:

- Inspection: Epigastric fullness (stomach distension) with visible peristalsis in the upper abdomen from the left to right.
- Palpation: firm olive shaped mass is felt in the epigastrium slightly to the right of midline (hypertrophy of the pylorus)

Complications of CHPS:

1. Dehydration and shock state in severe cases.
2. Electrolyte disturbances (hypokalemia, hyponatremia, hypocalcemia and metabolic alkalosis).
3. Recurrent aspiration and chest infections (pneumonia).

How to investigate?

1- Abdominal ultra-sound is usually diagnostic :

- Thickened pylorus (solid small mass)
 - Muscle wall thickness > 3 millimeters (mm)
 - pyloric channel length of > 15 mm
- 2- Plain X-ray abdomen: marked dilation of the stomach.
 - 3- Upper endoscopy and gastro-graffin meal help in the diagnosis.
 - 4- Lab. To detect electrolytes and acid base disturbances (Na, K, Ca, pH,...)

Treatment:

I. Correction of the general condition prior to surgery:

- Introduction of naso-gastric tube to decompress the stomach
- I.V. fluids administration to correct dehydration and electrolyte disturbances (lactated ringer solutions and normal saline)
- Administration of antibiotics to guard against infections.

II. Surgical management: Ramstedt's pyloro-myotomy:

- Laparotomy or laparoscopy
- Open the serosa at the pylorus.
- Dissect the longitudinal muscles without division.
- Expose the hypertrophied circular muscles.
- Partial division of the hypertrophied muscles to relieve the spasm.(care to avoid mucosal injury).

III- Endoscopic pyloro-myotomy: per-oral pyloro-myotomy:



Cancer stomach

- Age: old age
- Sex: males > females

Predisposing factors:

- Dietetic: spicy foods, excess salt intake, foods contain N-nitrosamine preservatives
- Smoking
- Any cause of chronic gastritis: as H.Pylori
- Pernicious anemia
- Genetic predisposition, blood group A
- Benign gastric ulcer – gastric polyp
- Following partial gastrectomy (bile reflux gastritis)

Site:

- Antrum, pyloric area → 60%
 - Body: 15 %
 - Fundus: 25%

Spread:

worsen the prognosis

a.Direct:

longitudinal, circumferential, and vertical within the layers of stomach.
When the tumor gets out of the serosa → it infiltrates the nearby structures as spleen, liver, colon and pancreas

b.Lymphatics:

to the peri-gastric L.N.

- Left gastric, right gastric, gastro-epiploic, pyloric L.N.
- Celiac L.N. → Thoracic duct → left supraclavicular L.N. = Virchow's L.N.

c. Blood spread mainly to the liver (portal circulation) **d. Transcelomic spread** → peritoneal nodules – infiltration of the ovary →

Krukenberg's tumor or DOUGLAS pouch nodules.





What are TNM definition for carcinoma of stomach?

Tumor (T)

- T0 – no evidence of primary tumor
- TIS – carcinoma in situ
- T1 – tumor involving mucosa and submucosa
- T2 – tumor invading muscularis propria and extending up to the subserosa
- T3 – tumor invading the serosa but not to adjacent organs
- T4 – tumor invading the adjacent organs.

N (Lymph nodes):

- N0 – no nodes
- N1 – Metastasis in 1–6 lymph nodes
- N2 – Metastasis in 7–15 lymph nodes
- N3 – Metastasis in >15 lymph nodes

M (Metastasis):

- M0 – no metastasis
- M1 – Distant metastasis present.

What are the common tumor markers in gastric cancers?

- **CEA, CA 19-9, CA 50, CA 12-5, CA 72-4** are the important tumor markers to evaluate recurrence and prognosis.
- Most important is **CA 72-4** which disappears with surgery and reappears with regional or distant recurrence.

Clinical presentations of cancer stomach:

A. Dyspeptic symptoms: in male above 50 y is important alarm, progressive:

- Anorexia: is marked and more towards meat
 - Nausea and vomiting
 - Epigastric pain
 - Hematemesis and melena

B. Asthenia, Anorexia, Anemia:

- Unexplained weight loss in old age
 - Marked weight loss
 - Unexplained anemia in elderly





C. Mass group:

epigastric mass ; firm to hard fixed in late cases

D. Obstructive group:

■ Mass at gastric outlet:

produces G.O.O. of rapid onset and progressive course with rapid marked weight loss .

■ Mass at the cardia :

produces progressive dysphagia

E. Metastatic group:

- Nodular hepatomegaly (secondaries), jaundice
 - Malignant ascites
 - Enlarged left supraclavicular L.N. (Troisier sign).

Investigations:

Upper endoscopy:

most valuable, visualize the pathology – take a biopsy

Endo-sonar:

determine infiltrations within the gastric wall

Barium meal:

Detect the lesion in the form of filling defect, pyloric obstruction, malignant stricture, linitis plastica.



Figure 3.11 — Large gastric cancer

- **Abdominal ultrasound – C.T.** to assess spread outside gastric wall and distant spread.

- **Lab. :** CBC – anemia

Liver and kidney functions

Tumor makers: for prognosis and follow up

Treatment:

I. Early cases:

curative treatment:

1. Minimal surgery in the form of endoscopic **submucosal resection (EMR)** is being tried in some centers; in:

- Tumor smaller than 2 cm
- Tumor confined to the mucosa.
- No lympho-vascular invasion
- No surface ulceration.

2. Tumor larger than 2 cm, submucosal invasion, and surface ulceration or with lympho- vascular invasion should undergo gastrectomy with appropriate lymph node dissection. • Gastric resection involving the tumor with safety margin at least 5 cm proximal and 2 cm of the duodenum distally

- Removal of the omentum
- Removal of the draining L.N.



a) Lower radical gastrectomy:

for tumors of lower 1/3 of stomach + gastro- jejunostomy = Billroth II

b) Total radical gastrectomy:

for proximal gastric tumors + roux en Y esophago- jejunostomy.

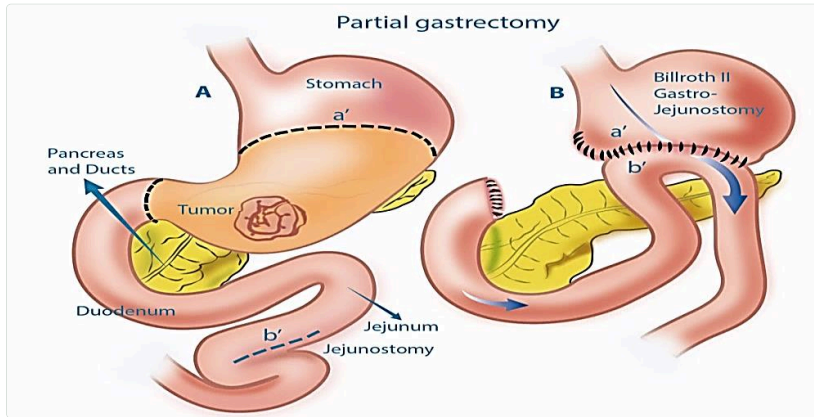


Figure 3.12 — Distal (lower radical) gastrectomy

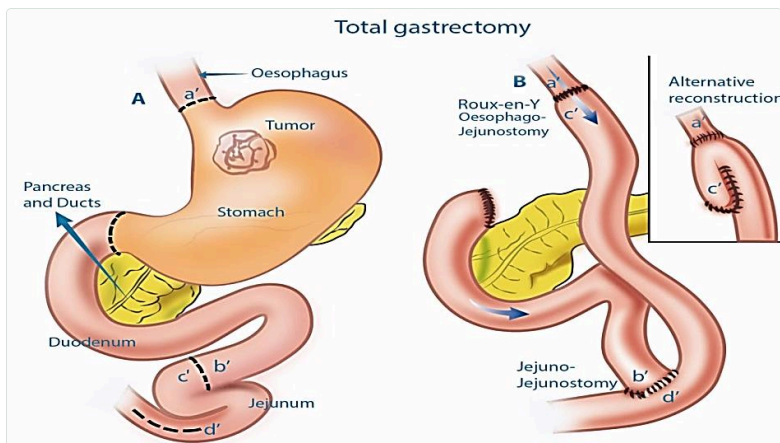


Figure 3.13 — Roux-en-Y anastomosis of the jejunum to the esophagus

II. Late cases of cancer stomach:

with advanced local spread or distant spread: • Radiotherapy and chemotherapy

- Palliative Gastro-jejunostomy in pyloric obstruction
- Feeding jejunostomy in advanced cases

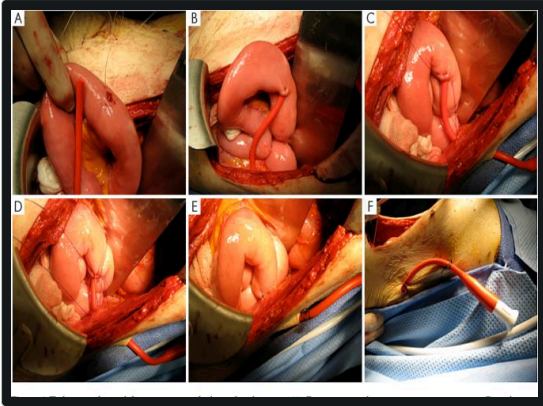


Figure 3.14 — Operative management of gastric cancer



Figure 3.15 — Feeding jejunostomy

Prognosis:

- Depends on tumor grading, stage, state of L.N. and distant spread
- The tumor usually discovered late so it has low 5 years survival rates.

Acute gastric dilatation

Definition

Acute Gastric Dilatation (AGD) is a rare, life-threatening emergency characterized by rapid, massive enlargement of the stomach due to the accumulation of fluid, gas, or ingested material.

Causes & Risk Factors

I- Mechanical Causes:

- Gastric Outlet Obstruction (GOO): Caused by severe peptic ulcer disease, pyloric stenosis, or periampullary/gastric malignancies.



- Superior Mesenteric Artery (SMA) Syndrome
- Gastric Volvulus

II- Functional Causes

- **Eating Disorders:** Anorexia nervosa, bulimia nervosa
- **Metabolic & electrolyte Disruption:** Acute severe hyperglycemia, & Diabetic Ketoacidosis. Electrolyte disturbances (severe hypokalemia, hypomagnesemia).
- **Iatrogenic & Trauma:** Cardiopulmonary resuscitation, post-operative ileus, abdominal trauma, or spinal cord injuries.
- High-dose anticholinergics, opioids, and anesthetics that suppress GI motility.

Complications: Gastric Perforation (Ischemic Necrosis) & Hemodynamic Collapse

Diagnosis:

Early clinical suspicion is vital, as mortality reaches 80–100% once ischemia or perforation occurs.

Physical Exam & History	Severe epigastric pain, marked abdominal distension, tympanitic percussion, succussion splash, and retching.
Abdominal Plain Film	A massively dilated gas- or fluid-filled stomach occupying the abdomen down to the pelvis, displacing bowel loops.
Abdominal CT with IV Contrast (Gold Standard)	Defines gastric dimensions, confirms NG tube position, evaluates wall enhancement

DIAGNOSIS OF ACUTE GASTRIC DILATATION

Treatment Protocol

Management depends on whether gastric perfusion is intact or compromised.

1. Immediate Resuscitation & Decompression

- **Nasogastric Decompression**
- **Fluid & Electrolyte Restoration**
- **Gastric Protection:** IV proton pump inhibitors (PPIs) to reduce gastric secretion volume and mucosal acid injury.

2. Monitoring: conservative management with serial abdominal exams and fasting:

If the stomach successfully decompresses via NG tube, pain resolves, and imaging shows no signs of ischemia or perforation

3. Surgical Intervention:

Emergency exploratory laparotomy is indicated if:

- NG tube decompression fails, CT demonstrates pneumoperitoneum
- Clinical signs of peritonitis, severe septic shock

**Surgical Procedures:**

- **Gastrotomy & Decompression:** Direct operative evacuation of unsuctionable food material or bezoars.
 - **Resection:** Debridement of all non-viable gastric tissue. Depending on the extent of necrosis, this ranges from partial gastrectomy to total gastrectomy
 - **Abdominal Washout & Drainage:** In cases of perforation, irrigate the peritoneal cavity and place abdominal drains.
-

Pathology of GIT tumours



04



TUMORS OF ESOPHAGUS

- Benign tumors of the esophagus are less common than malignant tumors. The most common are leiomyomas.
- Malignant tumors may be either
 - a. Carcinoma most common
 - b. Lymphoma & other sarcomas rare

They present by esophageal obstruction, dysphagia, bleeding, or rarely, paraneoplastic syndrome.

Prognosis of esophageal carcinoma is usually poor (5 year cure rate about 10%)

Carcinoma of esophagus

- Squamous cell carcinoma - More prevalent worldwide 50% in middle 1/3
Risk factors: long-standing esophagitis, achalasia, smoking, alcohol, diet (low vitamins and zinc), genetics
- Adenocarcinoma - More common in USA - More in lower 1/3., Occurs on top of Barrett esophagus, preceded by dysplastic changes

Gross Pathology

1. Exophytic (Polypoid or Fungating). Growth filling the lumen (Most common type).
2. Ulcerative type, which invade deeply into the surrounding structure. With raised reverted edge, rough necrotic floor and indurated base.
3. Infiltrating (Schirrhous) tumor encircling the tube, also called (Annular) & cause constriction (Is the least frequent type).

Histopathology

1. Squamous Cell Carcinoma: Occur in >90 % of cases.
2. Adenocarcinoma (5%): Originating in the cardiac end of the esophagus from dysplastic mucosa in the setting of Columnar Metaplasia
3. Undifferentiated Carcinoma: lacking squamous features & accompanied by lymphocytic infiltration.





Gastric tumors

Benign Neoplasms

Epithelial polyps are rare in the stomach. Three types occur:

- 1- Hyperplastic polyps (70%),
- 2- Fundic gland polyps (20%),
- 3- Adenomatous polyps (10%). Shows dysplasia. The risk of carcinoma is moderate in adenomatous polyp, slight in hyperplastic polyp, and nil in fundic gland polyps.

Gastric polyps appear as small pedunculated lesions often multiple, that can be removed endoscopically. Large polyps are very rare

Mesenchymal Neoplasms : Benign mesenchymal neoplasms are uncommon.

Malignant Neoplasms

Gastric Adenocarcinoma

Sex: male > female Geography: Japan, Chile, Iceland Age > 50

Predisposing factors

Blood group A = 20% increased risk , Adenomatous polyp , Pernicious anemia (atrophic gastritis type A) , Chronic gastritis associated with Helicobacter pylori infection

Grossly it may appear as: -

-Early gastric cancer. defined as gastric carcinoma restricted to the mucosa and submucosa - Appears as a small, flat mucosal thickening that may have a minimal polypoid and ulcerative component.

- Late gastric cancer... defined as a gastric carcinoma that has invaded the muscle wall. It may present in various ways:

- (1) Fungating mass that protrudes into the lumen.
- (2) Malignant ulcer with raised, everted edges
- (3) Diffusely infiltrating lesion that causes thickening and contraction of the stomach wall with relatively little mucosal involvement (linitis plastica, or leather bottle stomach).

It may take one of two types

1- Intestinal-type

- Risk factors: diet (nitrites, smoked food, increased salt), chronic gastritis (H. pylori), altered anatomy after resection
- On top of intestinal metaplasia



- Histopathology : Glandular morphology

Diffuse-type:

- Undefined risk factors (no known relation to H. pylori)
- Signet cell morphology

Spread of gastric carcinoma –

- Direct spread.... Pancreas, liver, esophagus. –
- Lymphatic spread : Draining lymph nodes Thoracic duct left supra-clavicular L.N. –
- Blood spread: . Liver
- Transcoelomic, ovary (Krukenberg) , peritoneal implants

Tumors of the Colon and Rectum

Neoplastic adenomatous polyp of the colon –

Adenomas of the colon are present in 20–30% of all individuals over the age of 50 years.

- 1- Tubular adenomas (pedunculated)
- 2- Tubulo -villous adenomas
- 3- Villous adenomas – (sessile)

Colonic adenomas are associated with genetic mutations identical to those seen in colon cancer. The number of mutations in adenomas is less than in carcinoma but increases as the adenoma enlarges and becomes more dysplastic.

Tubular adenoma

- Tubular adenomas account for over 90% of colonic adenomas.
- They are commonly multiple, with 10–20 lesions present in some patients,
- They are pedunculated with a well-defined stalk –

Histologically, a tubular adenoma is composed of benign neoplastic glands bunched together above the muscularis. The epithelial cells are dysplastic (hyperchromatic and stratified and show loss of normal mucin content)

Villous Adenoma

Villous adenomas are uncommon, comprising less than 10% of colonic adenomas. - They commonly occur in older individuals as a single large, sessile lesion have a broad base of attachment to the mucosal surface without a defined stalk.



Histologically, villous adenoma is composed of neoplastic proliferation of colonic epithelial cells organized into long finger-like papillary or villous processes. - The covering epithelium is dysplastic.

Inherited Polyposis Syndromes

(Adenomatous Polyposis Coli APC - Familial Adenomatous Polyposis (FAP), - Gardner Syndrome. - Turcot Syndrome. - Hereditary Non-polyposis Colorectal Cancer Syndrome (HNPCC)

Familial Adenomatous Polyposis (FAP)

Autosomal dominant - average onset is 25 years old -Adenomatous polyps, mostly colorectal. Patients, on average, have 1,000 polyps -Minimum of 100 polyps are necessary for diagnosis -The genetic defect is on chromosome 5q21, APC (Adenomatous Polyposis Coli) tumor suppressor gene - Precancerous: virtually 100% will develop cancer within 10 to 15 years

Malignant colonic tumors

I- Epithelial Adenocarcinoma and mucoid adenocarcinoma

II- Mesenchymal - Gastro-intestinal Stroma Tumor (GIST) - Lymphoma Colonic

Adenocarcinoma

Risk factors: low fiber, high fat, decreased vit A, C, E, idiopathic inflammatory bowel disease, familial adenomatous polyposis - Several hits to different genes: APC, k-ras, p53; or DNA mismatch repair genes

-Survival depends on stage (depth of invasion and node metastasis) -Incidence - Primarily occurs in patients between 60 and 79 years of age - Ratio of male to female patients 2:1 for rectal cancers, 1:1 before 50)

Gross appearance:- -

Sites:- Rectosigmoid , Coecum any other site

Shape: Fungating -Ulcerating or infiltrating (annular)

Histopathology

- Glands have lost polarity, disorganized and adhere to each other
- Epithelial cells are hyperchromatic, tightly packed, and have a higher nucleus-to cytoplasm content (appear more malignant).
- The malignant acini are separated by desmoplastic stroma.
- Tumor invades deeply into colonic mucosa
- Tumor cells may continue to produce mucin; in 10-15% of patients, mucin lakes can be seen.



Course: tumor invades bowel wall and lymphatics/blood vessels with metastasis to lymph nodes, liver, lungs, and bones.

25% of patients have metastatic disease at presentation.

Diagnosis is based on :

- Endoscopy and biopsy.
- Increased serum CEA which is more valuable in follow up after surgical removal of a tumor.

Intestinal carcinoid

Carcinoid tumors arise from neuroendocrine cells present in the mucosa throughout the gastrointestinal tract.

Grossly,

- Carcinoid tumors are firm and yellow. They arise in the submucosa, elevating and ulcerating the mucosa and locally infiltrating the wall. –

Multiple lesions are present in 25% of cases.

Microscopically - Sheets of monotonous, polyhyal cells with deeply stained nuclei. - Stained with silver or by IHC for chromogranin or neurone specific enolase.

Gastrointestinal stromal tumor (GIST)

Gastrointestinal stromal tumors are the most common mesenchymal neoplasms of the gastrointestinal tract. GISTs arise in the smooth muscle pacemaker interstitial cell of Cajal.

- Grossly: large, bulky, intramural masses with tannish brown cut section with hemorrhage, necrosis and cystic softening.
- Microscope: Mesenchymal tumor usually centered in the muscularis propria
- There are two types:

1-Spindle cell GISTs: intersecting fascicles of plump spindled cells with eosinophilic cytoplasm within variably hyalinized or edematous stroma

2- Epithelioid GISTs: rounded epithelioid cells

DIARRRHEA AND MALABSORPTION



05



Objectives

The student will

- 1) Know the definition of acute and chronic diarrhea
- 2) Understand the basic mechanisms and causes underlying acute and chronic diarrhea 3)
- Know the clinical presentation of various forms of acute and chronic diarrhea
- 4) Know the diagnostic studies for acute and chronic diarrhea
- 5) Understand the basic treatment options for acute and chronic diarrhea
- 6) Understand the basic concepts and mechanisms for malabsorption.
- 7) Know the various causes of malabsorption
- 8) Know the clinical features of various types of malabsorption
- 9) Know the diagnostic tools for malabsorption
- 10) Know the basic treatment options for different types of malabsorption.

Diarrhea

- Diarrhea is defined as an increase in the fluidity, frequency, and volume of daily stool output.
- Normal stool frequency ranges from three times a week to three times a day.
- Perception as normal bowel pattern is related to consistency and frequency.
- Consistency is defined as the ratio of fecal water to the water-holding capacity of fecal insoluble solids.
- Acute diarrheas are those lasting less than 2 to 3 weeks or, rarely, 6 to 8 weeks.
- Approximately 80% of acute diarrheas are due to infections with viruses, bacteria, and parasites.
- Diarrhea lasting >4 weeks warrants evaluation to exclude serious underlying pathology. • In contrast to acute diarrhea, most of the causes of chronic diarrhea are noninfectious.

Basic physiology

- During fasting, the intestine contains very little fluid.
- On a normal diet of three meals a day, about 9 L of fluid is delivered to the small intestine.
- 90% is absorbed in the small intestine.
- One to two liters is presented to the colon, and approximately 90% is absorbed.
- The colon has the capacity to absorb all of the fluid presented to it.
- This results in 100 to 200 mL of fluid in the stool.
- Feces on the average contain 100 mL of water, 40 mEq/L of sodium (Na⁺), 90 mEq/L of potassium (K⁺), 16 mEq/L of chloride (Cl⁻), and 30 mEq/L of bicarbonate (HCO₃⁻). • Water transport across the intestinal epithelium is passive.



- Na^+ is the major ion actively absorbed, and Cl^- is the major ion secreted.
- Water is absorbed along with Na^+ .
- Active Cl^- secretion or inhibited Na^+ absorption also creates an osmotic gradient favorable for the movement of fluids from blood to lumen and is the basis of the secretory diarrheas.

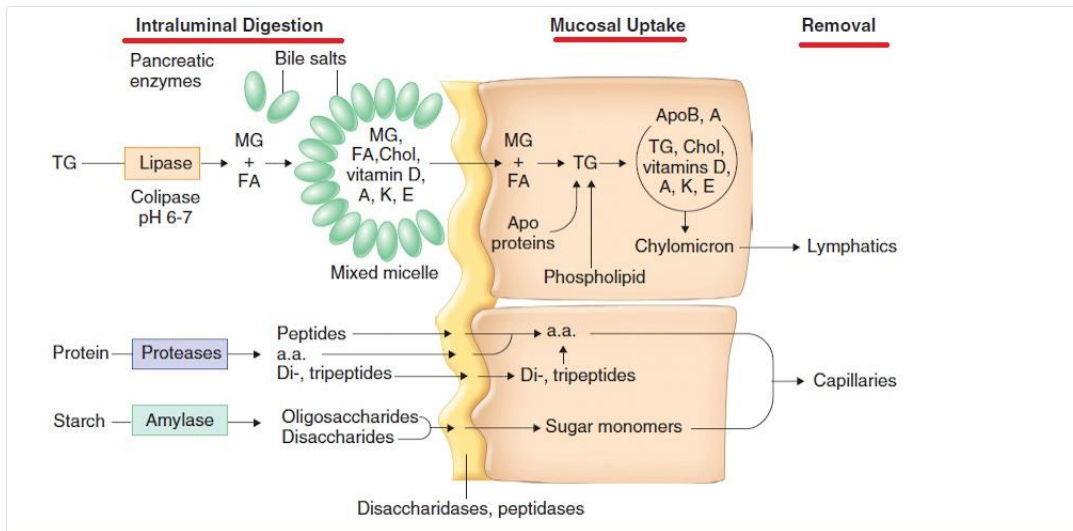


Figure 5.1 — Digestion and absorption

Role of bile acids

- Bile acids are not present in the diet but are synthesized in the liver by a series of enzymatic steps that also include cholesterol catabolism.
- Bile acids are either primary or secondary: primary bile acids are synthesized in the liver from cholesterol, and secondary bile acids are synthesized from primary bile acids in the intestine by colonic bacterial enzymes.
- The primary functions of bile acids are (1) to promote bile flow, (2) to solubilize cholesterol and phospholipid in the gallbladder by mixed micelle formation, and (3) to enhance dietary lipid digestion and absorption by forming mixed micelles in the proximal small intestine.

Defects Affecting Bile Acids

Process	Pathophysiologic Defect	Disease Example
Synthesis	Decreased hepatic function	Cirrhosis
Biliary secretion	Altered canalicular function	Primary biliary cirrhosis
Maintenance of conjugated bile acids	Bacterial overgrowth	Jejunal diverticulosis
Reabsorption	Abnormal ileal function	Crohn's disease

Figure 5.2 — Defects affecting bile acids





Classification of diarrhea

pathophysiologic classification

Osmotic diarrhea

Secretory diarrhea

Exudative diarrhea

Altered motility

Osmotic diarrhea:

There is increased amounts of poorly absorbable, osmotically active solutes in the gut lumen.

Secretory diarrhea:

There is, increased Cl⁻ and water secretion with or without inhibition of normal active Na⁺ and water absorption.

Exudative diarrhea:

There is mucus, blood, and protein from sites of active inflammation into the bowel lumen

Altered motility:

There is increased or decreased contact between luminal contents and mucosal surface.

clinical classification

acute

chronic

refractory

Causes of osmotic diarrhea

Carbohydrate malabsorption:

1) Disaccharidase deficiency in the small-intestinal mucosa:

Primary (e.g., idiopathic lactase, sucrase-maltase deficiency)

Secondary (e.g., after infectious gastroenteritis and with mucosal inflammation)

2) Ingestion of mannitol, sorbitol

3) Lactulose therapy

General malabsorption

Sprue (idiopathic gluten enteropathy, tropical)

Ingestion of sodium sulfate (Glauber's salt), sodium phosphate, magnesium sulfate





■ Secretory Diarrhea

- Diarrhea of 1 L or more per day results from secretion of fluid across the intestinal mucosa.
- In most cases, a pathophysiologic event causes small intestinal secretion by simultaneously stimulating active secretion and partial inhibition of intestinal absorption.
- Often, the intestinal mucosa is intact and has normal histologic findings
- Clinically, there are features that characterize secretory diarrhea:
- Stool volume is usually large (>1 L per day).
- Stools are watery in consistency.
- Stools do not contain pus or blood.
- Diarrhea typically continues while the patient fasts for 24 to 48 hours.

Causes of secretory diarrhea

Enterotoxins

Vibrio cholerae

Staphylococcus aureus

Hormonal secretagogues

Vasoactive intestinal polypeptide (pancreatic cholera syndrome)

Serotonin (carcinoid)

Gastric hypersecretion

Zollinger-Ellison syndrome

Laxatives

Fatty acids

Pancreatic insufficiency

■ Exudative Diarrhea

- If the intestinal mucosa is inflamed and ulcerated, mucus, blood, and pus leak into the lumen and are discharged as stool.
- This may also create an increased osmotic load.
- If a large surface area of the bowel lumen is involved, absorption of ions, solutes, and water will also be impaired, and patients may have large-volume diarrhea.
- Inflammation may generate prostaglandins, which stimulate secretion and may increase bowel motility, thus compounding the diarrhea.
- The severity of the diarrhea and the systemic signs and symptoms depends on the extent of bowel involvement.

Causes of exudative diarrhea

Idiopathic (e.g., Crohn's disease, ulcerative colitis)



Infectious (e.g., from invasive organisms or cytotoxins: Shigella, Salmonella, Campylobacter, Yersinia, tuberculosis, amebae, and Clostridium difficile)

Ischemic

Vasculitic

Due to radiation injury

■ Motility Disturbances

- Both reduced and increased motility of the intestine may result in diarrhea.
- Increased Motility Of The Small Intestine: Results in decreased contact time of chyme with absorptive surfaces.
- Large amounts of fluid delivered to the colon may overwhelm its absorptive capacity and result in diarrhea.
- The reduced contact time in the small intestine may interfere with absorption of fatty acids and bile salts, allowing them to reach the colon where they induce a secretory diarrhea.
- Decreased Motility Of The Small Intestine May allow colonization of the small intestine with colonic-type bacteria.

Clinical classification of diarrhea

Acute diarrhea

- Approximately 80% of acute diarrheas are due to infections with viruses, bacteria, helminthes, and protozoa.
- The remainder are secondary to the ingestion of medications, poorly absorbed sugars (fructose polymers or sorbitol).
- Most infectious diarrheas are acquired through fecal-oral transmission from water, food, or person-to-person contact.
- Patients with infectious diarrhea often complain of nausea, vomiting, and abdominal pain and have watery, malabsorptive, or bloody diarrhea and fever (dysentery).
- Food poisoning: is produced by a preformed bacterial toxin that contaminates the food. • Bacterial replication in the host is not necessary for the development of disease.
- Diarrhea resulting from multiplication of organisms in the intestine may be divided into inflammatory-invasive versus noninflammatory-noninvasive categories.

Causes of Acute Diarrhea According to the mechanism



Causes	Mechanism	Organisms
Viral Infections Small intestine	Mucosal invasion absent. Noninflammatory Watery diarrhea. Fecal leukocyte absent.	Rotavirus (children, adults) Norwalk virus
Bacterial infections Small intestine	Mucosal invasion absent Noninflammatory Watery diarrhea Fecal leukocytes absent	Vibrio cholerae Toxigenic Escherichia coli
Colon	Mucosal invasion present Inflammatory Bloody diarrhea Fecal leukocytes present	Salmonella Shigella Invasive E. coli
Parasitic infections Small intestine	Mucosal invasion absent Noninflammatory Watery diarrhea Fecal leukocyte absent	Giardia lamblia Cryptosporidium
Colon	Mucosal invasion present Inflammatory Bloody diarrhea Fecal leukocytes present	Entamoeba histolytica
Food poisoning Small intestine	Toxin induced Mucosal invasion absent Noninflammatory Watery diarrhea Fecal leukocytes absent	Staphylococcus aureus Clostridium botulinum

Remarks about treatment of acute diarrhea

- Acute diarrhea with fluid and electrolyte depletion is a major cause of mortality, especially in children in developing countries of the world.
- Fluid repletion by intravenous or oral routes can prevent death.
- Oral rehydration therapy can be accomplished with a simply prepared oral rehydration solution.
- The use of Opiate derivatives include paregoric (tincture of opium), diphenoxylate with atropine (Lomotil), and loperamide (Imodium) may help in certain selected cases.
- They should not be used in patients with fever, systemic toxicity, or bloody diarrhea.
- Antimicrobials should be only used when there is evidence of mucosal invasion.



Chronic Diarrhea

- Diarrhea lasting >4 weeks warrants evaluation to exclude serious underlying pathology. • In contrast to acute diarrhea, most of the causes of chronic diarrhea are noninfectious. **CAUSES**

Infections.

- Giardia lamblia.
- Entamoeba histolytica.
- Tubercle bacillus.
- Clostridium difficile.

Inflammatory bowel disease:

Ulcerative colitis and Crohn's disease may result in diarrhea of varying severity, depending on the extent and degree of bowel involvement. Diarrhea in Crohn's disease of the small bowel may be compounded by concomitant bile salt and fat malabsorption.

Malabsorption syndromes

Endocrine disorders:

Zollinger-Ellison syndrome. Hyperthyroidism. Carcinoid. Non-β- cell pancreatic tumor.

Neoplasms

Drugs and laxatives

Irritable bowel syndrome

Refractory diarrhea

Refractory chronic diarrhea unresponsive to specific antimicrobial therapy or standard unspecific medication may present a challenging and serious clinical problem.

Causes

- Protracted diarrhea in infants and children
- Tropical sprue
- Persistent diarrhea in travelers
- Protracted infectious diarrhea in adults: Organisms that are difficult to diagnose and are known to cause protracted or prolonged diarrheas include enteropathogenic

(enteroadherent) E. coli , Giardia , amoeba , Cryptosporidium , Aeromonas , Yersinia enterocolitica , Clostridium difficile infection is often difficult to clear, and five or more relapses have been observed .



- Post-infectious diarrhea induced irritable bowel syndrome: After documented infectious diarrhea, 25% of patients experience pain, bloating, urgency, a sense of incomplete evacuation, and loose stools for 6 months or longer.

This syndrome of infectious diarrhea-induced irritable bowel syndrome, also called

Brainerd's diarrhea if it is particularly prolonged and accompanied by severe diarrhea.

MALABSORPTION

- Malabsorption is caused by many different diseases.
- Dietary fat is the nutrient most difficult to absorb.
- Fat is predominantly insoluble in the aqueous milieu of the intestine and critically depends on all phases of digestion, absorption, and delivery for its assimilation.
- Steatorrhea (excess fat in the stool) is the hallmark of malabsorption.

Malabsorption Syndromes

- Malabsorption is characterized by suboptimal absorption of fats, fat-soluble and other vitamins, proteins, carbohydrates, electrolytes and minerals, and water.
- At the most basic level, it is the result of disturbance of at least one of these normal digestive functions:

Intraluminal digestion.

Terminal digestion.

Transepithelial transport.

Intraluminal digestion

The process begins in the mouth with saliva, receives a major boost from gastric peptic digestion, and continues in the small intestine, assisted by pancreatic enzyme secretion and the emulsive action of bile.



Terminal digestion

which involves the hydrolysis of carbohydrates and peptides by disaccharidases and peptidases, respectively, in the brush border of the small intestinal mucosa



Transepithelial transport

in which nutrients, fluid, and electrolytes are transported across the epithelium of the small intestine for delivery to the intestinal vasculature or to the intestinal lymphatic system

Overall digestive and absorptive process in small intestine



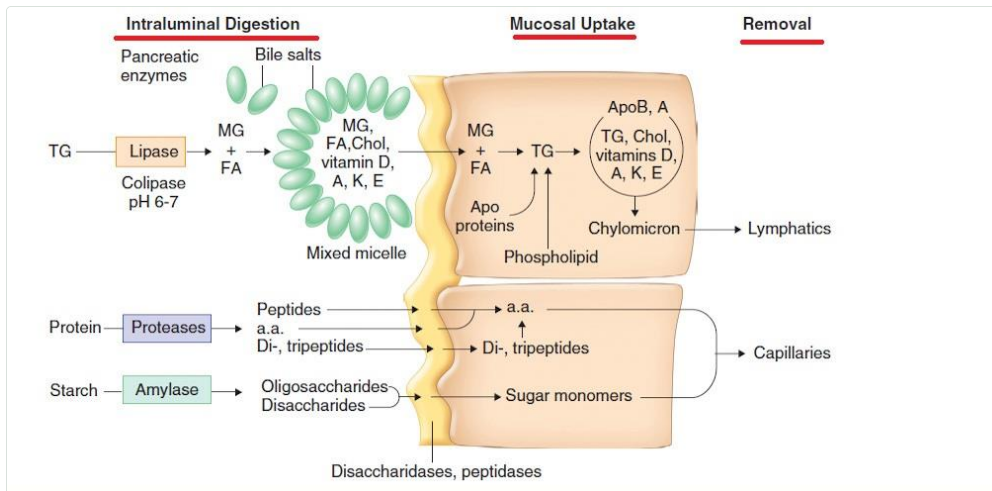


Figure 5.3 — Overall digestive and absorptive process

The Major Malabsorption Syndromes

- Disorders of absorption constitute a broad spectrum of conditions with multiple etiologies and varied clinical manifestations.
- Most, but not all, malabsorption syndromes are associated with steatorrhea, an increase in stool fat excretion of >6% of dietary fat intake.
- Some malabsorption disorders are not associated with steatorrhea.
- Disorders of absorption (malabsorption) must be included in the differential diagnosis of chronic diarrhea.
- Most patients will indicate that they have diarrhea, not that they have fat malabsorption.



MECHANISM OF MALABSORPTION	CONDITIONS
Impaired mixing	Partial/total gastrectomy Gastric bypass surgery
Impaired lipolysis	Chronic pancreatitis Pancreatic cancer
Impaired micelle formation	Severe chronic liver disease Cholestatic liver disease Bacterial overgrowth Crohn disease Ileal resection
Impaired mucosal absorption	Giardiasis Celiac disease Tropical sprue Radiation enteritis Whipple disease Lymphoma
Impaired nutrient delivery	Lymphoma Tuberculosis Constrictive pericarditis Severe congestive heart failure

CLINICAL CONSEQUENCES OF MALABSORPTION

Nutrient Malabsorbed	Clinical Manifestation
Protein	Wasting, edema
Carbohydrate and fat	Diarrhea, abdominal cramps and bloating, weight loss/growth retardation
Fluid and electrolytes	Diarrhea, dehydration
Iron	Anemia, cheilosis, angular stomatitis
Calcium/vitamin D	Bone pain, fractures, tetany
Magnesium	Paresthesia, tetany
Vitamin B12/folate	Anemia, glossitis, cheilosis, paresthesias, ataxia (vitamin B12 only)
Vitamin A	Night blindness, xerophthalmia, hyperkeratosis, diarrhea
Vitamin K	Ecchymoses
Riboflavin	Angular stomatitis, cheilosis
Selenium	Cardiomyopathy
Essential fatty acids	Dermatitis

SPECIFIC DISEASE EXAMPLES

Lactase Deficiency (lactose intolerance)

- Acquired lactase deficiency is the most common cause of selective carbohydrate malabsorption.



- The prevalence of lactase deficiency is highest (85 to 100%) in persons of Asian, African, and Native-American descent.
- Adults with lactase deficiency typically complain of gas, bloating, and diarrhea after the ingestion of milk or dairy products but do not lose weight.
- Unabsorbed lactose is osmotically active, drawing water followed by ions into the intestinal lumen.
- On reaching the colon, bacteria metabolize lactose to short-chain fatty acids, carbon dioxide, and hydrogen gas.
- Short-chain fatty acids are transported with sodium into colonic epithelial cells, facilitating the reabsorption of fluid in the colon.
- If the colonic capacity for the reabsorption of short-chain fatty acids is exceeded, an osmotic diarrhea results.
- The diagnosis of acquired lactase deficiency can be made by empirical treatment with a lactose-free diet, which results in resolution of symptoms; by the hydrogen breath test after oral administration of lactose.
- Many intestinal diseases cause secondary reversible lactase deficiency, including viral gastroenteritis, celiac disease, giardiasis, and bacterial overgrowth.

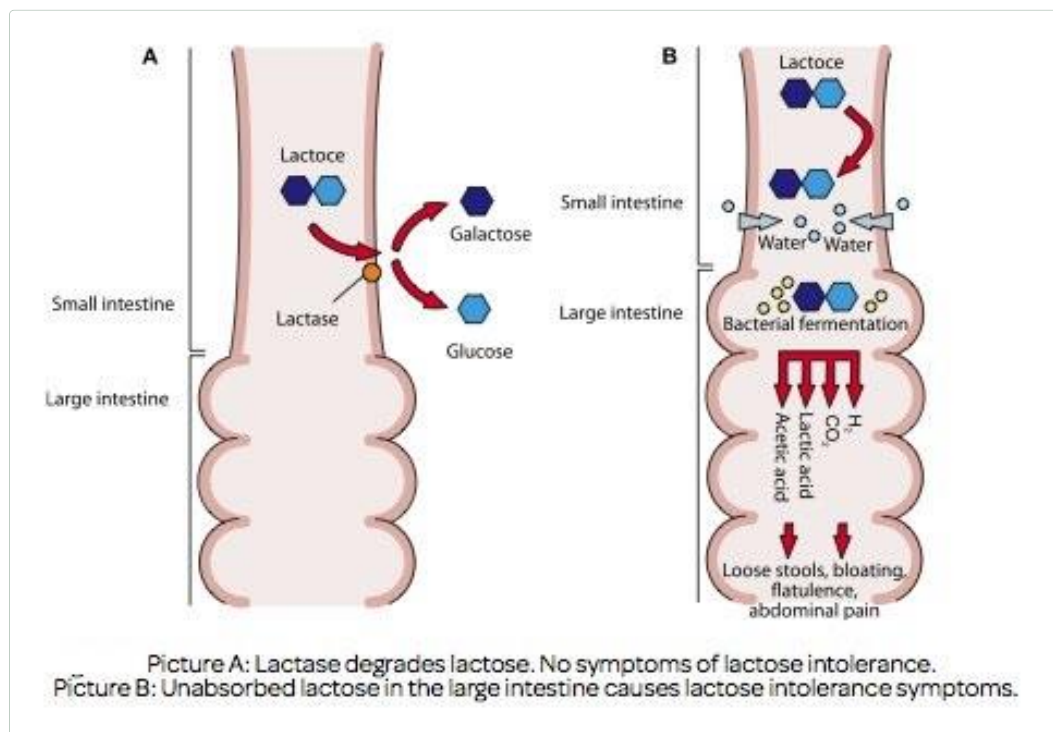


Figure 5.4 — Lactase deficiency and lactose intolerance



Intestinal Bacterial Overgrowth

- In health, only small numbers of lactobacilli, enterococci, gram-positive aerobes, or facultative anaerobes can be cultured from the upper small bowel lumen.
- Motility And Acid are the most important factors in keeping the number of bacteria in the upper small bowel low.
- Any condition that produces local stasis or recirculation of colonic luminal contents allows development of a predominantly “colonic” flora (coliforms and anaerobes, such as Bacteroides and Clostridium) in the small intestine.
- This will cause impaired micelle formation by releasing cholyamidases, which deconjugate bile salts.
- As a result, the concentration of bile salts decreases in the intestinal lumen causing malabsorption of fats and fat-soluble vitamins.
- Vitamin B12 deficiency and carbohydrate malabsorption also can occur with generalized bacterial overgrowth.
- Anaerobic bacteria ingest vitamin B12 and release proteases that degrade brush-border disaccharidases.
- Individuals with bacterial overgrowth usually have low serum vitamin B12 levels but normal or high folate levels.
- This helps distinguish bacterial overgrowth from tropical sprue.

Some Causes of Bacterial Overgrowth

STRUCTURAL

■ Surgical

Ileocecal valve resection

■ Anatomic

Duodenal and jejunal diverticula

Gastrojejunal fistulas

MOTOR

Scleroderma

Diabetes mellitus

HYPOCHLORHYDRIA

Atrophic gastritis

Proton-pump inhibitors





MISCELLANEOUS

Cirrhosis

Chronic renal failure

Celiac Disease

- Celiac disease, also called celiac sprue, nontropical sprue, and gluten-sensitive enteropathy, is an inflammatory condition of the small intestine precipitated by the ingestion of wheat, rye, and barley in individuals with certain genetic predispositions. • It is the prototype of a noninfectious cause of malabsorption resulting from a reduction in small intestinal absorptive surface area.
- The basic disorder in celiac disease is sensitivity to gluten, the component of wheat and related grains (oat, barley, and rye) that contains the water-insoluble protein gliadin
- The small intestinal mucosa, when exposed to gluten, accumulates large numbers of B cells and plasma cells sensitized to gliadin; accumulation of lymphocytes in gastric and colonic mucosa also may occur.
- Total flattening of mucosal villi (and hence loss of surface area) is the outcome, affecting the proximal more than the distal small intestine.
- Patients with celiac disease have increased levels of serum antibodies to a variety of antigens, including IgA antiendomysial, anti gliadin and anti tissue transglutaminase antibodies (TTG Ab).
- The age of presentation with symptomatic diarrhea and malnutrition varies from infancy to mid-adulthood.
- About half of adults with celiac disease in the present with anemia or osteoporosis without diarrhea or other gastrointestinal symptoms.
- These individuals most likely have proximal disease that impairs iron, folate, and calcium absorption but an adequate surface area in the remaining intestine for absorption of other nutrients.
- Other Extraintestinal Manifestations Of Celiac Disease include rash (dermatitis herpetiformis), neurologic disorders (peripheral neuropathy, ataxia, epilepsy), psychiatric disorders (depression, paranoia), reproductive disorders (infertility, spontaneous abortion), short stature, dental enamel hypoplasia, chronic hepatitis, or cardiomyopathy.
- Individuals with significant mucosal involvement present with watery diarrhea, weight loss or growth retardation, and the clinical manifestations of vitamin and mineral deficiencies.





- Diarrhea is caused by many mechanisms, including a 1) decreased surface area for water and electrolyte absorption, 2) the osmotic effect of unabsorbed luminal nutrients, and 3) increased surface area for chloride secretion (crypt hyperplasia), and the stimulation of intestinal fluid secretion by 4) inflammatory mediators and unabsorbed fatty acids.
- Recently The perception that gluten causes gastrointestinal and extra gastrointestinal symptoms in patients who do not have coeliac disease (CD), and are not allergic to wheat, is an increasing clinical problem.
- This has been called Non-coeliac Gluten Sensitivity (NCGS)
- It is purely a clinical diagnosis as, in contrast to CD, it has no biomarker or characteristic intestinal biopsy findings.

Celiac Disease vs. Wheat Allergy vs. NCWS

	Celiac Disease	Wheat Allergy (IgE-mediated)	Wheat Allergy (Non-IgE)	Non-Celiac Wheat Hypersensitivity
Route of exposure	Oral	Inhalation or oral	Oral	Oral
Onset after exposure	Hours / months	Immediate	Days / weeks	Hours / days
Extraintestinal features	Liver, joint/bone, skin, blood, CNS	Respiratory tract, skin, anaphylaxis	None	Skin, joint/muscle, CNS
Antibodies	TGA & DGP (IgA/IgG)	IgE	None	Possibly antigliadin
Complications	Poor growth, micronutrient deficiency, infertility, osteoporosis, lymphoma	None	Esophageal strictures (EoE)	—
Comorbidities	Type 1 diabetes, immune liver & thyroid disease	Other allergies	Atopic dermatitis	Thyroiditis

Figure 5.5 — Celiac Disease vs. Wheat Allergy vs. NCWS





Classical, Non-Classical & Subclinical Disease

Coeliac disease is a “clinical chameleon” that can present at any age with widely varying manifestations — a major reason for delayed or missed diagnosis.

Classical	Non-Classical	Subclinical (Silent)
<i>More common in children < 5 years</i> - Chronic diarrhea - Poor appetite - Weight loss - Abdominal distension - Muscle wasting - Mood changes	<i>The most common presentation overall</i> - Recurrent abdominal pain, bloating - Diarrhea or constipation - Persistent iron deficiency - Chronic fatigue - Elevated liver enzymes - Arthralgia, alopecia, urticaria	<i>Found only through screening</i> - No overt symptoms or signs - Identified via population screening - Or <u>casefinding</u> in at-risk relatives - Villous atrophy still <u>present</u> on biopsy

Figure 5.6 — Celiac disease: classical, non-classical and subclinical presentations

Manifestations Across the Lifespan

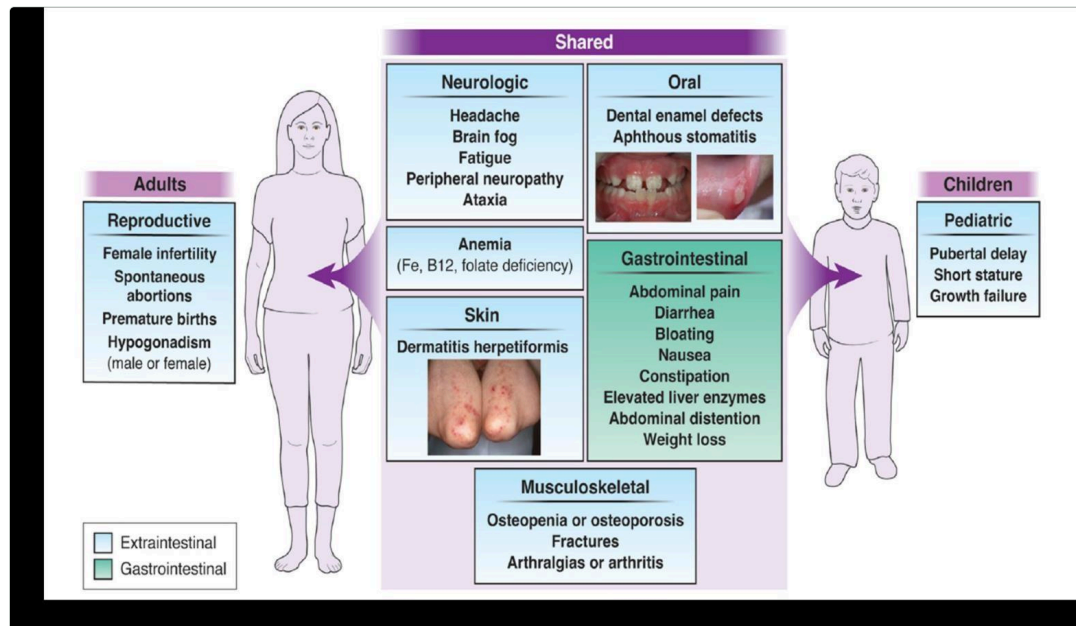


Figure 5.7 — Manifestations across the lifespan

Serologic Testing

tTG-IgA is the single preferred test in adults, measured concurrently with total IgA to catch IgA deficiency (then use DGP -IgG).



Test	Sensitivity	Specificity	Notes
tTG-IgA	90–98%	95–97%	Preferred single first-line test
EMA-IgA	85–98%	97–100%	Highest accuracy; costlier, less available
DGP-IgA	94%	99%	Useful adjunct / IgA-deficient patients
DGP-IgG	92%	100%	Used when IgA deficient
AGA (IgA/IgG)	Lower	Lower	No longer recommended — 15–20% false positive

Figure 5.8 — Serologic testing

Tropical Sprue And Whipple Disease

- Are two disorders that exemplify malabsorption syndromes arising from intestinal infection.
- Tropical sprue resembles celiac disease in symptomatology but occurs almost exclusively in persons living in or visiting the tropics.
- Small intestinal changes vary from near normal to a severe diffuse enteritis with villus flattening.
- In contrast to celiac disease, injury is seen at all levels of the small intestine.
- It is caused by overgrowth of predominantly coliform bacteria.
- Individuals with tropical sprue classically present with diarrhea and megaloblastic anemia secondary to vitamin B12 and folate deficiency, but some have anemia only.
- Intestinal biopsy characteristically shows subtotal and patchy villous atrophy in the proximal and distal small intestine.
- Treatment is a prolonged course of tetracycline (250 mg PO four times daily) or doxycycline (100 mg PO two times daily), folic acid (5 mg/day PO), and, with coexistent deficiency, vitamin B12 injections (1000 µg weekly)

Whipple disease is a rare, systemic infection that may involve any organ of the body but principally affects the Intestine, Central Nervous System, And Joints.

- The hallmark of Whipple disease is a small intestinal mucosa laden with distended periodic acid-Schiff (PAS)-positive macrophages in the lamina propria.
- The organism responsible for causing Whipple disease is a gram-positive actinomycete, *Tropheryma whippelii*.
- Approximately one third of patients have cardiac involvement, most commonly culture- negative endocarditis.
- Occasionally, individuals present with ocular or neurologic disease without gastrointestinal symptoms.
- Men are affected more commonly than women, particularly white men.
- Treatment is with a prolonged course of broad-spectrum antibiotics (e.g., ceftriaxone, 2 g/day intravenously (IV) or meropenem 1 g IV three times daily; then trimethoprim 160 mg and sulfamethoxazole 800 mg PO two times daily for 1 year.



DIAGNOSTIC TESTS FOR MALABSORPTION

GENERAL TESTS OF ABSORPTION

Quantitative stool fat test Gold standard test of fat malabsorption. Requires ingestion of a high-fat diet (70- 100 g) for 2 days before and during the collection. Stool is collected for 2-3 days. Normally, <7 g/24 hr is excreted on a high-fat diet

Qualitative stool fat test Sudan stain of a stool sample for fat.

Hydrogen breath test Most useful in the diagnosis of lactase deficiency. An oral dose of lactose (25 g) is administered after measurement of basal breath H₂ levels. The sole source of H₂ in the mammal is bacterial fermentation; unabsorbed lactose makes its way to colonic bacteria, resulting in excess breath H₂

SPECIFIC TESTS FOR MALABSORPTION

1) Tests for Pancreatic Function

Secretin stimulation test The gold standard test of pancreatic function

Fecal elastase-1 test Stool test for pancreatic function

2) Tests for Bacterial Overgrowth

Quantitative culture of small intestinal aspirate

Hydrogen breath test The 50-g glucose breath test has a sensitivity of 90% for growth of 10⁵ colonic-type bacteria in the small intestine

3) Tests for Mucosal Disease

Small bowel biopsy

4) Test of Ileal Function

⁷⁵SeHCAT test This is a test of bile acid absorption.

Seven days after ingestion of radiolabeled synthetic selenium-homocholic acid conjugated with taurine (⁷⁵SeHCAT), whole body retention is measured by a gamma-counting device. The result is expressed as a fraction of baseline ingestion



Practical approach to malabsorption

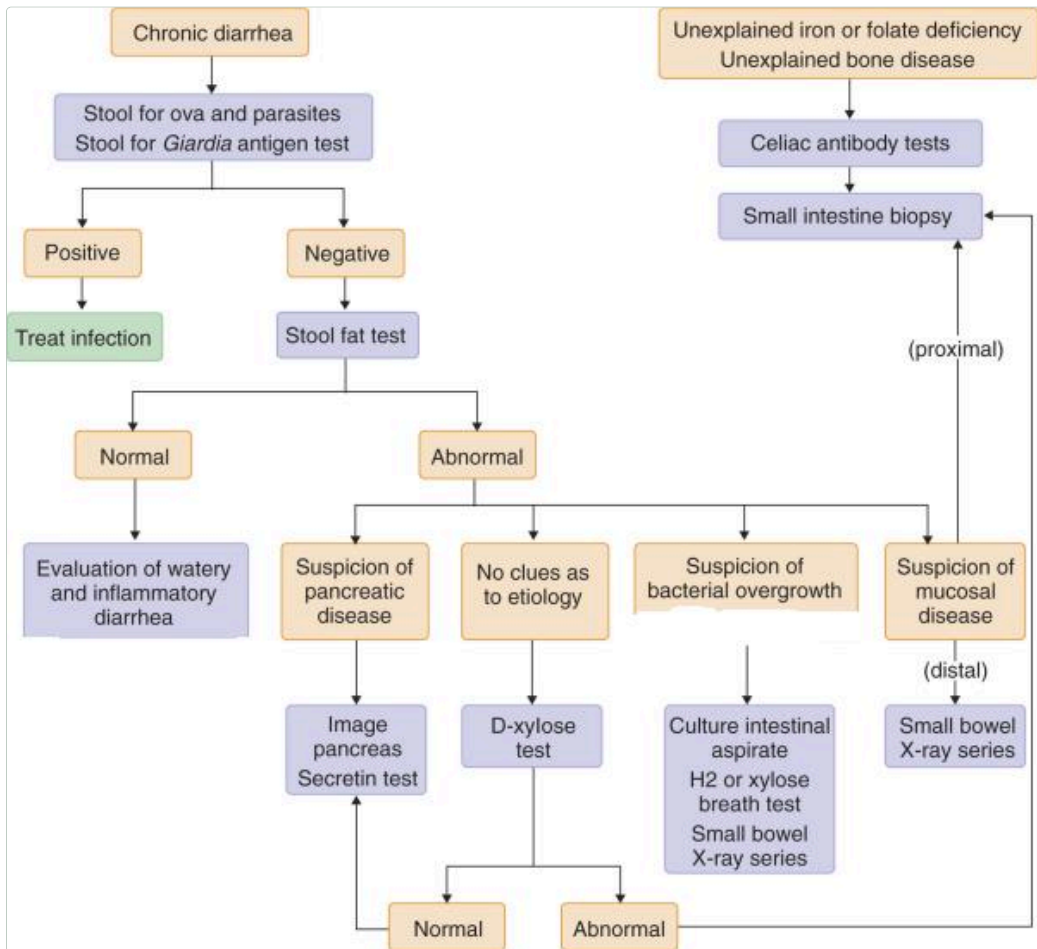


Figure 5.9 — Practical approach to malabsorption

Constipation



06



Definition

- Constipation is a common gastrointestinal symptom or disorder characterized by infrequent bowel movements, difficult stool passage, excessive straining, hard stools, or a sensation of incomplete evacuation. It results from impaired colonic transit, disorders of anorectal evacuation, or both.
- Prevalence and clinical importance: Constipation affects approximately 16% of adults worldwide and is one of the most common gastrointestinal disorders encountered in clinical practice. Although often regarded as a minor complaint, it may significantly impair quality of life and occasionally indicates serious underlying disease

1. Classification

Primary (functional) constipation	Secondary constipation
Normal-transit constipation; Slow-transit constipation; Defecatory disorders (pelvic floor dyssynergia)	Drugs; Endocrine and metabolic disorders; Neurological disorders; Structural diseases

3. Pathophysiology

Constipation results from disturbances in colonic transit, anorectal evacuation, or both. Lifestyle factors such as inadequate dietary fiber, insufficient fluid intake, and physical inactivity may also contribute to the development of constipation.

Normal colonic motility

Normal colonic transit depends on coordinated colonic contractions that propel fecal material toward the rectum while allowing adequate absorption of water and electrolytes. Defecation requires coordinated relaxation of the pelvic floor muscles and anal sphincters.

Slow-transit constipation

Slow-transit constipation results from reduced colonic motility, leading to prolonged stool transit. Excessive water absorption from the retained stool produces hard, dry stools that are difficult to pass.

Defecatory disorders

Defecatory disorders (pelvic floor dyssynergia) occur when there is impaired coordination between abdominal muscle contraction and relaxation of the pelvic floor muscles and anal sphincters during defecation. As a result, stool evacuation becomes difficult despite normal colonic transit.



4. Clinical Assessment

History

- Stool frequency
 - Stool consistency
 - Straining
 - Sense of incomplete evacuation
 - Manual maneuvers
 - Duration
 - Medication history
 - Dietary habits
 - Physical activity
 - Associated symptoms

Alarm features or Red flag features

Based on these history and physical exam findings, it's important to look for red flag features that could indicate underlying serious conditions like malignancy.

Hematochezia

Unintentional weight loss,

Family history of colorectal cancer,

Acute onset of constipation in an older adult,

Change in stool caliber,

Anemia,

The presence of a rectal mass

If any of these red flag features are present, we should consider colorectal cancer and proceed with an endoscopy, such as colonoscopy, with a biopsy to look for obstructive lesions.

Physical examination

- General examination : May be completely normal or there may be alteration of the general condition
 - Abdominal examination : May reveal bloating , tenderness or a mass



- Digital rectal examination: Assessment of perianal sensation and sphincter tone during digital rectal examination may help identify neurological or defecatory disorders . The examination may reveal a rectal mass or palpable stool ball , some patients may have anal fissures or hemorrhoids

5. Diagnostic Evaluation

General measures

For patients with no red flag features, exclusion of secondary causes of constipation is mandatory . These include medication side effects, commonly seen in patients taking opioid medications or iron supplements; endocrine or metabolic disorders like hypothyroidism, hypokalemia, or hypercalcemia; and neurogenic conditions, such as spinal cord injury or multiple sclerosis.

In primary constipation a trial of initial conservative treatment , which includes increasing dietary fiber or starting a fiber supplement, advising sedentary patients to increase their physical activity, and starting an over-the-counter laxative, such as an osmotic agent or a stimulant laxative. . If there is an adequate response to empiric therapy no further workup is needed.

On the other hand, if they have an inadequate response to empiric therapy, additional testing will be mandatory as anorectal manometry and a balloon expulsion test or defecography . These tests measure pressures inside the rectum and anus and the ability of the pelvic muscles to expel stool from the rectum. These test are abnormal in defecatory disorder. In this condition constipation arises from difficulty in evacuating stool from the rectum.

However, if the anorectal manometry and balloon expulsion test are normal, a colonic transit study will be recommended. During colonic transit studies, patients swallow radiopaque markers, which are then tracked through the gastrointestinal tract using X-rays. If the markers are not passed within the expected time frame, then the test is abnormal, so you can diagnose slow transit constipation.

On other side, if the colonic transit study results are normal, diagnose normal transit constipation, which is also referred to as a disorder of the gut-brain axis. There are different types of normal transit constipation. One type is irritable bowel syndrome, which might be marked by alternating episodes of diarrhea and constipation, or dominated by either symptom.

6. Management

Management should begin directly with



Life style measures:

Education

- Dietary fiber
- Adequate hydration
- Physical activity
- Scheduled toileting

Pharmacological treatment

Class	Examples	Main indication
Bulk-forming	Psyllium	Mild constipation
Osmotic	Polyethylene glycol, lactulose	First-line
Stimulant	Bisacodyl, senna	Rescue therapy
Stool softener	Docusate	Selected patients
Secretagogues	Lubiprostone, linaclotide	Refractory cases
5-HT4 agonist	Prucalopride	Chronic refractory constipation

Biofeedback

Biofeedback is the treatment of choice for pelvic floor dyssynergia.

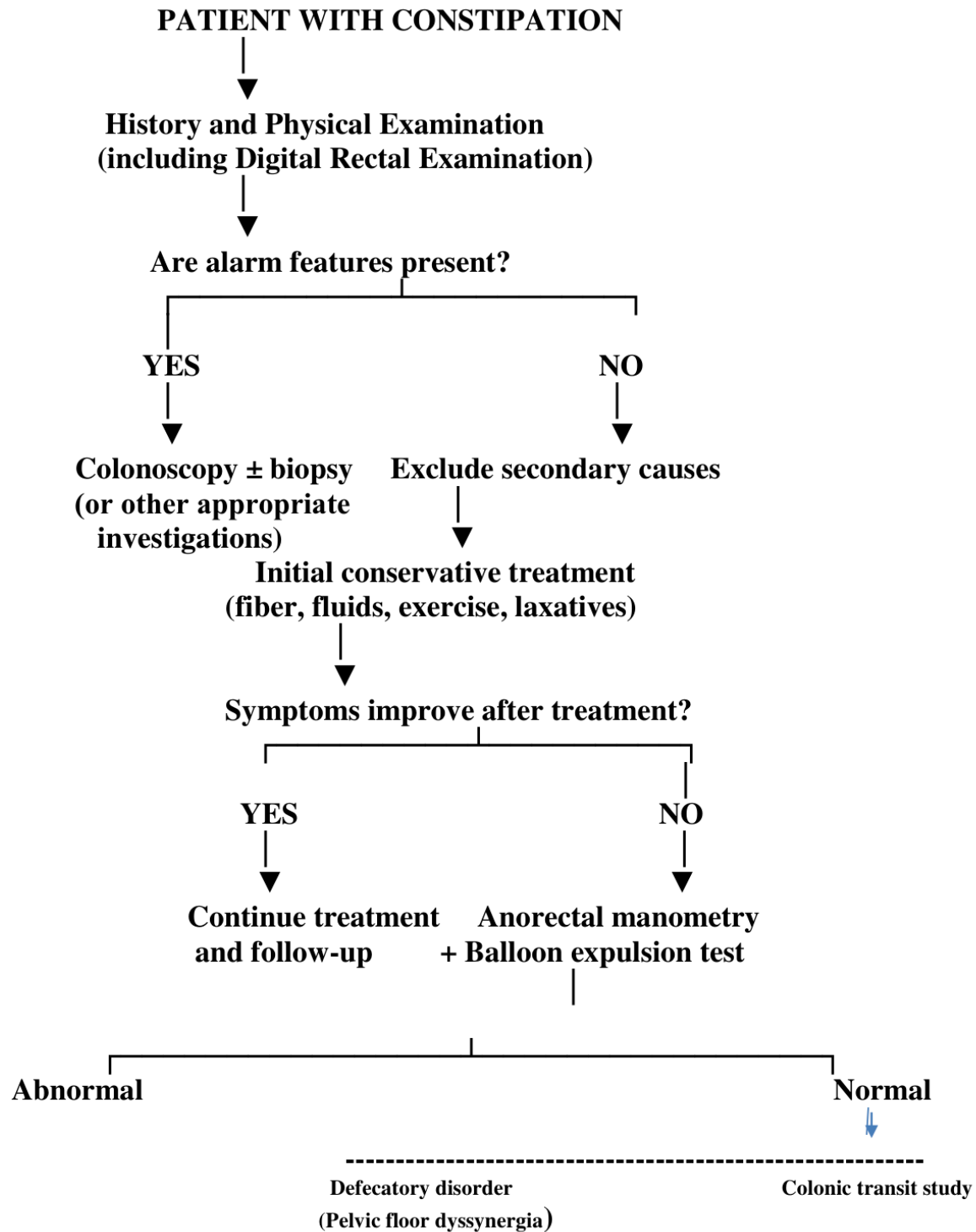


Figure 6.1 — Evaluation of constipation



Disorders of Gut–Brain Interaction (DGBI)

Definition : Disorders of Gut–Brain Interaction (DGBI) are a group of chronic gastrointestinal disorders characterized by persistent or recurrent gastrointestinal symptoms resulting from abnormal interactions between the gut and the brain in the absence of structural or biochemical abnormalities that fully explain the symptoms. They are common, frequently overlap, and may significantly impair quality of life.

Symptoms can be related to any combination of motility disturbances, visceral hypersensitivity, altered mucosal and immune function, altered gut microbiota, or disordered central nervous system processing (formerly known as "functional gastrointestinal disorders"). DGBIs are recognized through clinical symptom rather than structural biomarkers.

- Pathophysiology relies on the gut-brain axis, where psychological factors (like stress, anxiety, or trauma) directly alter gut motility and processing, also peripheral gut factors (such as post-infectious low-grade inflammation or dysbiosis) alter signaling pathways back to the brain.
- **Diagnostic Approach:** Standard medical tests—including endoscopies, blood work, and imaging—typically yield normal results, making clinical criteria (such as the Rome IV or Rome V guidelines) essential for diagnosis.

3. Common DGBIs

There are many recognized adult and pediatric DGBIs, categorized by the region of the gastrointestinal tract they impact:

GI Tract Region	Common DGBI Examples	Core Clinical Presentation
Esophageal	Functional Heartburn, Reflux Hypersensitivity	Chest burning or pain despite normal acid profiles and endoscopy.
Gastroduodenal	Functional Dyspepsia (FD)	Early satiety, postprandial fullness, or epigastric burning.
Bowel	Irritable Bowel Syndrome (IBS)	Abdominal pain related to defecation, accompanied by altered bowel habits (IBS-C, IBS-D, IBS-M).
Anorectal	Functional Constipation, Fecal Incontinence	Difficult, infrequent, or incomplete bowel movements unrelated to structural blockages.

4. Clinical Importance

- DGBIs represent a massive public health burden, affecting over 40% of the population worldwide
- **Diminished Quality of Life:** The chronic, unpredictable nature of symptoms significantly impairs daily functioning, causing a reduction in quality of life



- **Psychiatric Comorbidities:** There is a high overlap with anxiety, depression, and somatization. This requires a multidisciplinary approach
- **Healthcare Utilization Costs:** Because these disorders lack physical biomarkers, patients frequently undergo unnecessary, invasive diagnostic testing and surgeries (e.g., appendectomies or cholecystectomies), driving up healthcare expenses

Irritable Bowel Syndrome (IBS)

Irritable Bowel Syndrome (IBS) is a classic example of a Disorder of Gut-Brain Interaction (DGBI) characterized by recurrent abdominal pain associated with altered bowel habits.

1. Definition (Rome V Updates)

To make a clinical diagnosis of IBS, a patient must present with recurrent abdominal pain or discomfort that satisfies the following conditions:

- **Frequency Threshold:** No continuous abdominal pain but occurring, on average, at least 3 days per month over the last 3 months.
- **Duration Frame:** IBS requires an overall symptom onset of at least 6 months prior, with active symptoms over the last 3 months,
- **Bothersomeness:** Symptoms must be sufficiently bothersome to interfere with daily activities or prompt the individual to actively seek healthcare.

2. Required Symptom Associations

The abdominal pain or discomfort must be associated with two or more of the following characteristics:

1. **Relation to Defecation:** The pain/discomfort is modified (either relieved or exacerbated) by bowel movements.
2. **Change in Stool Frequency:** Onset of symptoms is linked to a change in how often the patient passes stool (e.g., more or less frequent).
3. **Change in Stool Form:** Onset of symptoms is linked to a change in the appearance or consistency of stool,

3. Mandatory Clinical Exclusions - Ruling Out "Red Flags"

A clinician can apply the Rome V clinical criteria only if standard medical history and a basic physical examination rule out these warning signs:

- Symptom onset after 50 years of age.
- Unintentional weight loss or unexplained fever.



- Rectal bleeding (hematochezia) or melena not attributed to hemorrhoids.
- Nocturnal symptoms that awaken the patient from sleep.
- Unexplained iron-deficiency anemia.
- Family history of colorectal cancer, celiac disease, or inflammatory bowel disease (IBD).

4. Epidemiology

- Prevalence Rebound: at approximately 8.5% to 10% and higher in women than men (approx. 2:1 ratio) and is most frequently diagnosed in individuals under the age of 50.

5. IBS Subtypes

- IBS with Predominant Constipation (IBS-C)
- IBS with Predominant Diarrhea (IBS-D):
- IBS with Mixed Bowel Habits (IBS-M):]
- IBS Unclassified (IBS-U): Meets diagnostic criteria for IBS, but stool consistency cannot be accurately categorized into any of the three main subtypes.

6-Pathophysiology of IBS

1. Altered Gut-Brain Interaction
2. Visceral Hypersensitivity
3. Altered Gastrointestinal Motility

Disruptions in the transit time and contraction patterns of the muscular layers of the GI

4. Altered Intestinal Microbiota (Dysbiosis)

Patients with IBS consistently exhibit a loss of microbial diversity and stability.

- 5.Low-Grade Immune Activation & Increased Intestinal Permeability

6. Psychosocial Factors

Psychological status acts as both an initiator and a modifier of IBS severity.

7- Clinical Assessment

History

- Abdominal pain
 - Relationship to defecation





- Stool frequency
- Stool consistency (Bristol Stool Form Scale)
- Bloating
- Symptom duration
- Dietary factors
- Psychological factors
- Previous gastrointestinal infection

Physical examination

- Usually normal
 - Mild abdominal tenderness and tympanitic resonance of the abdomen
 - Digital rectal examination when indicated

8-Rome V Clinical Diagnostic Criteria

IBS is a positive clinical diagnosis based on the Rome V clinical criteria rather than a diagnosis of exclusion.

We can rely only on limited laboratory investigations

- Fecal calprotectin (to exclude IBD)
- Celiac serology (mainly in IBS-D)
- Colonoscopy only when alarm features or other indications are present

Management of IBS

General measures

- Patient education
 - Reassurance
 - Regular physical activity
 - The patient should be advice for diet modification, which includes consuming soluble fiber, and a trial of a diet low in a specific set of carbohydrates, and low in high FODMAP including fructans and oligosaccharides found in onions, garlic, wheat, rye, and legumes.





- The use of peppermint oil, has been shown to relax smooth muscle and target spasming of the GI tract; as well as tricyclic antidepressants, like amitriptyline or nortriptyline, which are beneficial in relieving IBS- associated abdominal pain. Finally, gut-directed psychotherapy can be beneficial since IBS has a strong association with mental health.
- No routine recommendation of Probiotics
- For patient with the diarrhea-predominant type, the goal is to increase stool firmness or decrease bowel movement frequency. Treatment can be achieved by the non-absorbed antibiotic rifaximin, which works by addressing bacterial growth associated with diarrhea.
- On the other hand, there's the constipation-predominant type. Treatment for these individuals is geared towards stool softening and increasing bowel movement frequency. There are two options available for treatment. The first one includes chloride channel activators, like lubiprostone; while the second as prucalopride which is a selective, high-affinity serotonin type 4 (5-HT₄) receptor agonist. It acts as a prokinetic agent

Foodborn Infection



07



Learning objectives

- Explain the terminology related to gastrointestinal infection.
- Discuss the etiology, pathogenesis, epidemiology and prevention of microbial infection of gastrointestinal infections
- Discuss virulence mechanisms and correlate virulence mechanisms with specific pathogens
- Describe the laboratory diagnosis for bacteria and viruses
- Discuss different methods for preventing foodborne illness and food poisoning

Foodborne illness: are infections caused by a variety of microorganisms mainly *Salmonella*, *Clostridium perfringens*, *Campylobacter*, and *Staphylococcus aureus*

Norovirus

Salmonella is a member of Enterobacteriaceae family, it is the causative agent of:

- Enteric Fever (typhoid) or Salmonellosis, there is bacterial invasion of bloodstream by *Salmonella*.
- Acute gastroenteritis, resulting from a food-borne infection mainly contaminated **eggs** and intoxication.

Morphology: Gram negative bacilli, **motile**, non spore forming.

Culture: non-lactose fermenter on MacConkey agar. Special culture media include SS agar, deoxycholate citrate agar (DCA) and broth selenite and tetrathionate are used for **its isolation**.

Biochemical reactions: partial sugar fermentation, **urease negative**, oxidase negative, and H₂S positive

Antigenic structure: O (somatic) antigens which subdivide the *Salmonellae* into groups, and H antigen (flagella) antigens which determine the type within the groups. Virulent antigen (Vi Ag) is anti-phagocytic and is present only in *S. typhi*.

Pathogenesis is due to endotoxins and exotoxins. The exotoxins such as enterotoxins is responsible for diarrhea and body fluid loss.

Resistance: Killed by heating at 60°C for 30 min, antibiotics and other chemotherapy agents. Resist the action of bile salts.

■ Diseases caused by *Salmonella*:

1. Enteric fever is caused by *S. typhi* and the milder paratyphoid fever is caused by

Paratyphi A, B and C. Enteric fever is defined as generalized infection of reticuloendothelial system and lymphoid tissue accompanied by sustained fever and **bacteremia**. **Enteric fevers are endemic in Egypt.**

Pathogenesis:

.

After multiplication in reticuloendothelial system it is released into bloodstream and results signs of septicemia mainly fever. The organism is transported to the gall bladder to be excreted with bile. *Salmonella* multiply in intestinal lymphoid tissue and are excreted in **stools** at the second week.



Enteric fever is a strictly human disease, from case or carrier stool and urine. The organisms enter the human digestive tract with contaminated food or drink e.g raw shell fish ,milk products and raw vegetables. The organism adhere to the mucosa of small intestine by pili then penetrate to submucosal layer causing no lesion. In lymphoid tissue the organism invade macrophages where bacterial multiplication occurs . They are transported to mesenteric lymph nodes and lymphatics, released into the bloodstream and are carried to many organs such as bone marrow, kidneys intestine, spleen and liver where replication of salmonella inside the macrophages occurs.

Salmonella organisms **survives intracellularly** and **replicates in macrophages** due to decrease fusion of lysosome with phagosomes as well as due to the presence of surface proteins **defensins** which allow the bacteria to withstand oxygen-dependent and independent intra-cellular killing. When released from macrophages, the Vi capsular antigen (only in S. typhi) withstand complement - mediating killing

After multiplication in reticuloendothelial system it is released into bloodstream and results signs of septicemia mainly fever. The organism is transported to the gall bladder to be excreted with bile. Salmonella multiply in intestinal lymphoid tissue and are excreted in stools **at the second week**.

- The incubation period of the disease is 10-14 days .The fever is **gradually** elevated during the first week then become steady during the second week. There is enlargement of spleen and liver .The white cell count is normal **or low**. Salmonella excretion by patients may continue long after clinical cure. Outcome: 5% of patients cured from typhoid remain carriers for months or even years. Urinary carriers are common in Egypt because of bilharziasis.



- Laboratory diagnosis: Specimens are collected according to the stages of the disease. A- Isolation of Salmonella during the first week by doing blood culture
B- Second week, Salmonella is isolated from stool and after the second week from urine by direct culture on selective media.
C- Serological diagnosis by Widal agglutination test: This test demonstrates agglutination of serum antibodies of the patients against O-somatic antigens of Salmonella and H-flagellar antigens of S. typhi and Paratyphi A, B, C. Antibodies rise sharply during the second and third weeks of enteric fevers. Demonstration of a rising titers in antibodies in the paired samples indicates acute recent infections. The results should include the titer of both O and H antibodies and Vi antibodies. The presence of **IgM** is indicative of acute recent infections.

Interpretation of Widal test:

- 1- High or rising titer of antibody to O antigens ($\geq 1:160$) suggests that active infection is present.
- 2- High titer of antibody to one type of H antigen ($\geq 1:160$) suggest recent infection with this type of Salmonella.
- 3- High titer of antibody to the **Vi antigen** occur in some carriers.
- 4- High titer of antibodies to O antigens and more than one H antigen indicate recent immunization. With old immunization or past infection O antibodies disappear and only antibodies to H antigens remain.

Widal test has many limitations as a diagnostic serological test for enteric fever.

Several factors need to be considered for interpretation:

- 1- **Endemicity of enteric fever** in an area of investigation. Healthy persons have antibody titer due to subclinical infection. Titers up to 1:160 are considered normal in Egypt.

False positive results may occur in an endemic area, immunization with any typhoid vaccine or a previous infection or infection with other related members of **Enterobacteriaceae**

- 2- False negative in case of administration of antibiotics and if widal test is done in early stage of disease where low antibody titers are found.

- Antibiotics may be not effective on Salmonella carriage because the site of carriage may **not allow penetration by the antibiotic e.g due to the presence of fibrous tissue**

Immunity:

Persistent immunity is acquired after enteric disease. **CMI** is the main immunity. Humoral immunity destroy the organism in the blood. Antibody titer can continue for a long time after recover.

Prevention and Control by

- 1- Immunization: several typhoid vaccines are available for use :



- a. Parenteral heat-phenol-inactivated vaccine contains *S. typhi* and paratyphi A,B,C hence the name TAB vaccine . The vaccine is given as 2 subcutaneous (SC) doses. This is the current vaccine used in Egypt.
 - b. Other vaccines include :acetone-inactivated parenteral vaccine,oral live-attenuated vaccine for *S. typhi*. Living vaccine should not be given to children younger than 6 years of age. Vi capsular polysaccharide (ViCPS) is a newly licensed capsular polysaccharide vaccine for parenteral (intramuscular) use.
- No typhoid vaccine is 100% effective and is not a substitute for good hygiene

2.Salmonella food poisoning:

a- *Salmonella typhimurium* foodborne toxic infections. Symptoms (diarrhea, vomiting, fever) appear after 12-24 hours following ingestion of **meat** contaminated **with rat excreta which contain large number of organisms. It last for 2-5 days.**

b- *Salmonella enteritidis* infection or Egg-associated salmonellosis The egg can be inside perfectly normal. Incubation period from 12 to 72 hours after consuming a contaminated food. The illness usually lasts 4 to 7 days, and most persons recover without antibiotic treatment.

Campylobacter :12-15 species are associated with human disease:

Campylobacter jejuni (poultry; humans), *Campylobacter coli* (porcine; humans) and *Campylobacter fetus* (cattle; sheep).

Morphology: They are gram negative curved bacilli (S-shaped or “gull-winged” appearance),non-spore forming, have a characteristic rapid darting motility by means of long sheathed polar flagellum at one (polar) or both (bipolar) ends.

Motility slows **quickly in wet mount microscopy preparation due to aerotoxicity (oxygen sensitivity).**

a- Culture: Micro-aerophilic thermophilic and capnophilic (CO₂ :10%) on Selective media with added antibiotics as Skirrows medium at **42°C** which is the natural body **temperature of avian species.**

Biochemical reactions: **Oxidase positive**, catalase positive, do not utilize carbohydrates. Pathogenesis: Virulence factors include:

a- Cellular components: Endotoxin(LPS), flagellum,adhesins and invasins.

b- Extracellular components: Enterotoxins (LT) and cytotoxins.

Transmission: feco-oral route.*Campylobacter* is usually acquired from food (specially poultry),milk,or contact with infected animals (e.g. dogs). Children under 5 years and young adults are the most frequently affected age groups.

Incubation period of 1-7 days with acute onset, abdominal cramps, nausea but not vomiting. The bacteria damage the mucosal surfaces of jejunum; ileum and colon. Stools may contain mucus or gross blood. Generally self-limiting, symptoms may last for a week or longer. Bacteremia is common and *Campylobacter* is isolated from blood.



Important late onset complications following gastroenteritis include: Guillain-Barre

Syndrome; Reiter's syndrome; reactive arthritis and spontaneous abortion .

Laboratory diagnosis: Specimen collection and processing stool should be examined within few hours. Care to avoid oxygen exposure while taking sample or better a special transport medium is used.

A- Direct examination: by Gram stain and observation of darting motility in wet mount preparation

B-Culture: on Skirrows medium.

Antibiotic resistance: Fluoroquinolone resistance has developed rapidly related to unrestricted use and the use of enrofloxacin in poultry. The organism is inherently resistant to trimethoprim and most cephalosporins.

Brucella is the cause of brucellosis, a zoonotic disease. It is a virulent, highly infectious bacteria. They are facultative intracellular pathogen surviving in macrophages. There are different species of Brucella which include: Brucella abortus (cattle) Brucella melintensis (goats, sheep) Brucella suis (swine, pigs) Brucella canis (dogs). Species are differentiated by production of **urease and H₂S**, dye sensitivity, cell wall antigens and phage sensitivity.

Morphology: Gram-negative small bacilli, non-spore forming, non-motile and non- capsulated. Culture: Aerobic, require enriched media containing serum e.g. serum dextrose agar. Colonies appear after 48 to 72 hours. Biochemical reactions: Positive for catalase, oxidase, urease and reduce nitrates.

Brucellosis (Malta fever or undulant fever): This a true zoonotic disease as human to- human transmission has not been identified. The disease is characterized by acute undulating fever i.e bouts of fever 3-4 weeks then 3-4 weeks without fever.

Transmission: Direct contact through skin abrasions, mucous membrane with infected animals or animal products (farmers, butchers and veterinarians), Consumption of unpasteurized milk products and meat, inhalation of aerosols from infected animal or laboratory-associated.

Laboratory diagnosis:

Specimen: blood is taken during acute disease. Other sample according to site of infection such as joint fluid, CSF, pus. Gram stain is of no significance.

A- Culture: Blood culture is the method of choice early in the disease. Cultures need to be incubated for up to 4-6 weeks as they are slow -growing. It is important to differentiate Brucella from Salmonella by urease test as brucella produce urease (+ve) while Salmonella not (-ve).

B- Serology: Is the main method of laboratory diagnosis. Two serum samples is collected 14-21 days apart. A wide range of serum dilutions must be done in order to avoid false negative result due to prozone effect that is commonly



present in cases of brucellosis as large amount of antibodies are produced. Dilutions are done from 1:20 to 1:520. ELISA is used to differentiate between specific IgM and IgG antibodies.

D- Brucellin test: intradermal test based on delayed type hyper-sensitivity.

Interpretation of the result is difficult in persons frequently exposed to infection.

Prevention: Wearing protective impermeable clothing, rubber boots, gloves and face masks when exposed to animals. Pasteurization of milk. Treatment of domestic animals and Live attenuated vaccines for animals (cattle)

Treatment: using combinations of antibiotics for 4 to 6 weeks.

Food born poisoning:

Staphylococcus aureus : are catalase positive Gram-positive cocci

.Staphylococcal enterotoxins do not change the taste, color or odor of food. They are heat stable for 30 minutes of boiling and are not inactivated by brief cooking. The source of food contamination is a food handler carrying the organism in nose, hands, or having superficial lesions as boil (abscess). The common types of food are carbohydrate rich food e.g. cake, koshari and protein rich food e.g. milk and its product. The incubation period is short, 1-6 hours, due to consumption of preformed enterotoxins. The food poisoning is characterized by vomiting and watery non bloody diarrhea.

Laboratory diagnosis: Specimen include food, vomitus microscopic examination of

Direct examination: Gram positive cocci in grape like clusters. Culture is done on blood agar to isolate beta (complete) hemolytic colonies and by using selective mannitol-salt agar. Typing of S. aureus is done for epidemiologic purposes, in food poisoning outbreaks by phage typing and PCR.

Salmonella food poisoning: under Salmonella

Spore forming gram positive bacilli : The gram positive spore forming bacilli include Bacillus and Clostridium species. Bacillus species are aerobes while clostridia are anaerobes.

1. B. cereus cause two forms of food poisoning a. The "short-incubation" or emetic form of the disease is mediated by preformed heat stable enterotoxin usually in improperly stored, cooked rice. It has an incubation period of 1-6 hours. It resembles S. aureus food poisoning b. The "long-incubation" or diarrheal form: caused by heat labile enterotoxin present in meat or vegetables with an incubation period of 8-16 hours. This type of exotoxin activates adenylate cyclase, increases cAMP causing profuse fluid secretion. In either type, the illness usually lasts for less than 24 hours after onset.

2. Clostridium botulinum: is the causative agent of Botulism which is a rare but fatal form of food poisoning, is caused by a potent botulinum toxin.



Transmission and pathogenesis : Toxin is produced within the contaminated canned food and ingested preformed then becomes activated by proteases in the gastric fluid and by gastric acidity then absorbed in the intestine and is transported systemically via the blood stream. it binds to receptors on peripheral nerves, inhibits the release of acetylcholine and lead to flaccid paralysis. Botulinum toxin is rapidly acting and signs and symptoms appear within 12 hours after ingestion . Antitoxin is only effective before the toxin binds the neuromuscular junction within 12 h after ingestion. Prognosis is poor for patients diagnosed after this time period.

Laboratory diagnosis: Diagnosis depend on clinical manifestations .The presence of **exotoxins in canned food, is detected by by ELISA .**

Trivalent anti-toxin serum(types A,B and E) is given IV along with respiratory support are used as treatment .Antibiotics are not used to treat botulism, Prevention: Swollen cans must be discarded.

Other GIT pathogens include

Leptospira are spirochetes that live in stagnant water, lakes .

Morphology: Spiral-shaped, motile ,one or both ends are hooked. Because they are so thin, live Leptospira are best observed by dark-field microscopy.Culture: Aerobic, at 30 °C, need enriched media (Stuart 's media) to enhance growth that appear after 1-2 weeks.

Leptospirosis (Weil's disease,7-day fever): is a bacterial zoonotic disease caused by

Leptospira. The infection is commonly transmitted to humans by **allowing fresh water** that has been contaminated by animal urine to come in contact with breaks in the skin, eyes or with the mucous membranes. Occupational risk factors include veterinarians, slaughter house workers, farmers, mine, and sewer workers.The disease is not known to be spread from person to person.

Pathogenesis : Incubation period: 4-14 days.It is a biphasic disease, the patient is asymptomatic in between phases: Variable febrile onset: last for 4-7 days .There is hepatitis and jaundice and the second phase is manifested by meningitis (in 40%) with intense headache and stiff neck. Hepatitis, nephritis may recur.

Laboratory diagnosis: Specimen: Organisms are present in blood for the first 7 to 10 days. After this period the microorganism can be found in fresh urine, kidney biopsy.Direct examination: Dark-field microscope and PCR. and Culture: Colonies are identified by agglutination test, DNA probes.Serologic diagnosis: By ELISA for Ig M and total antibody.There are no human vaccines. Animal vaccines are effective for a few months.

Yellow Fever (YF): It is a haemorrhagic fever caused by yellow fever virus which is a single-stranded enveloped RNA arboviruses (arthropod borne virus) .The disease is present in equatorial countries in Africa and South America (Amazon basin). Man usually forms an accidental, non-normal host for the virus .**The liver and stomach are** highly impacted



Pathogenesis and disease: Incubation period:3-5 days.

- Human activity can bring man into contact with mosquitos eg. clearing of forests, agriculture. The virus multiply in the reticuloendothelial cells, then viremia occur to be localized in liver,spleen,kidneys and bone marrow. This is a severe systemic disease in which hemorrhages, degeneration of the liver, kidney and heart occur.

Jaundice is a prominent feature due to liver necrosis. The case-fatality rate can be 50%.

Life cycles:

1- Jungle (sylvatic) yellow fever : Monkey is the main host and the virus is transmitted between monkeys by Aedes Africans mosquito in Africa .Forest worker/visitor are infected accidentally and carries infection to urban areas.

2- Urban yellow fever: The disease is introduced into towns; so the cycle is: man - mosquito-man.The mosquito is Aedes egypti.

Laboratory diagnosis:

- 1- Intra-cerebral inoculation of mice with blood sample.
- 2- Direct detection of RNA virus by PCR.
- 3- Detection of rising antibody titre by neutralizing test, complement fixation,haemagglutination test.

Prevention and control:

1- Use of insecticide to kill mosquitoes and their eggs.

2- Vaccine: “17 D strain” live attenuated virus, given as a single dose subcutaneously. Safe long lasting protection is obtained for 10 years. Vaccination is required for **travelers to endemic equatorial zones**.

Dengue, and Ebola all affect the liver and the gastrointestinal (GI) system, but with varying severity.

- Dengue virus : RNA enveloped viruses .The natural hosts are monkeys and man, mosquitoes are the vector. Dengue virus has an urban and a jungle cycle. Symptoms include fever, headache, rash, bone pain (“breakbone fever”),myalgias and arthralgia. Dengue Hemorrhagic Fever is a deadly complication of dengue.Case fatality rate is 50%.
- Ebola and Marburg viruses are RNA enveloped viruses cause fatal disease in man, there is haemorrhage, fever, diarrhea, vomiting and muscle pain. No known reservoir/ **vector hosts. They are transmitted by direct contact with infected body fluids.**

INFLAMMATORY BOWEL DISEASE (IBD)



08



Objectives

The student will

- 1) Understand the definition , basic concepts of inflammatory bowel disease
- 2) Understand the differences between types of inflammatory bowel disease
- 3) Know the clinical features of inflammatory bowel disease
- 4) Know the diagnostic tools for inflammatory bowel disease
- 5) Know the basic treatment **options** for inflammatory bowel disease

Definition

- Chronic inflammatory diseases of the gastrointestinal tract.
- They are diagnosed by a set of clinical, endoscopic, and histologic characteristics, but no single finding is absolutely diagnostic for one disease or the other.
- Moreover, some patients have a clinical picture that falls between the two diseases and are said to have indeterminate colitis.

Epidemiology

- The highest rates occur in white populations in northern Europe and North America, where the incidence for each disease is about 5 per 100,000 and the prevalence is approximately 50 per 100,000.
- The peak age at onset is between 15 and 25 years of age, with a second, lesser peak between 55 and 65 years.

Clinical features

Ulcerative Colitis

- The dominant symptom in ulcerative colitis is diarrhea, which is usually associated with blood in the stool.
- Other symptoms include fever and pain, which may be in either lower quadrant or in the rectum.
- The initial attack of ulcerative colitis may be fulminant with bloody diarrhea, but more commonly the disease begins indolently, with non-bloody diarrhea progressing to bloody diarrhea.

Crohn's Disease

- Crohn's disease is marked by one of three major patterns: (1) disease in the ileum and cecum (40% of patients), (2) disease confined to the small intestine (30%), and (3) disease confined to the colon (25%).
- The predominant symptoms are diarrhea, abdominal pain, and weight loss.
- In patients with colonic disease, especially with rectal involvement, diarrhea is of small volume and associated with urgency and tenesmus.



- Prolonged inflammation and scarring in the rectum can leave it so rigid and nondistensible that the patient is incontinent.
- In disease confined to the small intestine, stools are of larger volume and not associated with urgency or tenesmus.
- Patients with severe involvement of the terminal ileum and those who have undergone surgical resection of the terminal ileum may have bile salt diarrhea or steatorrhea.

Feature	Crohn's Colitis	Ulcerative Colitis
Mucosal lesions	<u>Aphthous ulcers</u> are common in early disease; late disease is notable for stellate, rake, bear-claw, linear, or serpiginous ulcers and <u>cobblestoning</u> .	Micro-ulcers are more common, but larger ulcers are seen <u>Pseudopolyps</u> are more common
Distribution	Often discontinuous and asymmetric, with <u>skipped segments</u> of normal intervening mucosa, especially in early disease	<u>Continuous, symmetric, and diffuse</u> , with granularity or ulceration found in the entirety of involved segments; however, peri-appendiceal inflammation (cecal patch) is common, even when the cecum is not involved
Rectum	Complete, or <u>more often relative, rectal sparing</u>	<u>Typically involved</u> with variable proximal distribution
Ileum	<u>Often involved</u> (=75% of cases)	<u>Not involved</u> , except as backwash ileitis in panulcerative colitis
Depth of inflammation	<u>Mucosal, submucosal, and transmural</u>	<u>Mucosal</u> ; transmural only in fulminant disease
Serosal findings	Marked erythema and creeping fat (the latter is virtually pathognomonic)	Absent except in severe colitis or toxic megacolon
Perianal complications	<u>Often prominent, including large anal skin tags, deep fissures, perianal fistulas</u> , that are often complex	<u>Not prominent</u> (fissure or fistula if present, should be uncomplicated)
Strictures	<u>Often present</u>	<u>Rarely present</u> ; when present, suggests adenocarcinoma
Fistulas	<u>Perianal, enterocutaneous, rectovaginal, enterovesicular</u> , and other fistulas may be present	<u>Absent</u> , except for the rare occurrence of rectovaginal or perianal fistula
Histopathology	<u>Granulomas are present in 15%-60% of patients</u> (higher frequency in surgical specimens than in mucosal pinch biopsies) Crypt abscesses may be present Focally enhanced inflammation, often on a normal background, is the hallmark	Granulomas should not be present (microgranulomas may be associated with ruptured crypt abscess) <u>Crypt abscesses and ulcers are the defining lesion</u> <u>Ulceration on a background of inflamed mucosa</u>
Serology	pANCA in 20%-25%, <u>ASCA in 41%-76%</u>	<u>pANCA in 60%-65%</u> , ASCA in 5%

Figure 8.1 — Crohn's disease and ulcerative colitis: comparison



Extra-intestinal Manifestations

COMPLICATIONS	CROHN DISEASE	ULCERATIVE COLITIS
Ocular disorders (uveitis, episcleritis)	+	+
Arthropathy	+	+
Oral ulcers	+	-
Skin disorders (pyoderma gangrenosum, erythema nodosum)	+	+
Nephrolithiasis	+	+
Primary sclerosing cholangitis	+	+
Bone disorders (osteoporosis, osteomalacia)	+	-
Thromboembolic disease	+	+
B ₁₂ deficiency	+	-

Figure 8.2 — Extra-intestinal manifestations

Diagnosis

■ Laboratory Studies

A complete blood count (CBC), urinalysis, stool analysis and culture and serum chemistry tests are appropriate during the initial evaluation.

Serologic examination of blood for specific antibodies , perinuclear antineutrophil cytoplasmic antibodies (pANCA)[frequent in UC], antibodies to *Saccharomyces cerevisiae* (ASCA) [frequent in CD].

A newly recognized and clinically useful inflammatory marker for UC disease activity is fecal calprotectin, which is a protein secreted by neutrophils in the feces and is therefore a marker of intestinal inflammation.

- Barium enema and barium follow through
- Endoscopy



Differential diagnosis

Diagnosis	Clinical Features	Radiologic and Colonoscopic Features	Histologic Features
UC	<u>Bloody diarrhea</u>	Extends proximally from rectum; fine mucosal ulceration	Distortion of crypts; acute and chronic diffuse inflammatory cell infiltrate; goblet cell depletion; crypt abscesses; lymphoid aggregates
Crohn's colitis	<u>Perianal lesions are common; may be associated with ileitis; frank bleeding is less common than in UC</u>	Segmental disease; rectal sparing; strictures, fissures, ulcers, fistulas; small bowel involvement	Focal inflammation; submucosal involvement; granulomas; goblet cell preservation; transmural inflammation; fissuring
Ischemic colitis	<u>Occurs in older adults; sudden onset, often painful; usually resolves spontaneously in several days</u>	Segmental splenic flexure and sigmoid involvement are most common, with thumbprinting early and ulceration after 24-72 hr; rectal involvement is rare	Mucosal necrosis with ghost cells; congestion with red blood cells; hemosiderin-laden macrophages and fibrosis (when disease is chronic)
Microscopic colitis	<u>Watery diarrhea; normal-appearing mucosa at colonoscopy</u>	Usually normal	Chronic inflammatory infiltrate; increased intraepithelial lymphocytes (lymphocytic colitis) and/or subepithelial collagen band (collagenous colitis)
Infectious colitis	<u>Sudden onset; identifiable source in some cases (e.g., <i>Salmonella</i> spp.); pain may predominate (e.g., <i>Campylobacter</i> spp.); pathogens are present in stool</u>	Nonspecific findings	Crypt architecture is usually normal; edema, superficial neutrophilic infiltrate, crypt abscesses
Amebic colitis	History of travel to endemic area; amebae may be detected in a fresh stool specimen but ELISA for amebic lectin antigen is the preferable diagnostic test	Discrete ulcers; ameboma or strictures	Similar to UC; amebae present in lamina propria or in flask-shaped ulcers, identified by periodic acid-Schiff stain
Gonococcal proctitis	Rectal pain; pus	Granular changes in rectum	Intense polymorphonuclear neutrophil infiltration; purulent exudate; Gram-negative diplococci
Pseudomembranous colitis	<u>Often a history of antibiotic use; characteristic pseudomembranes may be seen on sigmoidoscopy; <i>Clostridium difficile</i> toxin is detectable in stools</u>	Edematous; shaggy outline of colon; pseudomembranes may be identified radiologically or seen at colonoscopy	May resemble acute ischemic colitis; summit lesions of fibrinopurulent exudate

Figure 8.3 — Differential diagnosis of inflammatory bowel disease

Treatment

Diet and nutrition

- Patients with mild symptomatic IBD usually are able to take food orally.
- The diet should be nutritious.
- Patients with Crohn's disease who have terminal ileal involvement and steatorrhea may require supplemental fat-soluble vitamins, medium-chain triglycerides, and parenteral vitamin B12.
- Replacement iron may be indicated in patients who are iron-deficient.

Corticosteroids

- Historically, corticosteroids have been used in patients with severe UC or Crohn's disease to induce a remission.
- Intravenous (IV) steroids (i.e., hydrocortisone 100 mg IV q6h or methylprednisolone 10-30 mg IV q6-8h) are usually used in such patients.



- When patients can take oral medications, prednisone at doses of 40 to 60 mg per day is usually given.

Antibiotics

- Bacteria is known to play an important role in the pathogenesis of Crohn's disease and it may play a role in UC.
- In Crohn's diseases, the indications for the use of antibiotics include perianal disease, localized peritonitis due to microperforation, bacterial overgrowth secondary to a chronic stricture, and as an adjunct to drainage therapy for abscesses and fistulas.
- Metronidazole, Ciprofloxacin and Rifaximin can be used.

Immunomodulators and biologic therapy

- Immunomodulator drugs act by blocking lymphocyte proliferation, activation, or effector mechanisms
- Azathioprine, Methotrexate, Cyclosporin can be used
- Anti-Tumor Necrosis Factor Antibody (biologic therapy)
- TNF is a key inflammatory cytokine and mediator of intestinal inflammation. The expression of TNF is increased in IBD.
- Infliximab is a chimeric mouse-human monoclonal antibody against TNF that is effective in CD.

Anti-Adhesion Molecules

- Adhesion molecules are important in cellular trafficking in IBD and other diseases, in which immune and inflammatory cells from the periphery are recruited into sites of inflammation.
- Gut-selective anti-adhesion molecule, vedolizumab shows promising results.

Kinase Inhibitors

- The Janus kinase (JAK) family of kinases, which include JAK1 and JAK3 which are critical for lymphocyte proliferation, function, and activation.
- Tofacitinib, a novel oral inhibitor of JAK1 and JAK3, was recently shown to be safe and effective for the treatment of moderate to severe active UC.



Indications for surgical intervention

Ulcerative Colitis	Crohn's Disease
Intractable disease	Small Intestine
Fulminant disease	Stricture and obstruction unresponsive to medical therapy
Toxic megacolon	Massive hemorrhage
Colonic perforation	Refractory fistula
Massive colonic hemorrhage	Abscess
Extracolonic disease	Colon and Rectum
Colonic obstruction	Intractable disease
Colon cancer prophylaxis	Fulminant disease
Colon dysplasia or cancer	Perianal disease unresponsive to medical therapy
	Refractory fistula
	Colonic obstruction
	Cancer prophylaxis
	Colon dysplasia or cancer

Figure 8.4 — Indications for surgical intervention

Complications

Toxic megacolon

- Toxic megacolon is a condition in which the colon becomes atonic and dilated because of transmural inflammation.
- It is characteristically associated with severe ulcerative colitis, but it may complicate any severe inflammatory condition of the bowel, including Crohn's disease, bacterial colitis, amebiasis, pseudomembranous colitis, and ischemic colitis
- Worsening of the patient's clinical condition and the development of fever, tachycardia, and leukocytosis.
- Medical therapy is designed to reduce the likelihood of perforation.
- If the patient does not begin to show signs of clinical improvement during the first 24 to 48 hours of medical therapy, the risk for perforation increases markedly, and surgical intervention is indicated.

Malignancy

- Patients with extensive ulcerative colitis have a markedly increased risk for colon cancer in comparison to the general population beginning 8 to 10 years after diagnosis and increasing with time.
- There should be surveillance colonoscopy with random biopsies in patients with long-standing ulcerative colitis beginning 8 to 10 years after the onset of disease and repeated every 1 to 2 years.
- If the specimens show dysplasia the patient is sent for colectomy.

Parasitic GI diseases



09



BOOD FLUKES

SCHISTOSOMA

Schistosomes are the causative agents of the disease schistosomiasis. Three species of schistosomes that parasitize humans are of major importance: *S.mansoni*, *S. haematobium* and *S.japonicum*.

The morphologic characteristics of the three species are given in the following table:

	S.mansoni	S.japonicum
Disease	Intestinal bilharziasis	Intestinal bilharziasis
Distribution	Nile Delta, Africa, South America, Middle East.	Far East.
Intermediate host	Biomphalaria alexandrina	Onchomelania species.
Reservoir host	Monkeys, rodents.	Dogs, cats, rodents, pigs and cattle
Habitat	Inferior mesenteric plexus of veins	perior mesenteric plexus of veins





Lifecycle:

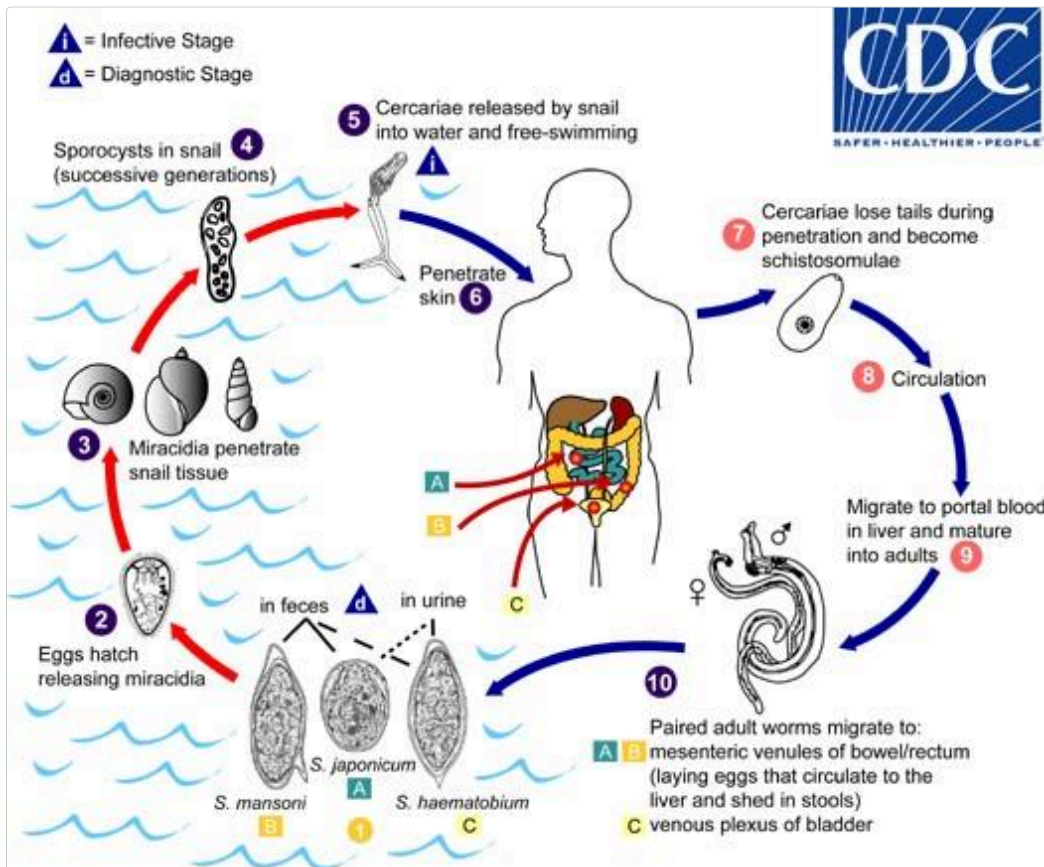


Figure 9.1 — Schistosoma life cycle

Eggs escape to the perivascular tissue due to:

- pressure within the venule.
- effect of the spine.
- the oozing lytic secretion of the miracidium within the egg. From the perivascular tissue eggs pass to the lumen of the intestine in case of *S. mansoni* and *S. japonicum* and to the lumen of the urinary bladder in case of *S. haematobium*.

When the embryonated eggs (diagnostic stage) discharged in faeces enter fresh water, miracidia hatch from the eggs, rapidly swim by means of cilia (survive for 24 hours). Then they penetrate the body of snail (intermediate host).

The cercariae infect the definitive host by penetrating the skin, aided by:

- proteolytic enzymes secreted from penetration glands.
- the surface tension of drying droplets of water.
- the strong lashing movements of the tail pressing the body of the cercaria into the skin. Drinking water may lead to infection when cercariae penetrate the mucosa above gastric acidity that kills them.

They lose their tails out and transform into the schistosomula.



Schistosomula migrate to the lungs via the venous circulation. They break out from the - lungs Through the pulmonary capillaries and are carried through the left heart into the systemic circulation. They finally reach the intra-hepatic branches of the portal veins in the liver where they settle down and mature.

Within several weeks, pairing of the worms takes place and they migrate to the mesenteric - veins (*S.mansoni* and *S. japonicum*) or to the vesical plexus (*S. haematobium*), where the females lay eggs.

The prepatent period between penetration by cercariae and the appearance of eggs in the - excreta is around 35 days for *S.munsoni* and *S.japonicum* and about 70 days for *S.haematobium*.

PATHOGENESIS AND CLINICAL PICTURE:

Intestinal schistosomiasis is 7-8 weeks (for *S. mansoni* infection) and 5-6 weeks (for *S. japonicum* infection).

Pathological changes are caused mainly by schistosome eggs deposited in various tissues. The antigens excreted from embryonated eggs result in formation of granulomas that lead eventually to fibrotic changes in the tissues involved in chronic infections. Thus, the lesions produced by the infections are essentially the same among the 3 major species of schistosomes. There are 4 progressive stages:

1. Stage of invasion (1 - 4 days):

- Skin lesion due to cercarial penetration.
- Local dermatitis, irritation, papular rash.

2. Stage of migration (3 — 4 weeks):

- Lung: verminous pneumonitis, minute haemorrhages, cough and haemoptysis.
- Liver: enlarged and tender.

Metabolic products of migrating parasites: result in toxic and allergic manifestations de - urticarial, eosinophilia, leukocytosis, fever, and headache and muscle pain.

3. Stage of egg deposition and extrusion (acute stage: 1-2 month):

Eggs deposited in the venous plexus escape from the veins into the perivascular tissue and finally find their way to the outside with urine and stool (open lesion stage). With extrusion of eggs through the wall of the intestine or bladder, there may be again generalized malaise, fever, rigors, urticaria, abdominal pain and liver tenderness.

- **In schistosomiasis mansoni and schistosomiasis japonicum (intestinal bilharziasis):**

Active egg deposition results in tissue damage and haemorrhage.



symptoms:

dysentery with blood and mucus in stool

- abdominal pain.
- Sjapronicum may present with Katayama fever

4. Stage of tissue reaction (chronic stage):

Eggs trapped in the tissue secrete soluble egg antigen that stimulate inflammatory reactions (eosinophilic granulomas) and fibrosis leading to formation of bilharzial nodules, papiliomata, sandy patches and reduced egg output (closed - lesion stage).

• In schistosomiasis mansoni and schistosomiasis japonicum:

The intestinal wall becomes fibrosed, thickened and may be complicated with strictures, sinuses, fistulae and prolapse.

Eggs that fail to be fixed to venules of intestinal walls fall in the lumen and are swept by the blood to the liver. This results in periportal fibrosis, portal hypertension, hepatosplenomegaly, followed by shrunken fibrosed liver.

This results in portal hypertension with oesophageal varices, haemorrhoids, and ascites. This is manifested by haematemesis, melena and distended abdomen.

This stage ends in hepatocellular failure (impaired liver function) and fatal haemorrhage. The enlarged spleen and the hypersplenism lead to destruction of the red blood cells that add to the existing anaemia. Later on it leads to pancytopenia.

II. Intestinal schistosomiasis:

1. Clinical diagnosis:

Living in or visiting endemic areas with history of exposure to canal water and complaining of dysentery or hepatosplenomegaly is suggestive.

2. Parasitological diagnosis:

Stool examination for recovery of the eggs (diagnostic stage) by:

- Direct smear.
- Concentration techniques may be required when infection is light.

3. Immunodiagnosis:

Detection of antigens and antibodies as in urinary schistosomiasis.

4. Other tests:

- Complete blood count (CBC): Shows eosinophilia up to 40 % and anaemia.
- Rectal swab
- colonoscopy and sigmoidoscopy for visualizing lesions e.g. papillomata, colonic polyps, and for biopsies.



d) Ultrasonography and other imaging techniques: show periportal fibrosis and heptesplenomegaly.

TREATMENT:

1. Praziquante! (Biltricide), Oxmoquine (Vansil) and Metriphonate

PREVENTION AND CONTROL:

1. Mass treatment and follow up of infected persons.
2. Health education
- 3- Snail control

TAENIA SAGINATA

(BEEF TAPE WORM, BALD TAPE WORM)

GEOGRAPHICAL DISTRIBUTION:

Cosmopolitan especially in cattle raising countries.

1. Adult lives in the small intestine of man who is the only definitive host attaching itself to the mucosa.
2. Mode of infection: by ingestion of undercooked beef containing viable cysticerci usually in the form of hamburgers or grilled meat (Kebab).

In the small intestine, the scolex is evaginated and gets attached to the mucosa and matures in about 3 months.

Habitat: small intestine.

DH: man.

IH: cattle.

Infective stage: cysticercous bovis in cattle meat.

Diagnostic stage : mature egg in stool.

PATHOGENESIS AND CLINICAL PICTURE:

1. Intestinal disturbances, hunger pains and diarrhea.
2. Neurologic manifestations.
3. Intestinal obstruction.
4. Loss of weight as the parasite consumes much of the food of the patient leading to malnutrition and avitaminosis.
5. Appendicitis or cholangitis caused by stray segments of the worm.



6. Migrating segments creeping out of the anus cause irritation, itching and worry of the patient

DIAGNOSIS:

1. Clinical diagnosis: Suggestive symptoms as diarrhoea and weight loss, with passage of gravid segments through the anus.

2. Parasitological diagnosis:

- Stool examination to find eggs and proglottids (diagnostic stages) Recovery of eggs from perianal region using a swab.
- Gravid segments in faeces, if not found give a saline purge.

Compress the segments between two slides and count the main lateral branches.

3. Immunodiagnosis: Detection of coproantigen in patients' stool by ELISA.

TREATMENT:

1. Praziquantel is the drug of choice for both *Taenia saginata* and *Taenia solium*; it causes worm paralysis, in a dose of 10mg/Kg as a single oral dose.
2. Yomesan in a single oral dose of 2 gm on empty stomach.

CONTROL:

1. Treatment of the infected persons.
2. Avoid grazing cattle on fields fertilized by human manure.
3. Proper meat inspection in slaughter houses.
4. Refrigeration of meat at -10°C for 5 days.
5. Thorough cooking of beef.

TAENIA SOLIUM

(Pork tape worm)

GEOGRAPHICAL DISTRIBUTION:

Cosmopolitan where pork is eaten. Thus it is rare in Islamic and Jewish communities.

■ Similar to *T.saginata* but differs in:

1. Pig is the intermediate host.
2. Larval stage is the cysticercus cellulosae which differs from cysticercus bovis in having invaginated scolex with four suckers and hooks (infective stage).
3. Man is infected by ingestion of undercooked pork



PATHOGENESIS AND CLINICAL PICTURE:

- 1- Intestinal disturbances, neurologic manifestations, intestinal obstruction or loss of weight.
- 2- When man ingests eggs of *T.solium*, the larval stage (*cysticercus cellulosae*) develops in the extraintestinal tissues leading to a condition known as cysticercosis. This condition is likely to occur with *T.solium* not with *T.saginata*, a fact which makes *T.solium* runs a more serious course.

DIAGNOSIS:

- 1- Clinical diagnosis.
- 2- Laboratory diagnosis:
 - a) Examination of stool to detect the gravid segments .The eggs are rarely present in stool and are similar to those of *T.saginata*.

TREATMENT:

1. Similar to that of *T.saginata*.
2. Yomesan in a single oral dose of 2gm on empty stomach. It is recommended to give a saline purge 1-2 hours later to avoid cysticercosis from eggs of disintegrated segments. Also Quinacrine hydrochloride (atebrine) can be added as it leads to expulsion of the parasite in an intact form.
3. Giving antiemetic before treatment to avoid vomiting.



Life cycle: (Taenia saginata and solium)

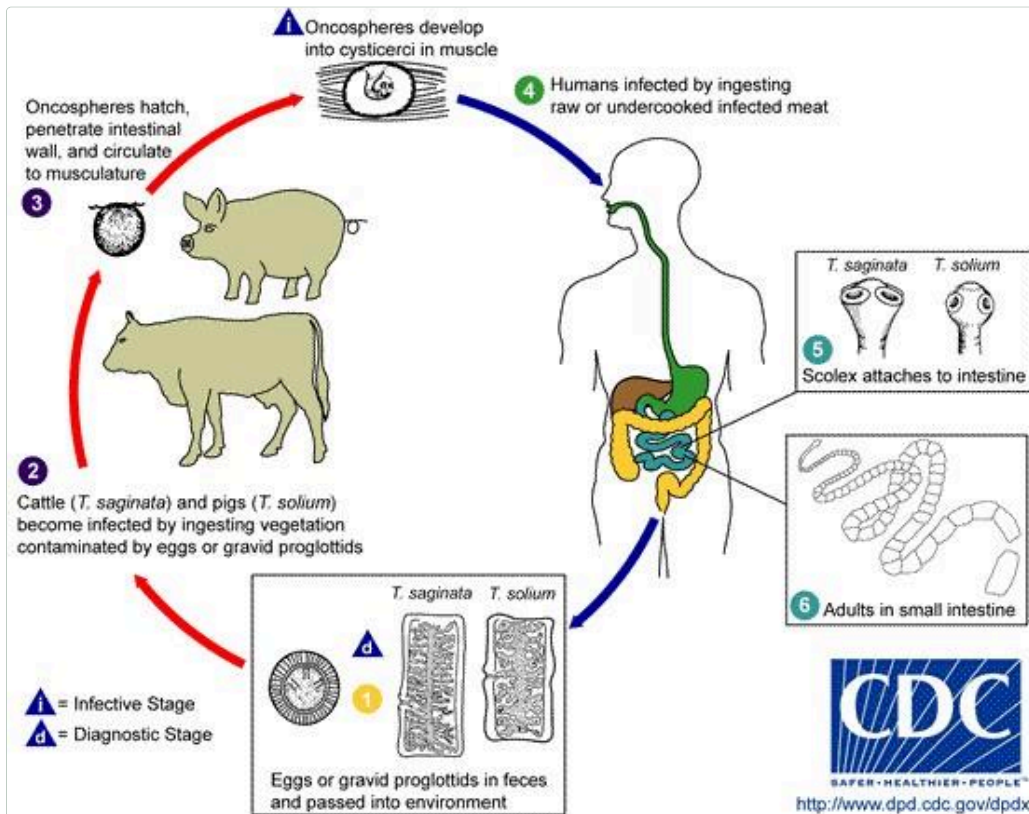


Figure 9.2 — Taenia saginata and solium life cycle

intestinal Nematodes

Ascaris lumbricoides

(Giant intestinal round worm)

GEOGRAPHICAL DISTRIBUTION:

Cosmopolitan. More prevalent in warm countries with bad sanitation.
 Common in Egypt (especially children).

- Host: Man.
- Habitat: Small intestine (free in the lumen).

Life cycle:

(Geohelminthic infection)

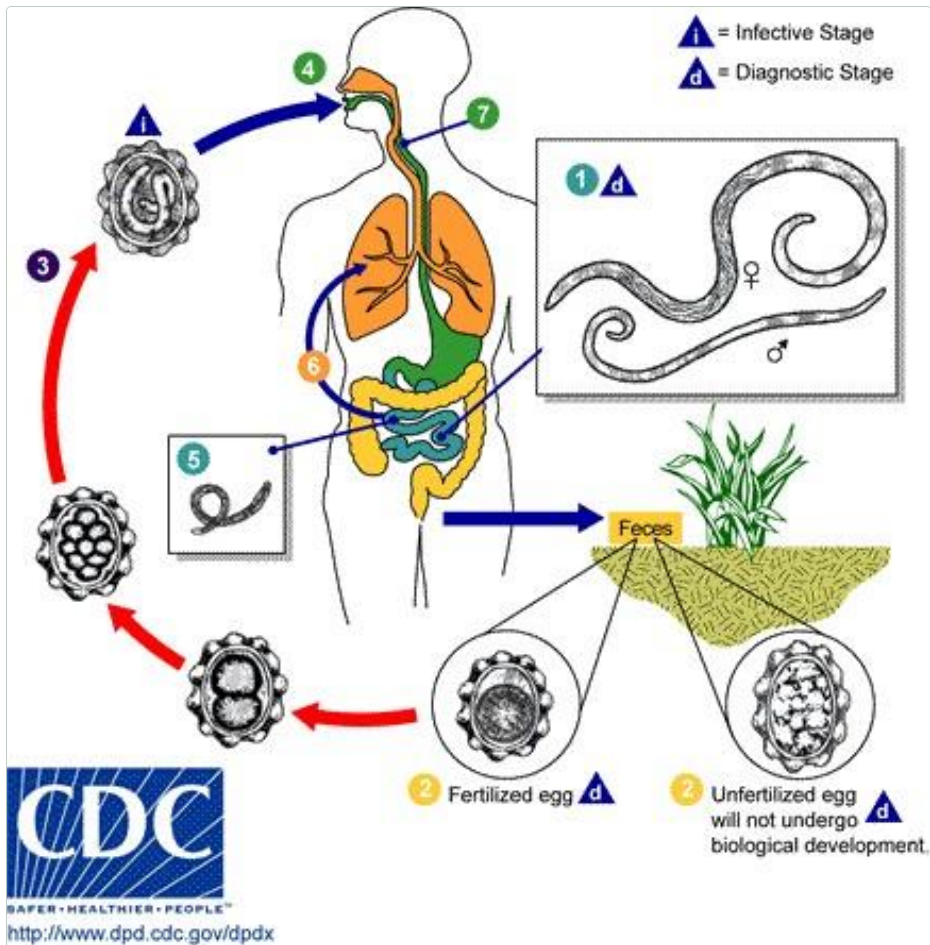


Figure 9.3 — Ascaris life cycle



Figure 9.4 — Ascaris: male and female worms



Figure 9.5 — Ascaris egg

Ascaris egg infection of man takes place by swallowing water or raw vegetables polluted with embryonated eggs containing the infective larva or through contaminated hands by polluted soil.

PATHOGENESIS AND CLINICAL PICTURE:

Tissue damage may be due to:

1. Large size of Ascaris (largest intestinal nematode).
2. Adults do not attach to intestinal wall, thus they may go to ectopic places.
3. Toxic products stimulate immune response.

CLINICAL PICTURE:

The usual infection (10 - 20 worms) often passes unnoticed. In heavier infection the following occurs:

1. During larval migration:

There is intense local inflammatory reaction around the larvae leading to pneumonitis manifested by fever, cough, dyspnea and eosinophilia.

■ In the intestine:

Abdominal pain, nausea, vomiting, colic, distention or dyspepsia due to production of - anti. enzymes that interfere with protein digestion leading to malnutrition and underdevelopment of children.

- Changes of the bowel movements (diarrhoea or constipation).

2. Complications:

Traumatic effects:

due to irritation of adult worms by fever, drugs or anaesthesia. -



Intestinal obstruction, volvulus and intussusception occur by a mass of worms.

Worms are not fixed to intestine so may be sloughed to:

- Bile ducts causing obstructive jaundice.
- Liver (carrying intestinal bacteria) resulting in liver abscesses.
- Appendix leading appendicitis.
- Ampulla of vater leading acute haemorrhagic pancreatitis.
- Peritoneum leading to peritonitis.
- Stomach leading to vomiting or escape through external nares.
- Very rarely to the trachea leading to suffocation.
- **Toxic effects:**

The by-products of living and dead worms may produce toxic symptoms in sensitized persons such as oedema of the face, urticaria, asthma, insomnia, nervous irritability and convulsions.

DIAGNOSIS:

1- Clinical diagnosis: non specific.

2- Parasitological diagnosis:

Detection of eggs in the stool.

Detection of larvae in the sputum and Charcot Leyden crystals early in the disease.

3- other tests:

- High eosinophilia.
- X-ray on abdomen after barium meal reveals worms as filling defects

TREATMENT:

Mebendazole or flubendazole: 100 mg twice per day for 3 days.

PREVENTION AND CONTROL:

- 1- Purification and chlorination of water.
- 2- Proper washing of raw vegetables and washing hands before meals.
- 3- Proper disposal of sewage (human faeces).
- 4- Mass treatment, personal hygiene and education of people.





Enterobius vermicularis

(Pin worm, Seat worm)

Geographical distribution:

Cosmopolitan, more in temperate and cold climates.

Final host:

Man especially children.

Habitat:

Large intestine mainly caecum, colon, rectum and appendix.

Mode of infection:

- 1- Ingestion of infective eggs through:
 - Autoinfection (contaminated hand to mouth).
 - Contaminated food or drinks.
- 2- Air-borne infection through inhalation of eggs in dust particles.

Pathogenesis and symptoms:

- 1) Irritation, itching and scratching of the peri-anal skin (pruritis ani), lead to insomnia, nervous manifestations and secondary bacterial infection.
- 2) Worms in young girls may deposit eggs on vulva producing pruritis vulvae.
- 3) Worms may migrate to the vagina (vaginitis), uterus, urinary bladder (urethritis) leading to irritation and inflammation.
- 4) Appendicitis is due to migrating adults.
- 5) Due to the easy spread of infection from one member of the family to another, all the family should be examined and all infected ones should be treated at the same time to get rid of the infection among the whole family.
- 6) Sometimes, the vagina and bladder are invaded in female patients causing irritation.
- 7) Migration to the peritoneal cavity (through uterine tubes) will cause pelvic peritonitis.

Diagnosis:

- 1- Clinically: Infection is suspected in children with Pruritus ani at night.

2-Laboratory:

- Adult worms may be seen in stool or anal area.



- Eggs are rarely found in stool (about 5 % only).

Swabbing of anal or perianal area by:

- a) N.I.H. swab (National Institute of Health) **b)** Scotch adhesive tape swab:

Treatment:

1. Albendazole, Mebendazole
2. Local application of white oxide of mercury (white precipitate ointment) around the anus at night (relieves itching, kills female worms coming out and prevents dispersal of eggs).

Prevention and control:

- Personal hygiene.
- Infected persons should sleep alone, their underwear and bedding carefully handled & washed.
- Food and drink should be protected from dust & hand.



Life cycle:

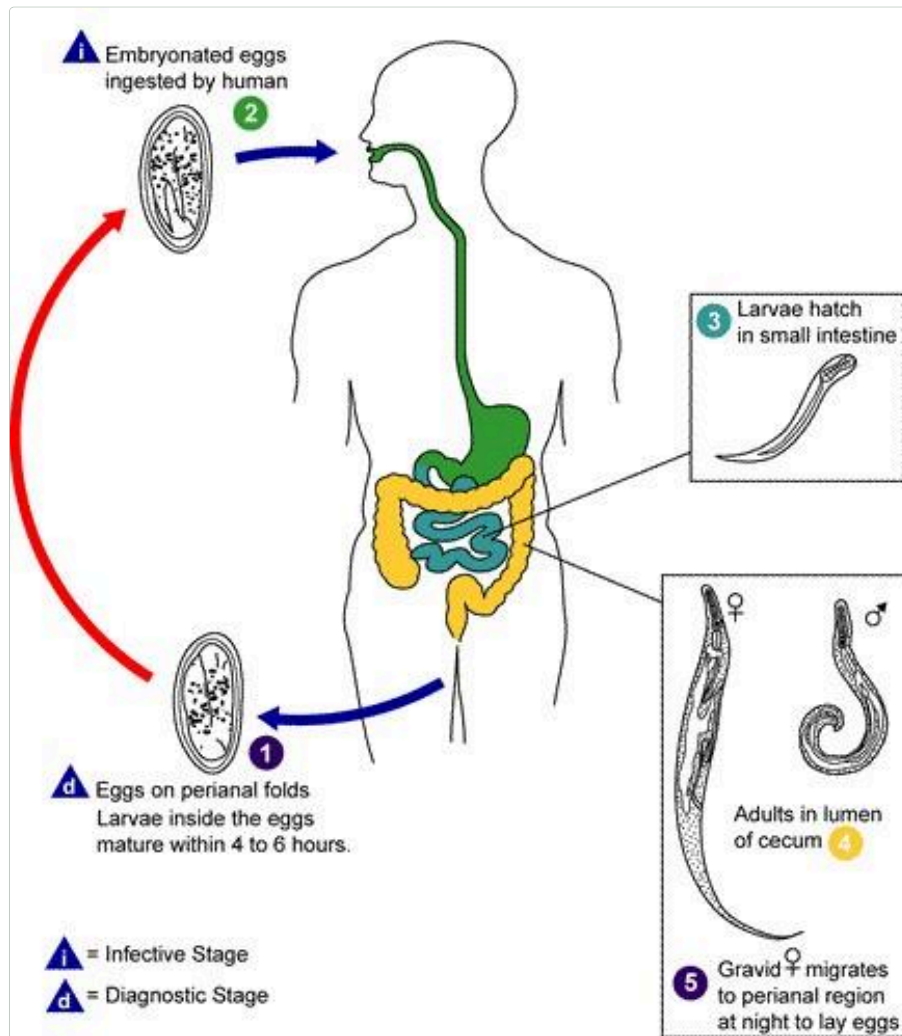


Figure 9.6 — Enterobius vermicularis life cycle

EXTRA-INTESTINAL AMOEBIASIS

A. HEPATIC AMOEBIASIS

PATHOGENESIS

Hepatic amoebiasis may occur due to either:

a. **Direct spread** of amoebic trophozoites from a primary intestinal lesion to surrounding tissues.

Or by

b. **Haematogenous** spread through invasion of blood, Portal circulation draining the intestine carries amoebic emboli mainly to the right lobe of the liver.



Hepatic amoebiasis:

liver affection may be in the form of diffuse amoebic hepatitis or liver abscess.

a. Diffuse Amoebic Hepatitis:

In considerable number of patients with amoebic colitis there is an associated enlargement of the liver. This condition starts with constant invasion of the liver by amoebae from an active intestinal process via the portal circulation.

b. Amoebic liver Abscess:

In foci where amoebic colonization continues, there is lytic necrosis of hepatic parenchyma. The lesions coalesce forming typically a solitary lesion. It begins as a small area of necrosed hepatic tissue, which increases in size with liquefaction in the centre and thickening of the wall. Its contents are lysed liver cells, bile, fat, RBCs and collagen strands; all forming a viscid chocolate red or creamy mass (anchovy-sauce pus). Amoebic trophozoites are only found in the edge of the lesion where they multiply. The commonest site is the posterior part of the right lobe (silent area) causing upward bulging of the diaphragm. The hepatic abscess may become chronic, absorbed, fibrosed, calcified or may extend to other organs.

Clinical picture:

a. Symptoms:

There is usually an acute onset of pain in the upper abdomen with high fever. The pain is intense and constant, radiating to the scapular region and right shoulder; it increases with deep breathing or coughing.

Fever is varying from 38.0 and 40.0 C. There is anorexia, nausea, vomiting and fatigue.

b. Signs:

1. Enlarged tender liver
2. Pallor
3. Earthy look
4. Fever
5. Tinge of jaundice

Differential diagnosis

Other causes of hepatomegaly and cystic lesions in the liver (hydatid disease)



Diagnosis:

a. Clinical diagnosis:

clinical picture

b. Laboratory diagnosis:

1. Indirect by serological tests: to detect specific antibodies. The indirect haemagglutination test (IHAT) and ELISA technique and both sensitive and specific but cannot be used to assess cure because antibodies persist for a long time.

2. Direct diagnosis by:

- i. Aspiration of the abscess: specimen is taken from the border of the abscess for detection of trophozoite.
- ii. Stool examination: may be positive in only one third of cases.
- iii. Detection of copro-antigen (in stool) by enzyme immunoassay.

c. Radiology:

- i. X-ray shows high density in hepatic shadow and abnormal elevation of the right dome of diaphragm
- ii. Ultrasonography, computed tomography (CT) and magnetic resonance may visualize the amoebic liver abscess.
- d. **Complete blood count (CBC)** shows polymorphonuclear leucocytosis in Amoebic Liver Abscess (differentiated from viral hepatitis).

Prevention and control:

1. personal hygiene
2. Washing of fresh-eaten vegetables and fruits.
3. Safe water supply, water from polluted source should be boiled.
4. Insect control or protection of food from insects
5. Treatment of patients and examination of contacts

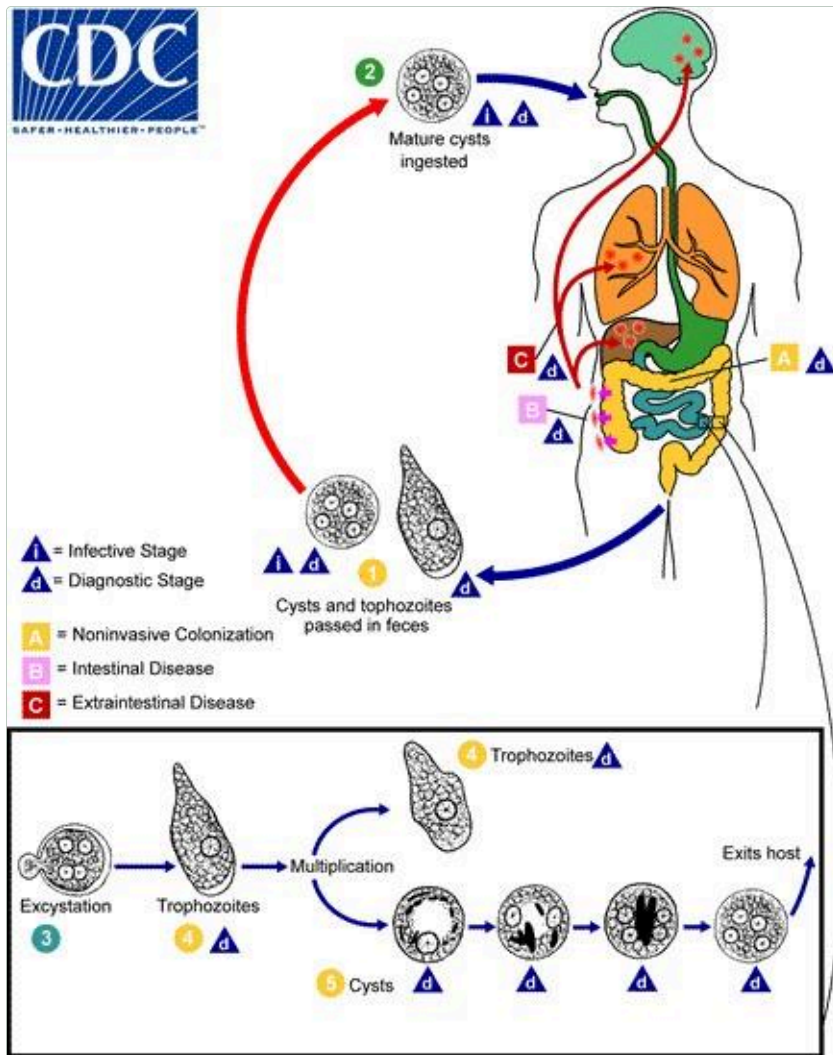


Figure 9.7 — *Entamoeba histolytica* life cycle



Surgery of the Small Intestine and Colon



10



Diverticular diseases

- 1- Meckel's diverticulum
- 2- Diverticulosis of the colon

Meckel's diverticulum

- It is the most common congenital abnormality of the small intestine.
- It is caused by an incomplete obliteration of the vitello-intestinal duct.

Patho - physiology:

1- In embryonic life, the fetal mid-gut receives its nutrition from the placenta via the vitello- intestinal duct.

2- The duct then undergoes progressive obliteration and usually disappears by 7 weeks' gestation.

3- When the duct fails to fully obliterate, different types of vitelline duct anomalies appear.

- (a) Umbilical fistula: a patent whole vitelline duct
- (b) A fibrous band that connects the ileum to the inner surface of the umbilicus.
- (c) Enterocoele or cyst in the middle part
- (d) Umbilical polyp patent distal part.
- (e) **Meckel diverticulum**, most common (97 %):
 - Patent proximal part of the duct.
 - A blind-ending true diverticulum that contains all layers of the ileum.
 - Meckel diverticulum almost always occurs on the anti-mesenteric border of the ileum.
 - **Rule of 2 :**
 - It is usually 2 feet (40-60 cm) proximal to the ileocecal valve,
 - 2 cm wide
 - 2 inches long
 - found in 2 % of the population,
 - often presents before age 2 years
 - May contain 2 ectopic mucosa (gastric – pancreatic).
 - Symptomatic in 2 %
 - 2 times common in males than females



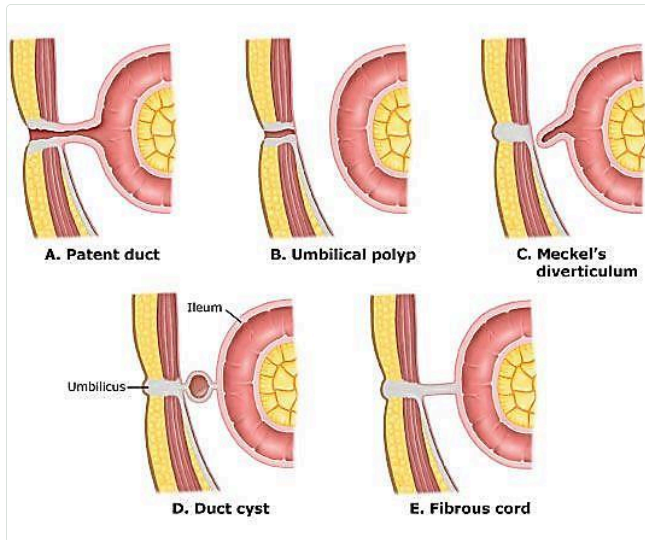


Figure 10.1 — Anomalies of vitello-intestinal duct

Clinical presentation and complications:

I- Asymptomatic in 98 %

II -

Lower GI bleeding:

- 25-50% (of symptomatic cases)
- Due to ileal ulceration by heterotopic **gastric mucosa**:
- presented with painless bleeding per rectum mostly in children below 5 y III - **Local inflammation = Meckel's diverticulitis** : 10 - 20%
- Clinically resembling appendicitis.
- Usually in adult patients
- Usually due to ectopic acid producing **gastric mucosa**.

IV - Intestinal obstruction :

- Due to internal segmental volvulus or intussusception or by congenital band.
- Picture of acute intestinal obstruction (Colics – distension –vomiting – absolute constipation)

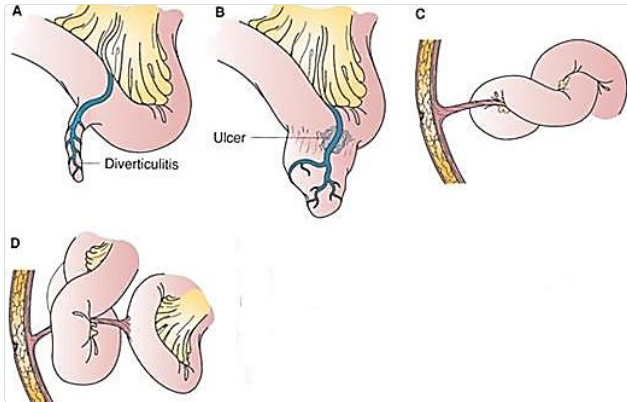


Figure 10.2 — Complications of Meckel's diverticulum

Investigations:

- 1- Difficult diagnosis preoperatively
- 2- “**Meckel's scan**” by ^{99m}Tc :
 - Most accurate test, especially in children.
 - Taken up by gastric mucosa.
 - (● Accuracy 90% in pediatric patients.
- 3- Abdominal CT scan is often helpful in cases of obstruction
- 4- Barium meal and follow through.

Treatment

- ● **If symptomatic**, surgical intervention to resect the diverticulum or segment of ileum containing the diverticulum.
- ● If bleeding, the source of bleed is often in the segment of ileum adjacent to the diverticulum.
- ● **If asymptomatic** and found incidentally at surgery:
 - In children, resection is recommended.
 - In asymptomatic adults, resection is controversial since only about 2% of these patients will become symptomatic and there is about a 2% incidence of complications (stenosis, adhesions) after prophylactic resection.



Figure 10.3 — Meckel's diverticulum



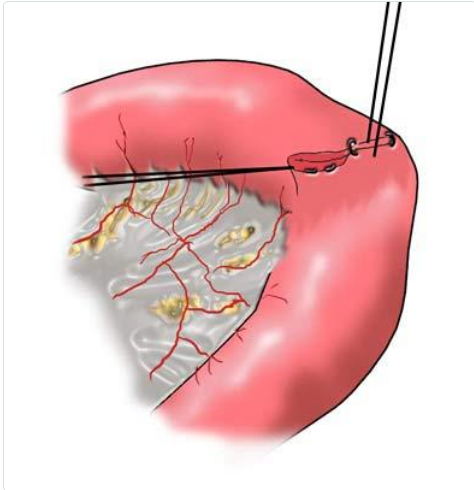


Figure 10.4 — Diverticulum resection

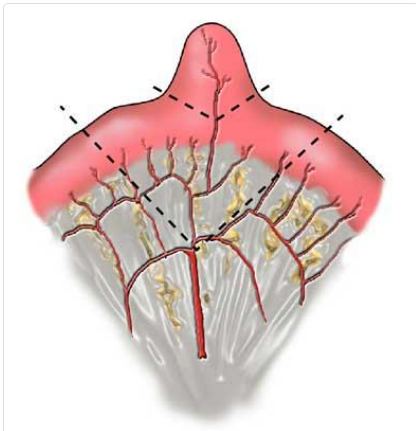


Figure 10.5 — Resection of the diverticulum

.....

Diverticulosis of the colon



11

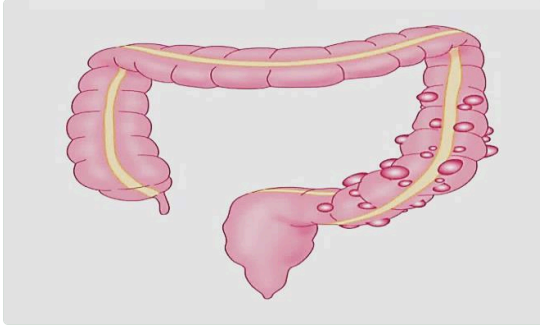


Figure 11.1 — Diverticulosis of the colon

Definition:

A diverticulum is a sac like protrusion in the colonic wall that develops as Result of herniation of the mucosa and submucosa through points of weakness in the muscular wall of the colon.

Predisposing factors: (increased intra-luminal pressure)

- Low-fiber diet and high carbohydrates
- Obesity
- Decreased physical activity
- Cigarette smoking
- Disordered motility of the colon
- Chronic constipation
- Increasing age

How it is formed ? Increased intra-luminal pressure as in chronic constipation → protrusion of the mucosa through the points of entry of the arteries in the muscle wall (weak points).

Which parts of the colon affected?

- Any part could be affected except the rectum
- The rectum has complete muscle coat (no taenia coli) prevent protrusion of the mucosa

Clinical presentation and complications

- I- Aymptomatic (non-complicated)
- II- Acute diverticulitis
- III- Fistula (colo-vesical or colo-vaginal)
- IV- Colonic bleeding
- V- Obstruction



A- Acute diverticulitis

■ Stages:

- 1- Acute diverticulitis → just inflammation of the diverticulum
- 2- Perforation of the diverticulum → peri-colic collection of pus and fecal matter
- 3- peri-colic abscess formation
- 4- Generalized peritonitis

General Symptoms and signs:

- Fever (high with perforation and abscess)
 - Toxemia in case of peritonitis.

Local symptoms:

1- Pain

- Mainly in left side of the abdomen (more in left iliac fossa).
- Dull aching and fixed
- 2- Anorexia and vomiting
- 3- History of chronic constipation of long duration

Local signs:

- 1- Abdominal distension
- 2- Tenderness in left side of the abdomen and left iliac fossa (diverticulitis)
- 3- Tenderness, rebound tenderness and guarding rigidity in case of perforation and abscess formation
- 4- Picture of generalized peritonitis may occur in neglected cases.
(same as in 3 but generalized).

B- Picture of fistula

- Colo-vesical fistula presented with pneumaturia, and recurrent urinary infections.
- Colo-vaginal fistula presented with recurrent vaginitis.

C- Bleeding per-rectum

D.D : common causes of bleeding per/rectum

- 1- Tumours of the colon
- 2- Angiodysplasia
- 3- Inflammatory bowel diseases
- 4- Fissures and piles
- 5- Dysentery





Investigations of diverticulosis:

- Lab.** 1- CBC → Leucocytosis
2- Liver, renal functions

Radiology:

- 1- Plain X-ray abdomen erect → air under diaphragm in perforation
- 2- Abdominal ultra-sound to detect fluid collections in perforation
- 3- C.T. Scan → detect the diverticulae, collection and peri-colic abcess
- 4- Barium enema is done in non-complicated diverticulosis → Saw-tooth appearance

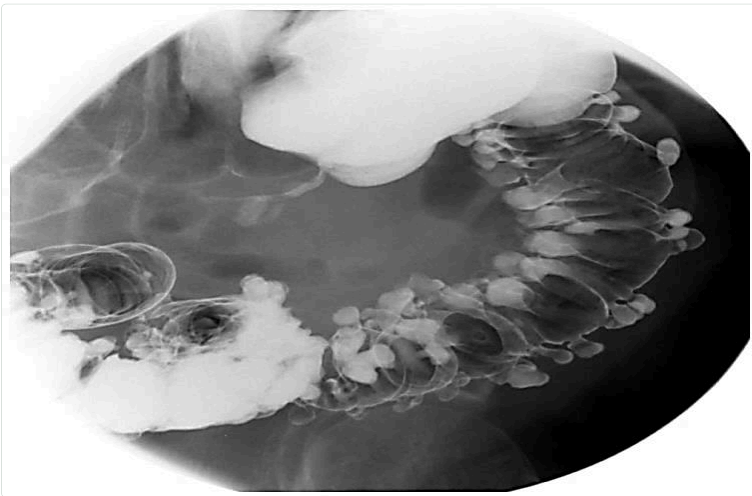


Figure 11.2 — Barium enema: multiple diverticulae

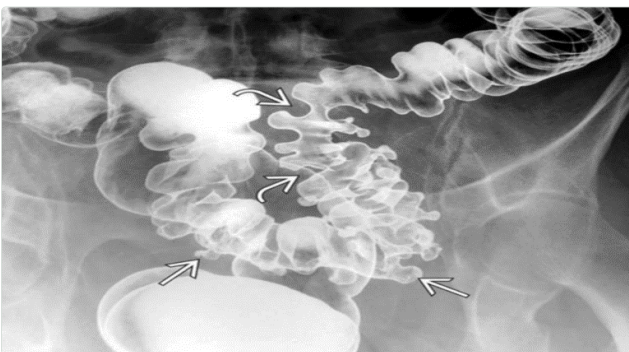


Figure 11.3 — Barium enema: multiple diverticulae, saw tooth

Treatment:

I- Conservative treatment: in NON-Complicated cases

- 1- Treatment of chronic constipation
- 2- High fiber diet
- 3- Antibiotics in acute diverticulitis without peritonitis or abcess will succeed in more than 85 % of cases



I-Per-cutaneous drainage:

ultrasound or CT guided in localized small abscess with drainage catheter left inside.

III- Surgical Treatment

■ Indications:

- 1- Diffuse peritonitis (signs & symptoms).
- 2- Failure of conservative treatment
- 3- Failure of percutaneous drainage of peri-colic abscess.

■ Types of surgery:

1- Hartmann" procedure:

1 st stage:

resection without anastomosis + end colostomy + closure of rectal stump •

2 nd stage:

colorectal anastomosis after 6 - 8 weeks

2-1 st stage:

resection + colo-rectal anastomosis + temporary ileostomy

2 nd stage:

closure of ileostomy.

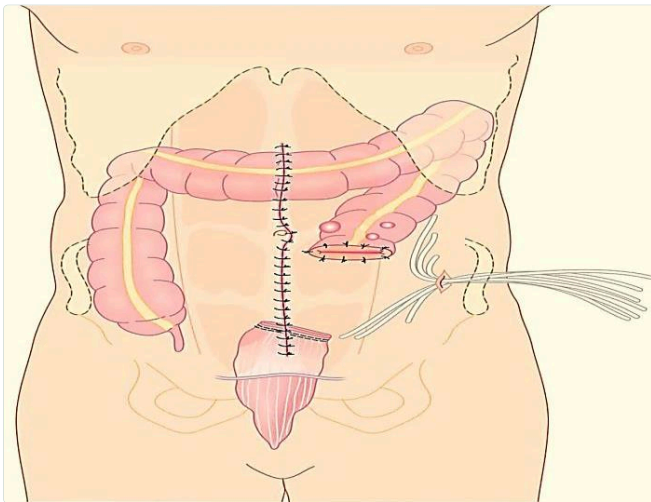


Figure 11.4 — Hartmann procedure

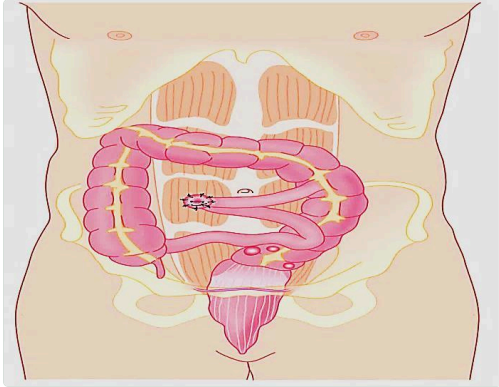


Figure 11.5 — Hartmann procedure (resection + temporary colostomy) Resection + colo-rectal anastomosis + diverting ileostomy



Appendix



12



- The human appendix ranges from **2 to 20 cm**.
- The diameter of the appendix is usually between **7 and 8 mm**.
- The appendix is connected to the mesentery in the lower region of the ileum, by a meso-appendix containing appendicular artery and vein

Parts of appendix:

The appendix has a base, body and tip. The mesentery attached to the appendix is known as mesoappendix.

■ Where the base of appendix is located?

- The base of the appendix is attached to the posteromedial wall of the cecum 2 cm below the ileocecal junction.
- If the taenia coli in the cecum are traced downwards all the three taenia coli converges to the base of the appendix and continue as the longitudinal muscle coat of the appendix.
- On the skin, the base of the appendix lies at Mac-Burney's point which is located at the right spino-umbilical line (Line joining between the anterior superior iliac spine and the umbilicus) at the junction of medial two-thirds and lateral one-third.

■ What is meso-appendix ?

- This is the mesentery of the appendix attached to the mesenteric border of the appendix. • The mesoappendix contains the appendicular vessels.
- The appendicular vessels stop just before the tip of the appendix.
- The tip of the appendix is the least vascular area, and in obstructive type of appendicitis the commonest site of gangrene is the tip of the appendix.
- The appendix is supplied by appendicular artery which is a branch of inferior division of ileo-colic artery. The appendicular artery is an end artery.
- Sometimes an accessory appendicular artery may arise from the posterior cecal artery and may supply the appendix in addition to appendicular artery.
- The vein drains into the superior mesenteric vein (portal).

■ Why appendix is known as abdominal tonsil?

The submucous coat of appendix contains lots of lymphoid follicles. The presence of lymphoid follicles is one important etiological factor for development of appendicitis.

■ What are the different locations of appendix?

The base of the appendix is usually located at the Mac-Burney's point. The body and the tip of the appendix may lie at different locations and hence named according to the location.

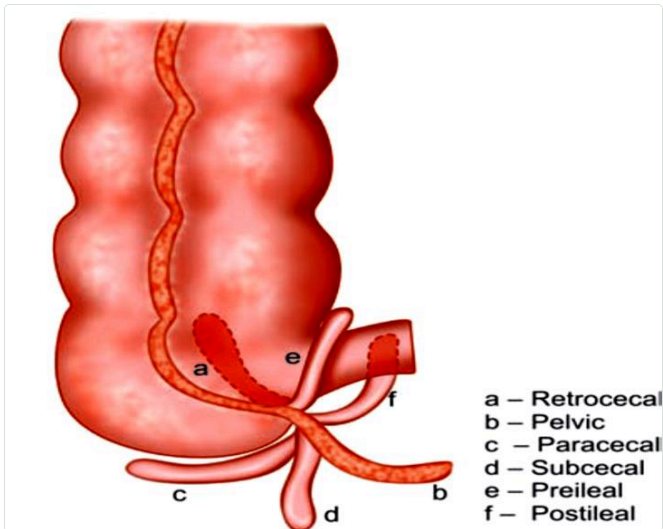


Figure 12.1 — Positions of the appendix

Retrocecal (60–70%):

commonest location, lies behind the cecum.

Pelvic (20–30%):

Second commonest location lies towards the pelvis.

Paracecal (1–2%):

Lies along the side of the caecum.

Subcecal (1 %):

Lies below the cecum.

Ileal (1–2%):

Lies towards the terminal ileum. May be:

- **Pre-ileal:** Runs in front of the terminal ileum
- **Post-ileal:** Runs behind the terminal ileum.

Ectopic appendix:

Due to mal-rotation of the gut, the appendix along with cecum may be located in **left iliac fossa** or in the **right sub-hepatic region**.

Acute appendicitis

Acute inflammation of the appendix



Pathophysiology:

1- Appendicitis is caused by obstruction of the appendiceal lumen.

2- The most common causes of luminal obstruction include:

- lymphoid hyperplasia secondary infections (more common during childhood and in young adults), • Faecoliths, foreign bodies, worms, adhesions causing kink of the appendix and neoplasms as carcinoid tumor or cancer cecum.

3- Bacterial infection:

The commonest organism is Escherichia coli.

Other organisms: Include enterococci, bacteroides, anaerobic streptococci, clostridia.

4- Dietary factors:

Low residue, high protein diet is associated with increased incidence of acute appendicitis. This has been related to constipation because of such diet.

Pathological types:

A-Acute catarrhal appendicitis:

Obstruction causes an increase in pressure within the lumen due to continuous secretion of fluids and mucus from the mucosa.

B-Mucocele of appendix:

A condition where appendix is swollen with mucoid secretion due to occlusion of its lumen proximally.

C-Acute suppurative appendicitis:

If appendicular obstruction persists, intra-luminal pressure rises. As a consequence, appendicular wall ischemia begins, resulting in bacterial invasion of the appendicular wall.

D-Gangrene and perforation:

due to thrombosis of the appendicular artery and veins.

E- **Appendicular abscess and peritonitis** in neglected cases

N.B.

The mucocele may rupture and the epithelial cells may be implanted on the peritoneal surface with formation of large mucoid mass. This condition is known as **pseudo-myxoma peritonii**.

Risk factors for perforation of acute appendicitis

- Extremes of age.



- Fecolith obstruction.
- Diabetes mellitus.
- Administration of purgative.
- Immunosuppressed patients—chronic renal failure, steroid

Clinical picture:

General symptoms:

1- Fever → low grade unless complicated by suppuration and peritonitis. 2- Toxic look

Local symptoms:

1- Pain

- Often begins in the center of the abdomen, (peri-umbilical)
- This pain is typically a dull, and poorly localized,
- As the inflammation progresses, the pain begins to localize more clearly to the right iliac fossa, as the peritoneum becomes inflamed .
- Typically, point pain is not present until the parietal peritoneum is Inflamed. **2- Anorexia, nausea and vomiting.**

General signs:

- Fever – tachycardia
- Toxic look

Local signs:

1- Localized tenderness, rebound tenderness and guarding rigidity:

- In the right iliac fossa mainly at **Mc-Burney's point** (1/3 of the way along a line drawn from the anterior superior iliac spine to the umbilicus).

2- Rovsing's signs:

deep palpation in left iliac fossa elicit pain in right iliac fossa

3- Psoas sign:

pain in right iliac fossa on extension of the right hip due to nflammation of peritoneum over psoas muscle

4- Obturator sign:

internal rotation of the flexed hip causing pain in right iliac fossa

5- Silent abdomen due to ileus.





- **Psoas sign** – Hip extension
- **Obturator sign** – Rotation of right flexed hip



Figure 12.2 — Psoas and obturator signs

D.D. of pain in right iliac fossa

1. Acute appendicitis
2. Meckel's diverticulitis
3. Right-sided disturbed ectopic pregnancy
4. Acute salpingitis
5. Twisted right ovarian cyst
6. Acute intestinal obstruction
7. Ileo-cecal tuberculosis
8. Acute Crohn's ileitis
9. Perforated peptic ulcer
10. Ureteric colic
11. Acute pyelonephritis
12. Psoas abscess.
13. Cancer cecum

Complications of acute appendicitis:

- 1- Appendicular mass
- 2- Appendicular abscess
- 3- Perforated appendix and generalized peritonitis
- 4- Portal pyemia and liver abscess.
- 5- Septic shock

Management of non-complicated acute appendicitis:

Investigations helpful for diagnosis of acute appendicitis:

1. CBC: polymorphonuclear leukocytosis.

2. Urine analysis:

for suspected urinary causes as UTI → **PYURIA**



3. **C-reactive protein** may be elevated.
4. **Ultrasonography**—sensitivity of 86% for diagnosis of acute appendicitis and exclude other causes as gynecological causes.
5. **CT scan of abdomen**—sensitivity of 94% for diagnosis of acute appendicitis and exclude other causes.
6. In women **of child-bearing age**— **beta-HCG to exclude ectopic pregnancy**.

Treatment:

A. Initial treatment:

- Nothing by mouth.
- Intravenous fluid: Ringer's lactate and normal saline
- Broad spectrum antibiotics—2nd generation cephalosporin and metronidazole.

B. Emergency Appendicectomy:

- The operation should preferably done within 24 hours of admission.
 - The risk of perforation increases with time.
 - After **36 hours** the risk of perforation is up to **36%** and the risk increases by **5 %** for every 12 hours passed.
 - The procedure either done laparoscopically or by open technique.

Appendicular mass

It is an inflammatory mass composed of:

- 1- Edematous omentum,
- 2- ileal loops and cecum
- 3- Inflamed appendix.

Clinically

- If the patient presents with classic features of acute appendicitis and then presents with a lump in right iliac fossa, this is likely to be an appendicular mass.
- Detected as tender mass in the right iliac fossa with guarding rigidity.
- Toxic look, high fever,...

How is the appendicular mass formed?

- Appendicular lump is a conglomerate consisting of inflamed appendix, adjacent terminal ileum, cecum and the greater omentum.

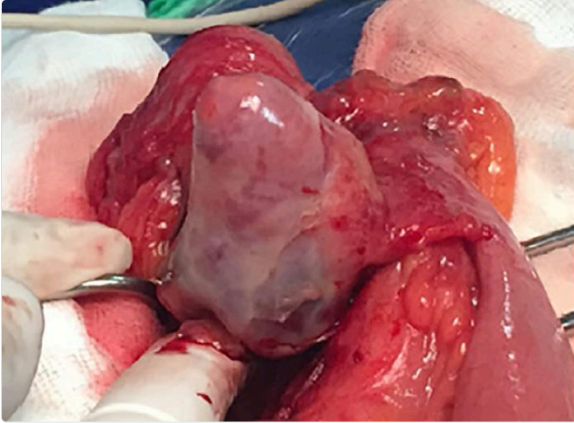


Figure 12.3 — Appendicular mass (operative finding)

How to investigate?

1- pelvi-abdominal ultrasound to exclude other causes (adnexal mass):

it reveals inflammatory mass in the right iliac fossa+ pelvic fluid collections

2- CT SCAN abdomen and pelvis with contrast:

more valuable in diagnosis and differentiate between appendicular mass and other pathologies

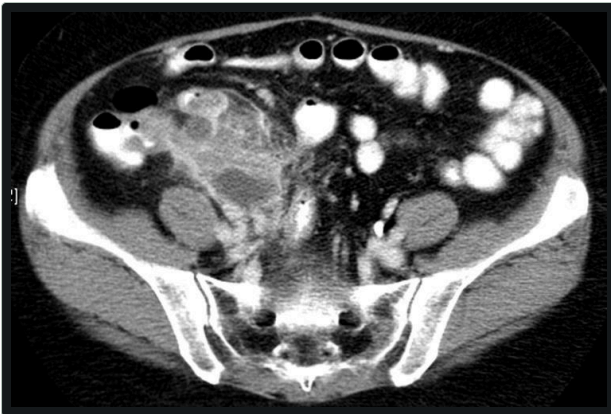


Figure 12.4 — Appendicular mass

How to differentiate an appendicular mass and an abscess?

- In a classical case the appendicular mass forms around the third day of acute attack of pain and develops into an abscess around fifth to tenth day.
- Pyrexia, aggravation of the local signs and a rising leukocyte count are indicators of abscess formation.
- Ultrasound and CT scan may demonstrate pus within an appendicular mass.

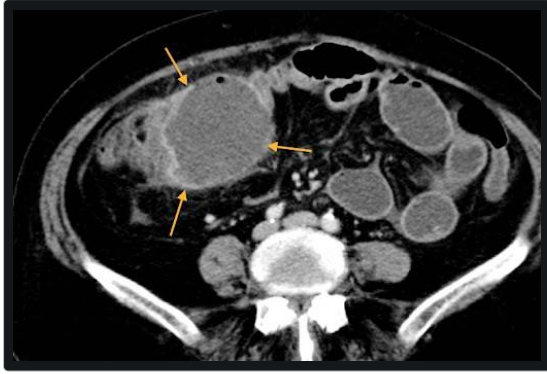


Figure 12.5 — Appendicular collection: CT view

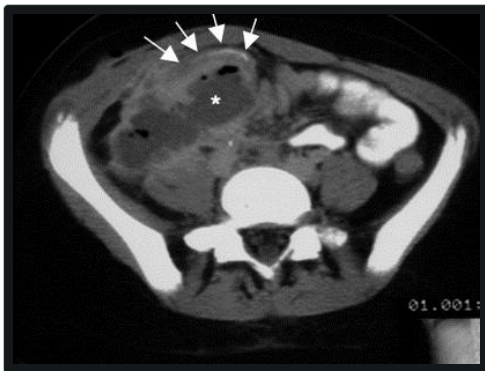


Figure 12.6 — CT appendicular abscess

How do you plan management of appendicular mass?

- If appendicular mass is palpable on admission: conservative treatment (**Ochsner-Sherren regimen**) is the preferred treatment
- Immediate appendicectomy is technically difficult.
- There is increased chance of fecal fistula postoperatively due to bowel injury
- **Operation should be done if conservative treatment fails with persistent picture.**
- Components of conservative treatment Ochsner's Sherren regime:

For pain:

Nonsteroidal anti-inflammatory drugs or narcotic analgesics as Pethidine. •

For vomiting:

Stop oral fluids, and insert nasogastric tube for suction.

Fluid administration:

Intravenous fluids to maintain fluid electrolyte balance.
(Ringer's and normal saline)



- **If patient is not vomiting**—oral fluids may be administered.

Control infection:

3rd generation cephalosporin along with metronidazole administered parenterally.

Monitoring is the most important component of conservative treatment:

Symptomatic improvement—pain, vomiting.

- Hourly monitoring of pulse, blood pressure and respiration.
- Monitoring of temperature 4 hourly.
- Tenderness in right iliac fossa
- Progress of the mass (is to be marked)
- If the patient responds well to conservative treatment: Lump may regress, pain decreases, pulse and temperature normalizes → Continue antibiotics for 7–10 days - Allow oral fluid and then food when patient has the desire to eat (4th– 5th day).
- Complete disappearance of the mass takes 6–8 weeks.
- **Interval appendicectomy** should be considered after 6–8 weeks.
- If with the above regimen; the mass increasing in size, pain and tenderness aggravating, rising pulse rate and fever with chills and rigor—this indicates **appendicular abscess** formation and early intervention for drainage is indicated.

Management of a case of appendicular abscess

- The patient may show features of septicemia.
 - Formation of pus is confirmed by USG or CT scan.
- I- Resuscitation followed by drainage of abscess is the treatment of choice.**

1. Intravenous fluids.
2. Nasogastric suction
3. Catheterization (for monitoring urine output)
4. Intravenous antibiotics—3RD generation cephalosporins+ Metronidazole.

II- Steps of drainage of appendicular abscess:

- a) General anesthesia with endotracheal intubation.
- b) A muscle cutting incision. Skin and external oblique incision like Mc-Burney's incision. But the internal oblique and transversus abdominis muscles are cut in the same line of incision for better drainage.
- c) The abscess wall is incised and abscess drained without opening the peritoneum.



d) A tube drain is inserted through separate wound and the main wound is closed in layers. e) if the appendix is accessible in the operative field it should be removed.

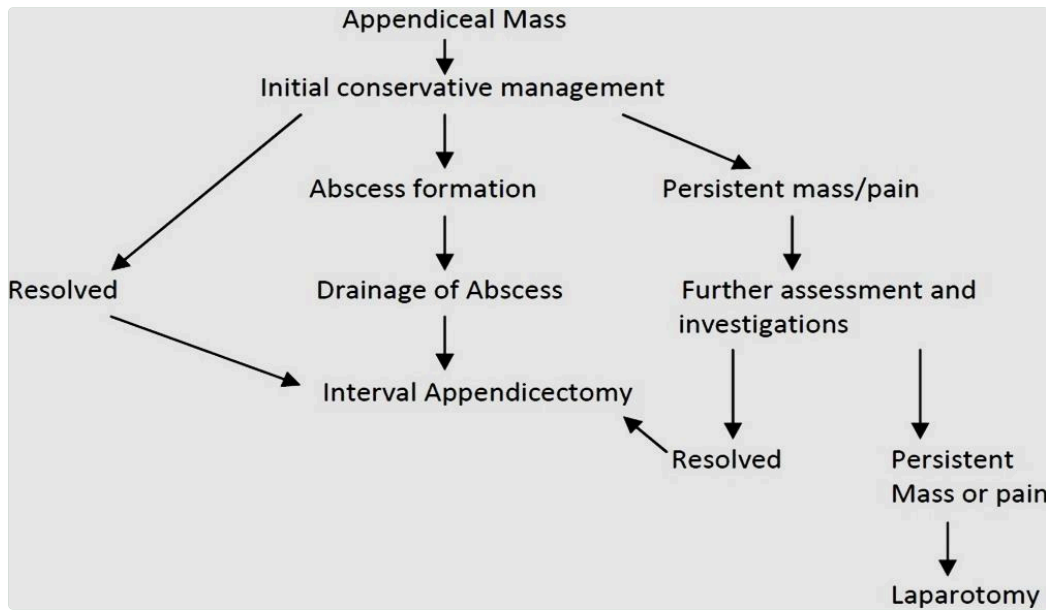


Figure 12.7 — Management of appendicular mass

Important causes of mass in right iliac fossa:

1. Parietal swelling:

- Lipoma
- Desmoid tumor

2. Intra-abdominal swelling:

- Appendicular lump.
- Appendicular abscess.
- Ileocecal tuberculosis.
- Crohn's ileitis.
- Carcinoma cecum or ascending colon
- Lymph node mass
- Adnexal mass

3. Retroperitoneal swelling:

- Hydronephrosis or tumor in an unascended or dropped kidney.
- Retroperitoneal sarcoma.
- Iliopsoas abscess.
- Tumor in an undescended right testis.

Intestinal obstruction



13



Definition:

Any condition interferes with normal forwards passage of intestinal contents.

Classification:

Cause of obstruction: mechanical or functional (Ileus).

Duration of obstruction: acute or chronic.

Degree of obstruction : partial or complete

Type of obstruction (complicated or not): (simple or strangulation) .

Mechanical obstruction

It means obstruction due to occlusion of the lumen of the bowel

According to the duration :

1- Acute (small intestine)

2- Chronic (colon)

Complicated or simple :

1- **Simple** → intact blood supply and wall of the intestine.

2- Closed loop as in volvulus = **complicated**

3- Strangulated (interfere with the blood supply) = **complicated**

4- Perforation = **complicated**

Important Facts:

- Adhesion is the most common cause of intestinal obstruction
- 80 % of bowel obstruction is due to small bowel obstruction ; 20% colonic obstruction • The most common causes of small bowel obstruction are **adhesions and hernias** and less common neoplasms
- Commonest causes of colon obstruction are **Colo-Rectal cancer** in 60-70% while 30% are **diverticular disease, volvulus and others.**
- Mortality rate ranges between 3 % for simple bowel obstruction to 30% when there is strangulation or perforation.

Etiology:

Mechanical bowel obstruction:

A. Small bowel obstruction:

1. Adhesion 60% (postoperative – congenital)
2. Hernia 20% (femoral, inguinal H.)
3. Neoplasm 5%



4. Volvulus 5%.
5. Others: intussusception, foreign bodies, gall stones

B. Large bowel obstruction :

- 1) Cancer colon 60 %.
- 2) Diverticular disease 15 %.
- 3) Volvulus (sigmoid or cecum) 15 %.
- 4) Others: hernia -fecal impaction - inflammatory strictures.

Functional bowel obstruction:

Adynamic ileus = Paralytic ileus

1. Postoperative mostly after abdominal surgery is the most common cause
2. Metabolic causes: DKA - hyponatremia - hypokalemia
3. Intra-abdominal inflammation—sepsis → peritonitis
4. Retroperitoneal hematoma
5. Mesenteric vascular occlusion
6. Fracture spine
7. Uremia.
8. Drugs: morphine - anticonvulsant.

Clinical picture of mechanical obstruction

Symptoms:

1. **Colicky abdominal pain** (due to contraction of the proximal intestinal segment against the obstruction)
 - Pain is central abdominal
 - Pain occurs in attacks few minutes in between
 - Pain gradually increasing in severity
2. **Vomiting**
 - Common in proximal intestinal obstruction and late in colonic obstruction
 - The colour of vomitus is whitish in gastric obstruction, greenish in upper intestinal obstruction and brownish or feculent in colonic obstruction.
3. **Distension** due to proximal loop dilatation
4. **Absolute constipation** in complete obstruction.
5. History of previous abdominal operation (**Adhesions**) in **adhesive I.O.** 6. Red currant jelly stool in **intussusception**
7. Symptoms of associated cancers (in colonic obstruction) in **obstructed carcinoma**.



8. History of tender swelling in anatomical site of hernia (Femoral- inguinal) in **obstructed hernia**.

General Signs:

1. Signs of dehydration in prolonged cases (dry tongue, sunken eyes)
2. Fever, tachycardia, toxic look in cases complicated with strangulation and peritonitis.

Local signs:

1- Inspection:

- Scars of previous operation
- Hernia swelling
- Abdominal distension

2- Palpation:

- Examine for hernias
- Tenderness, rebound tenderness and rigidity in cases of complicated peritonitis.
- Palpable mass in obstructed carcinoma

3- Percussion:

hyper-resonance

4- Auscultation:

- Exaggerated intestinal sounds (mechanical obstruction)
- Silent in associated peritonitis

5- P/R:

empty rectum

Investigations:

Laboratory:

- 1- Serum electrolytes (Na & K) decreased
- 2- Renal function: impaired in neglected cases (pre-renal failure).
- 3- CBC → Leucocytosis in strangulations or perforations and peritonitis

Radiology:

1- Plain X-Ray abdomen erect and supine

- Supine → dilated loops: detect the level of obstruction
- Concertina effect (jejunal)
- Featureless tubes (ileum)
- Haustrations (colon)



- Erect →
- Multiple air and fluid levels
- Air under diaphragm in perforated viscus (closed loop obstruction or strangulation)

2 - Computed tomography

- Used to detect obstruction, strangulation and fluid collection following perforation and peritonitis Accuracy 94% in diagnosis of obstruction

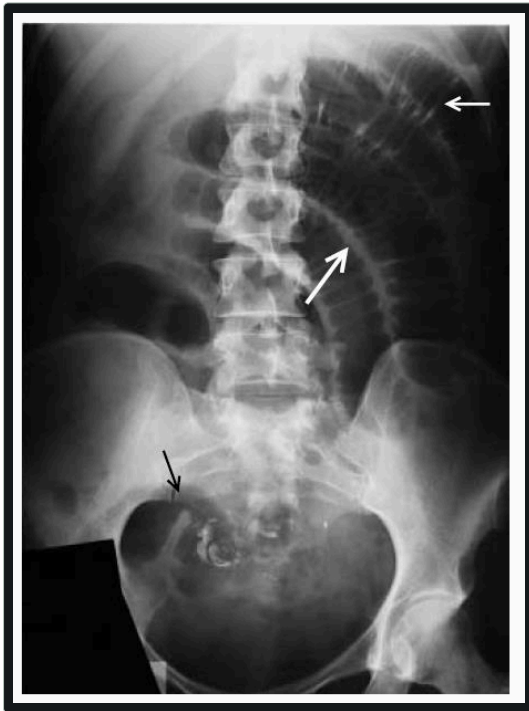


Figure 13.1 — Plain abdominal radiograph in obstruction



Figure 13.2 — Plain x-ray Supine and erect films

Treatment of mechanical intestinal obstruction (adhesive)

I- Conservative treatment

Indications:

- 1- All cases with adhesive intestinal obstruction
- 2- Preoperative preparation

Contraindications:

- 1- Evidence of peritonitis or strangulation
- 2- Obstructing carcinoma
- 3- Obstructed hernia

Lines of treatment:

I. Conservative treatment:

This treatment continues for 48 hour

- 1- **Intravenous fluids** administration to correct dehydration (Ringer solution) and protect the kidneys

2 - Naso-gastric tube:

- Suction of the fluids from the proximal segment



- Decreases the distension
- Decreases the colicky pain
- Gives an idea about the level of obstruction through the colour of aspirate

3- **Antibiotics** mainly against gram - ve and anaerobes (3rd Generation Cephalosporin and Metronidazole I.V.)

4- **Urinary catheter** to monitor urine output

■ **Indications for surgery in obstruction:**

1. Failure of conservative treatment 48 hours
2. Development of peritonitis
3. Proved or suspected strangulation
4. Obstructed hernia
5. Obstructed cancer
6. Gut perforation
7. Closed loop obstruction

■ **Types of surgical procedures:**

1- **In adhesive intestinal obstruction** → **Exploration and adhesiolysis**

- Exploration of the intestine from the duodeno-jejunal flexure till ileo-cecal junction
- The adhesive band is identified and divided.
- The band is identified at the junction of the dilated proximal segment and narrow collapsed distal segment.
- In cases of strangulation, or gut perforation → proceed for resection and anastomosis in case of small bowel, right and transverse colon
- In case of perforation of left colon → resection and temporary colostomy (1 st stage) and Closure of colostomy (2nd stage)

2-Obstruction with specific causes:

■ **Obstructed hernia** → **exploration of the hernia content**

- If viable return inside the abdomen and repair of the hernia
- If non-viable resection and anastomosis and repair of the hernia
 - **Obstructed carcinoma** of colon → look cancer colon
 - Intussusception
 - Volvulus of the sigmoid colon
 - Hirschsprung disease



Intussusception

- A condition in which part of the intestine slides into an adjacent part of the intestine • Intussusception is the most common cause of intestinal obstruction in children younger than 3 years.

Types:

- 1- ileo-cecal → the commonest
- 2- ileo-ileal
- 3- colo-colic

Etiology and Pathophysiology:

- The etiology is unknown in children, sometimes hypertrophied lymphoid tissue in the intestinal submucosa initiate the process
- In adults it might be due to tumor in the intestine

Parts of the Intussusception:

1. The entering = invaginating loop = Intussusceptum
 2. The outer = recipient loop = Intussusceptient
- The intussusceptum proceed along the lumen of the intussusceptient from proximal to distal.
 - The mesentery of the intussusceptum is invaginated with the intestine, leading to bowel obstruction.
 - Compression of the vessels in the mesentery leading to bowel ischemia
 - The intestinal mucosa sloughs and ulcerate due to ischemia, leading to bleeding and “currant jelly stool” (a mixture of sloughed mucosa, blood, and mucus).
 - Neglected cases leads to gangrene and perforation at the head of Intussusceptum

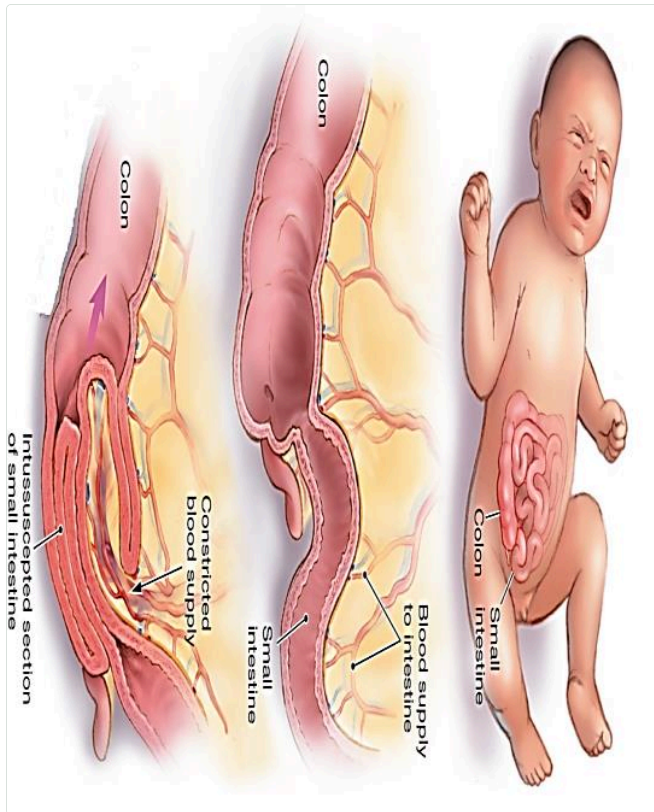


Figure 13.3 — Intussusception

Clinical picture:

Symptoms

- 1- The patient with intussusception is usually an infant,
- 2- Has an upper respiratory infection
- 3- Vomiting: Initially, vomiting is non-bilious, but when the intestinal obstruction occurs, vomiting becomes bilious
- 4- Abdominal pain: colicky, severe, and intermittent
- 5- Passage of blood and mucus: Parents notice the stools like currant jelly. (This is a mixture of mucus, sloughed mucosa, and blood.).
- 6- Palpable abdominal mass

Signs:

- 1- Signs of dehydration in neglected cases + fever in case of peritonitis
- 2- Abdominal distension
- 3- Right hypochondrium sausage-shaped mass and empty right iliac fossa





■ (Dance sign)

4- Tenderness, rebound tenderness and rigidity: If intestinal gangrene and peritonitis occur

— **Investigations:**

Lab. : CBC → Leucocytosis in strangulation or perforation

Rad.:

1- Plain x-ray abdomen erect and supine

Dilated proximal intestinal loops and air/fluid level

2 - Ultrasound:

a) Reveal intestinal obstruction caused by intussusception.

b) **Target sign** : Typically show a “bull’s-eye,” representing the intestine coiled within intestine.

3 - Barium enema Claw sign

→

.



Figure 13.4 — Claw sign

— **Treatment:**

A - Initial treatment: as in adhesive I.O.

1- Intravenous fluids to correct dehydration

2- Nasogastric tube for suction

3- Start antibiotic therapy

B - Hydrostatic reduction using barium enemas succeed in many cases

→

Indications: All early and uncomplicated cases (no signs of peritonitis and or strangulation)

■ Success known by passage of the contrast to the ileum



C - Surgery

- 1- When hydrostatic reduction fails
- 2- Presence of complications as peritonitis or gut perforation.

■ Procedures:

- 1- In non-complicated cases: manual reduction of the intussusception by compression not pulling to avoid tearing of the mesentery
 - 2- In case of perforation or gangrene → resection and anastomosis:
- In ileocecal intussusception : right hemicolectomy and ileo-transverse anastomosis

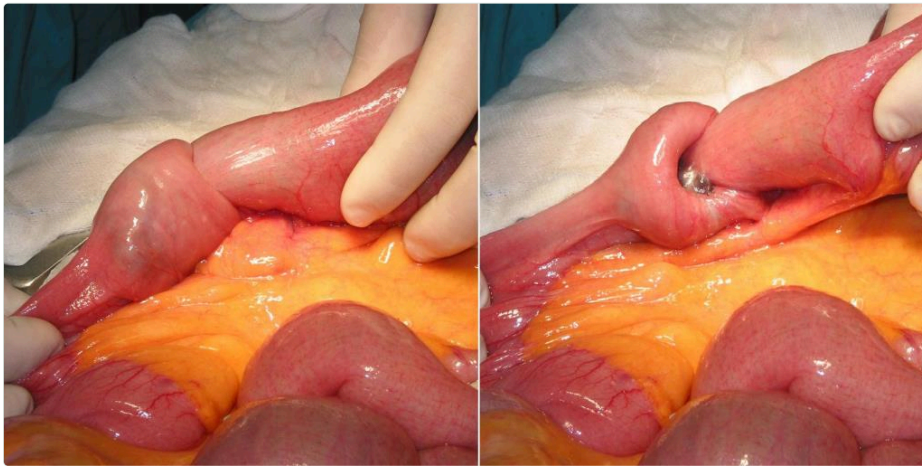


Figure 13.5 — Manual reduction of the intussusception

Colo nic Volvulus

- The term volvulus = twist
- A colonic volvulus occurs when a part of the colon twists on its mesentery, resulting in acute, sub-acute, or chronic colonic obstruction.

Types of colonic volvulus:

- 1- Sigmoid volvulus (most common = longer mesentery)
- 2- Caecal volvulus. (less common)

Etiology of Sigmoid volvulus:

1. The presence of a long mesentery with a narrow base of fixation to the retroperitoneum
2. Elongated, redundant colon predisposes to the formation of volvulus.



3. Chronic constipation → loaded sigmoid → elongation → twist along its mesentery
4. Old age → colonic hypotonia → chronic constipation → increases intra-luminal pressure.
5. Postoperative adhesions
6. Intestinal malrotations
7. Pelvic tumors
8. Mega-colon

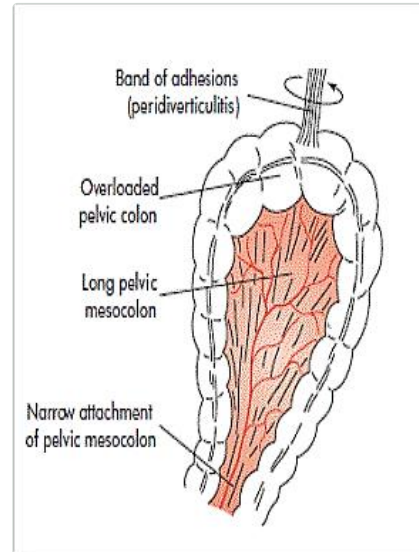
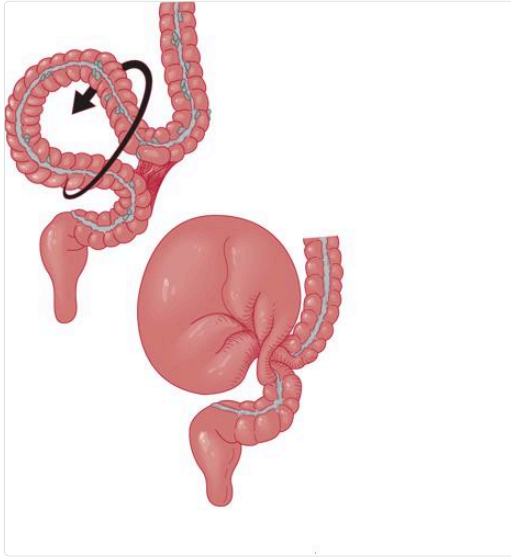


Figure 13.6 — Etiology of volvulus

Pathophysiology:

- The presence of long mesentery, narrow attachment and elongated loaded colon → anti-clockwise rotation of the sigmoid along its mesentery → occlusion of the lumen → obstruction
- Occlusion of the blood vessels → ischemia, gangrene and perforation

Clinical picture:

Symptoms :

- 1- History of chronic constipation
- 2- Colicky abdominal pains – constant severe abdominal pain indicates closed loop obstruction and impending perforation
- 3- Progressive abdominal distension
- 4- Progressive to absolute constipation
- 5- Late vomiting

**Signs:**

- 1- Fever, tachycardia and toxic look in cases with peritonitis
- 2- Massive abdominal distension
- 3- Tenderness, rigidity and rebound tenderness in associated perforation and peritonitis
- 4- Percussion → tympanitic resonance
- 5- P/R → empty rectum

Investigations :

A- Lab. → CBC for leukocytosis in peritonitis

B- Rad.

1- Plain X-Ray abdomen → huge distended sigmoid colon filling most of the abdomen

(omega sign = coffee bean sign)

2- Barium enema → “Birds beak”

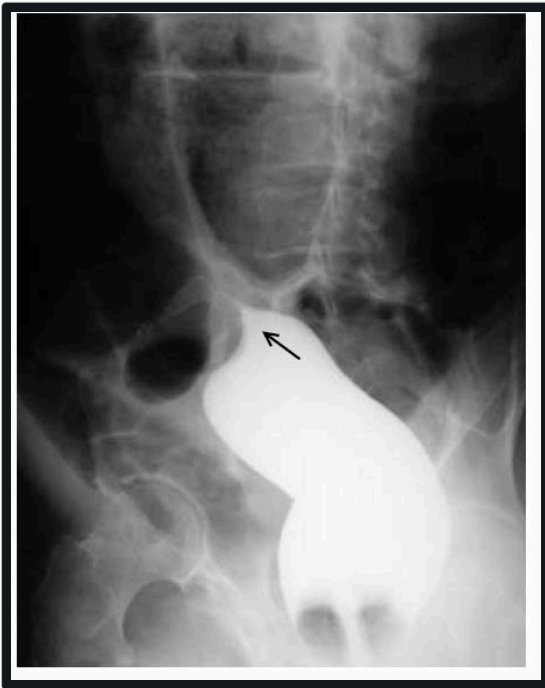


Figure 13.7 — Contrast study in sigmoid volvulus

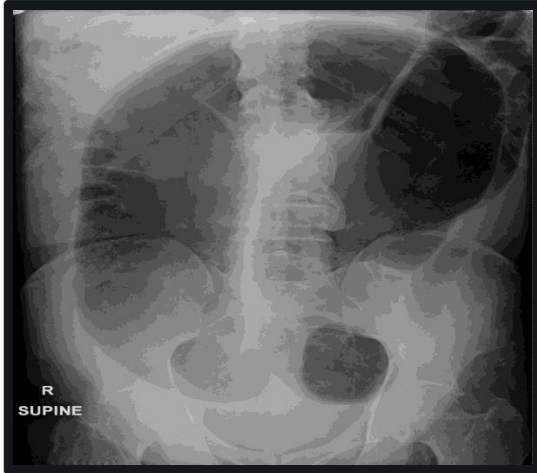


Figure 13.8 — Sigmoid volvulus (omega sign)

3- Abdominal ultra-sound → Abdominal collection in perforation

Treatment Sigmoid Volvulus

1 - Sigmoidoscopic de - torsion :

clock-wise and decompression and placing rectal tube succeeds in most cases

- First choice initial treatment in uncomplicated cases
- Contraindicated in suspected perforation

2- Surgery:

- ● **Elective:** Resection of redundant colon and re-anastomosis after colonic preparation
- ● **Urgent:** in case of gangrene or colon perforation: **Hartmann' procedure**

1 st stage:

Resection and colostomy

2 nd stage:

Closure of colostomy

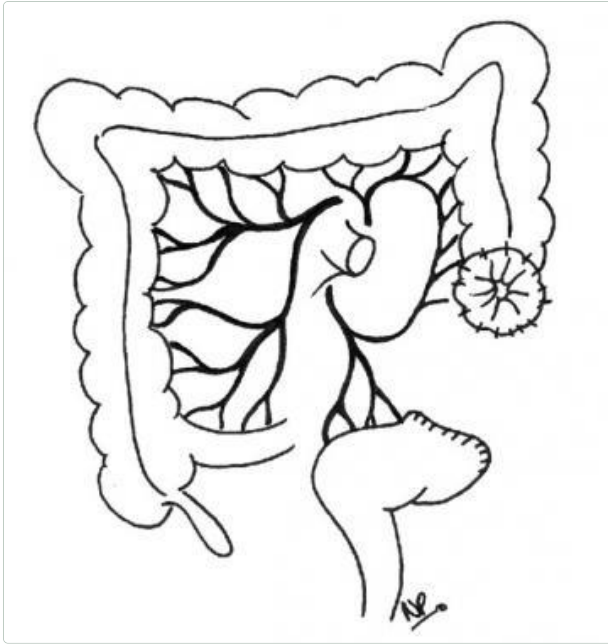


Figure 13.9 — Hartmann's procedure

Hirschsprung Disease = Congenital megacolon

Hirschsprung disease is a congenital disorder characterized by the absence of ganglia in the distal colon, resulting in a functional obstruction

Etiology :

genetic predisposition

Pathophysiology:

- The wall of the colon contains 2 enteric plexus
- Meisner plexus in the submucosa and Auerbach plexus in the muscle layer
- Both causing colonic wall relaxation
- The wall of the colon supplied by parasympathetic and sympathetic
- Parasympathetic is predominant and causes wall contraction
- In Hirschsprung Disease: the ganglion cells in the meisner and Auerbach's plexus are absent (aganglionic segment)
- So there is predominant action of the parasympathetic nerves leading to spasm in the segment of the colon affected → functional obstruction

■ Parts of the colon

1. Proximal dilated colon = normal



2. Transition zone = hypo-ganglionic

3. Distal zone = spastic aganglionic

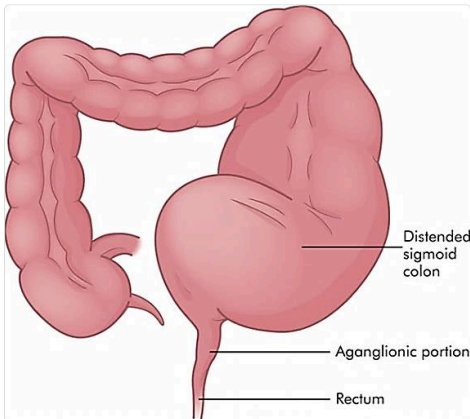


Figure 13.10 — Hirschsprung disease: bowel segments

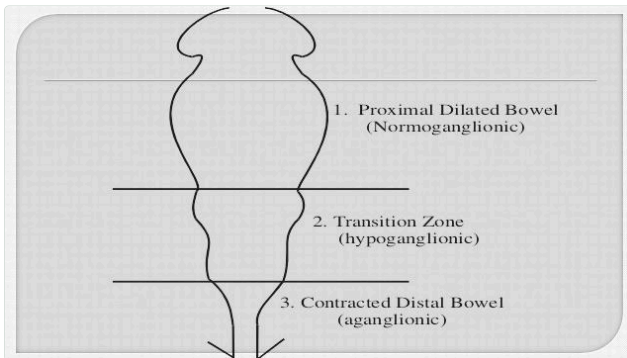


Figure 13.11 — Hirschsprung disease: transition zone

Sites:

H.D usually starts in the rectum → sigmoid → and proceeds proximally to involve variable parts of the colon

- Male is affected more than females (4: 1)

Clinical presentation:

Most cases of Hirschsprung disease are diagnosed in the **newborn** that **fails to pass meconium** within 24-48 hours of birth.

Abdominal distension

Vomiting

Poor nutrition and **failure to gain weight**

Older infants and children typically present with **chronic constipation** poorly responds to laxatives and distension

Hirschsprung enterocolitis : it is a severe form of entero-colitis:

1. Bloody diarrhea- foul smelling



2. Abdominal pain
3. Fever
4. Vomiting
5. Fatal if untreated

On examination:

1. Marked abdominal distension
2. Tympanitic resonance
3. Increased intestinal sounds
4. P/R: reveals an empty rectum and may result in gush of stool upon withdrawal of the finger.

Investigations

A- Lab. 1- CBC → anemia and infection

2- Serum electrolytes (Na and K)

B - Rad. : Plain X - Ray abdomen erect and supine **dilated left colon** &

→

- No rectal gas

1-Barium enema:

Shows **narrow distal segment, funnel-shaped dilatation** at level of transition zone with **marked dilatation of** the proximal colon.



Figure 13.12 — Barium enema Congenital mega-colon

3- Rectal biopsy:

Rectal biopsy is the **best diagnostic** test and demonstrates absence of ganglion cells (done through endoscopy),



4- Electro-manometry:

The classic finding is the absence of the recto-anal inhibitory reflex when the rectum is distended.. (Transient involuntary relaxation of the internal anal sphincter in response to distention of the rectum help to discriminate between the gas, fluid and stool at anal canal)

Management:

Management of HD differs according to the presentation:

1- If the patient presents with acute intestinal obstruction in the early life the management will be:

Resuscitation , IVF

Naso-Gastric Tube

Antibiotics

The initial surgery requires performing **a temporary colostomy**

2- Definitive procedures = pull through operation

- Resection of aganglionic segment and anastomosing the two normal ganglionic ends. They give excellent result in 90%.
- **Swenson.** The aganglionic segment is resected down to the sigmoid colon and rectum, and anastomosis is performed between the normal colon and the low rectum.

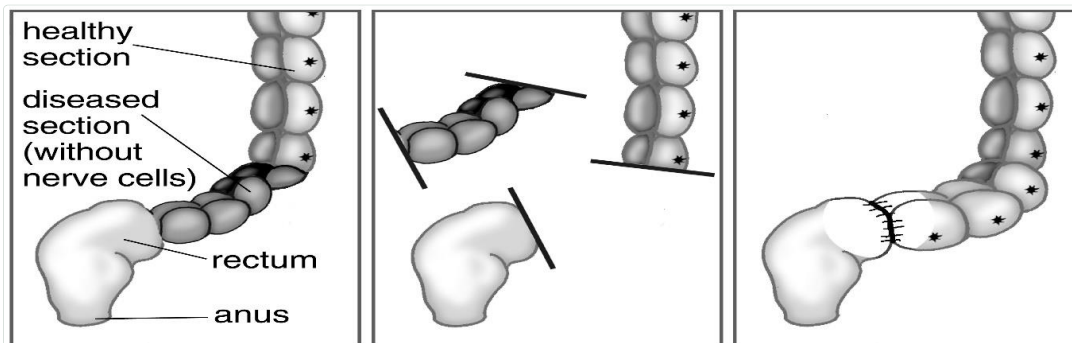


Figure 13.13 — Pull-through operation

...

Imperforate anus

Embryology of the rectum and anal canal:

1- The lowest part of the hindgut (cloaca) is divided by uro-rectal septum into:

A- Anorectal canal → posteriorly



B- Uro-genital sinus → anteriorly

2- This Ano-rectal canal will form the rectum and upper ½ of anal canal

3- The anus and lower ½ of anal canal develop by an external invagination from skin, known as the proctodeum, which deepens toward the upper part of anal canal to join it but remains separated from it by the anal membrane. This membrane should disintegrate at 8 weeks' gestation.

The causes of Imperforate anus are unknown

Classifications: in relationship to the level of the pubo-rectalis portion of the levator ani muscles

I Low anomaly :

Ano - rectal canal descends below the pubo - rectalis muscle a- Covered anus
b- Ectopic anus (perineum- low vaginal fistula) c- Anal stenosis

II - High anomaly :

The ano-rectal canal ends above pubo-rectalis muscle with or without fistulae formation a- Ano-Rectal agenesis + recto vaginal, or recto-vesical fistula

b- Ano-Rectal agenesis (atresia) → the rectum is blind ended above pubo-rectalis with no fistula formation.

Clinical presentation:

The term imperforate anus is clinical term: it means that the anus is not evident in its normal position at birth

I low anomaly:

a- The anus is missing or stenotic b- Failure to pass meconium after birth (picture of intestinal obstruction). c- Meconium evident at the perineum → a recto-perineal fistula = (Ectopic anus)

d- Meconium bulge at skin at site of anus → covered anus e- If meconium is in the urine, the diagnosis of a recto-urethral fistula f- Anal dimple and median raphe usually present.



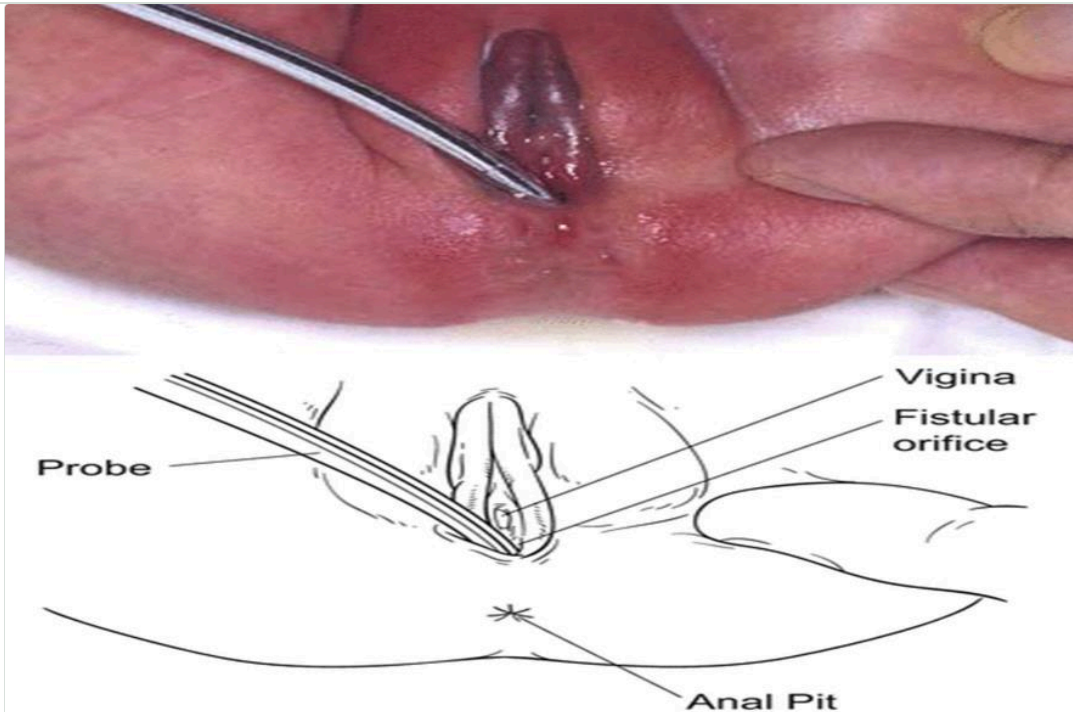


Figure 13.14 — Rectovestibular fistula

Rectovestibular fistula

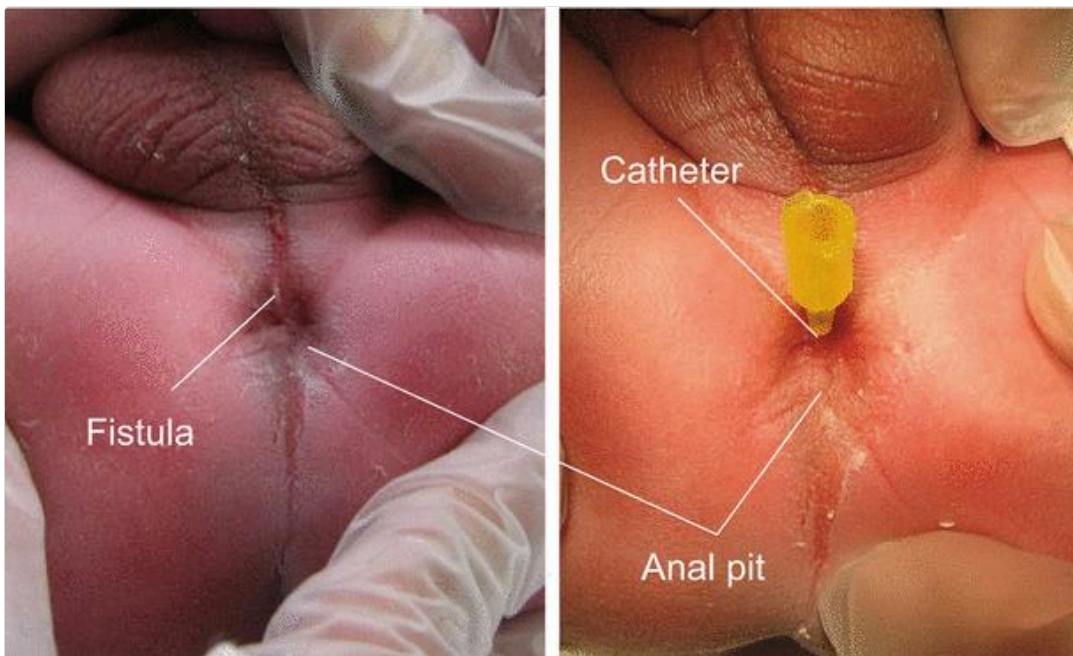


Figure 13.15 — Perineal fistula

Perineal fistula





Figure 13.16 — Covered anus

Covered anus

II- High anomaly:

a- The anus is missing b- Failure to pass meconium after birth (picture of intestinal obstruction) c- No Meconium evident at the perineum d- Meconium in urine → high recto- vesical fistula e- Recto-vaginal fistula → meconium comes out of the vagina. f- No anal dimple & No median raphe

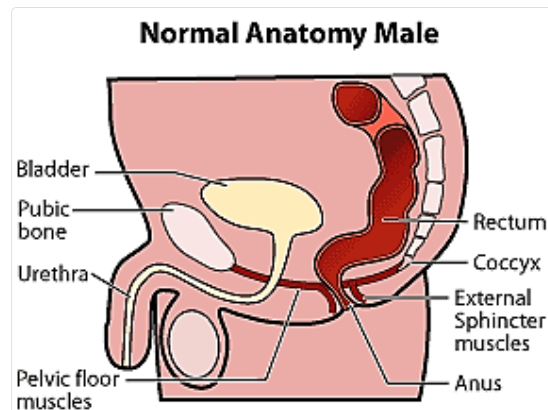


Figure 13.17 — Normal male anatomy

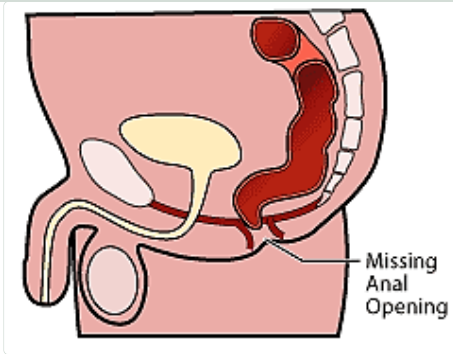


Figure 13.18 — Low anorectal anomaly
Low anomaly

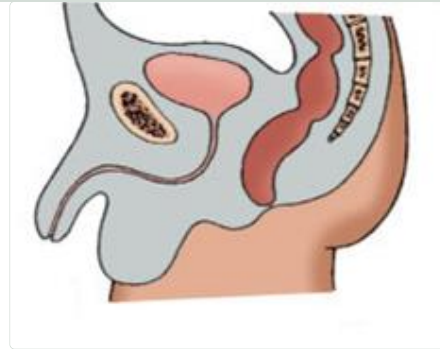


Figure 13.19 — Low anorectal anomaly

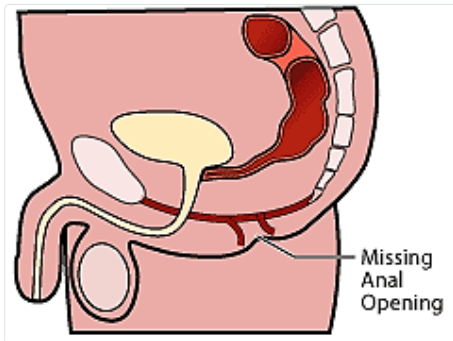


Figure 13.20 — Agnesis with fistula

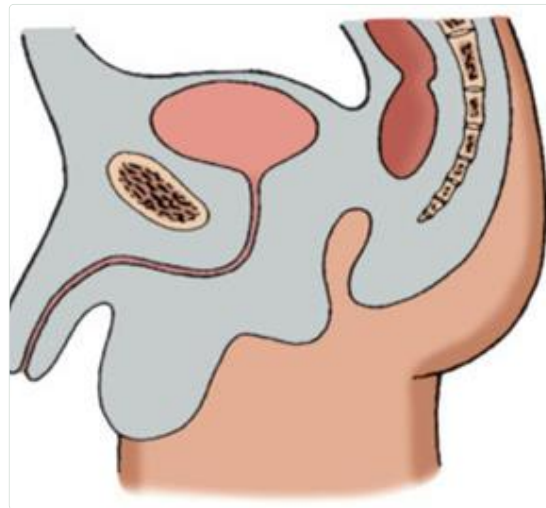


Figure 13.21 — Rectal atresia with no fistula

High anomaly

agenesis with fistula Rectal atresia with no fistula

Investigations:

1. Invertogram:

- Holding the baby upside down,
 - a lateral plain x-ray is performed to observe the distance between the level of gas in the distal rectum and a metal piece placed over the expected anus.
 - And the relation of recta gas to PC line (pubo-coccygeal line).
 - If the rectal Gas is below the P.C line or < 2.5 cm from the skin $\rightarrow$ it is low anomaly
 - If the rectal gas ends above the P.C. line or > 2.5 cm from the skin $\rightarrow$ it is high anomaly



2. C.T scan invertography: better evaluating tool

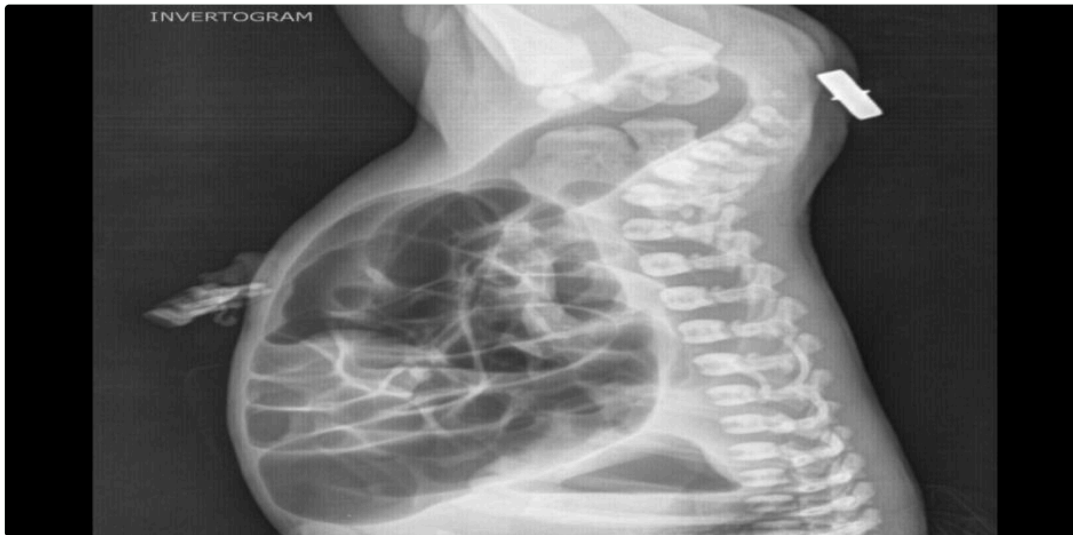


Figure 13.22 — Invertogram

Low anomaly



Figure 13.23 — Distal rectal gas level

High anomaly

Treatment:

- Colostomy for cases presented with intestinal obstruction
- Ano-plasty



Mesenteric vascular occlusion

It is a syndrome caused by inadequate blood flow through the mesenteric vessels, resulting in ischemia and gangrene of the bowel wall.

Types:

A- Mesentric Arterial ischemia

B- Mesentric Venous thrombosis

- **Acute mesenteric arterial embolism = 50 %**
- Mural thrombus after myocardial infarction, • Atrial thrombus associated with mitral stenosis and atrial fibrillation
- Septic emboli from infective endocarditis
- Emboli from fragments of proximal aortic thrombus (**30 %**)

B- Mesentric Venous thrombosis = 10 %

- Hypercoagulability
- Venous congestion from cirrhosis (portal hypertension)
- Infection

Pathophysiology:

- S.M. Vessels are more frequently involved than the I.M. vessels (larger diameter and better collaterals with inferior vessels).
- **Mucosal sloughing** may cause **bleeding** into the GIT
- If ischemia persists **disruption of mucosal barrier** occurs and bacteria, toxins, and vasoactive substances are released into the systemic circulation
- Damage may range from ischemia to necrosis, gangrene and perforation
- **Bowel necrosis** occurs in **8-12 hours** from onset of symptoms
- High mortality up to **90 %**



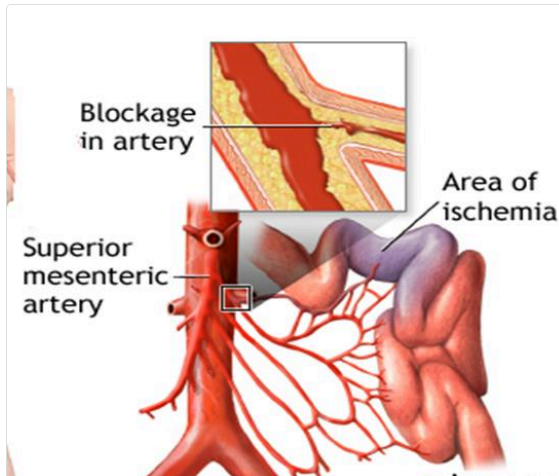


Figure 13.24 — Mesenteric arterial occlusion

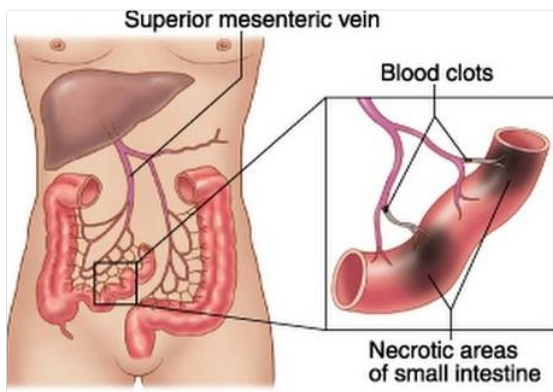


Figure 13.25 — Mesenteric venous occlusion

SM Venous occlusion SMA occlusion

Clinical presentation:

- History of cardiac disease (AF – Mitral stenosis)
- Usually old age
- The most important finding is **pain** that is disproportionate to physical examination findings.
- Typically, **pain is severe**, and diffuse not responding to Opioid
- Nausea and vomiting
- **Diarrhea** at first followed by **constipation**
- Abdominal distention
- Gastrointestinal **bleeding**
- As the bowel becomes gangrenous, rectal bleeding and sign of sepsis (tachycardia, hypotension, fever)



Abdominal examination:

In early cases no specific findings present although the severe abdominal pain

In cases complicated with **gangrene and peritonitis** → tenderness, rebound, rigidity distension and silent abdomen

Complications of MVO:

Bowel gangrene and peritonitis

- septic shock

Investigations:

I- Lab :

CBC – Serum electrolyte (Na & K) - Renal functions

II-Radiology:

A- Plain X-Ray abdomen erect and supine

- 1- Dilated intestinal loops + air and fluid levels
- 2- **Air under diaphragm** in perforation

B- C.T. or MRI angiography:

detect occluded mesenteric vessels and edematous bowel loops



Figure 13.26 — Thickened edematous bowel loops

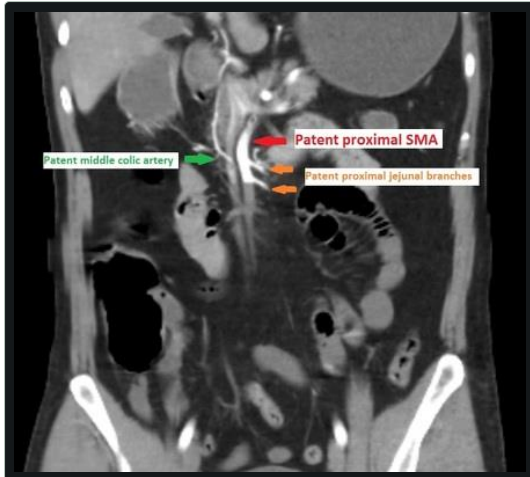


Figure 13.27 — Occlusion of the distal SMA

C- **Duplex scan** on mesenteric vessels

D- ECG: detect the cause as AF.

Treatment

I- Initial treatment:

- 1- Intra-venous fluids
- 2- Naso-gastric tube and urinary catheter
- 3- Antibiotics

II- Surgery: time factor is very important

- 1- Exploration and **mesenteric embolectomy** (in acute embolism)
- 2- Assess bowel viability: in ischemic, gangrenous cases → **Resection anastomosis** according to extent of ischemic bowel. Extensive resection may lead to short bowel syndrome

Polypsis of Colon



14



What is a polyp?

Polyp is a tissue growth protruding from the mucous membrane into the lumen of the gastrointestinal tract.

Types of colonic polyps

A. Benign non-neoplastic polyp:

- Juvenile polyps
- Peutz Jegher's syndrome
- Inflammatory polyp
- Hyperplastic polyp.

B. Benign neoplastic polyp:

- Adenomatous polyp
- Familial adenomatous polyposis = Familial Polyposis Coli
- Gardner's syndrome

Familial Polyposis Coli (F P C)

- This is an inherited disorder transmitted by autosomal dominant genes
- There are more than 100 polyps in the colon or any number of polyps in patients with family history of familial adenomatous polyposis
- It appears in second decade of life but may appear earlier
- If left untreated the chance of malignant transformation is 100%
- The mean age of cancer diagnosis is around 40 years
- These are adenomatous polyps.

Types of adenomatous polyps:

Tubular adenoma:

commonest type. Either pedunculated or sessile.

Villous adenoma:

These are usually sessile and has a velvety surface and there are finger like processes on the surface

Tubulo-villous adenoma:

characteristics in-between the tubular and villous adenoma.



Clinical presentation:

1. Asymptomatic ; discovered on routine screening of a family with a known patient with familial polyposis coli
2. Bleeding per rectum.
3. Lower abdominal pain.
4. Mucus discharge.
5. Diarrhea.

What is Gardner's syndrome?

Familial polyposis coli with some extra-colonic manifestation.

- a) Osteomas, epidermoid cyst and congenital hypertrophy of retinal pigment epithelium.
- b) Polyps of stomach and small intestine.
- c) Retroperitoneal fibrosis and desmoid tumor

Investigations of F P C:

1.Colonoscopy:

- is the most valuable investigation with biopsy to confirm the diagnosis and detect malignant changes in any suspicious polyp
 - The disease involves the colon, rectum and anal canal with innumerable polyps within the lumen.
 - The intervening mucosa appears normal. The serosal surface of the colon is also normal. • Screen family members of patients with familial polyposis coli at the age of 10 years, clinically and with colonoscopy.
 - Colonoscopy should be done yearly till the age of 20 years as most of the patients develop these polyps by the age of 20 years.
 - If there are no polyps at the age of 20 years, 2 yearly colonoscopy is to be done till the age of 40 years.



Figure 14.1 — Colonoscopy in familial polyposis





2. Barium enema:

shows multiple small filling defects

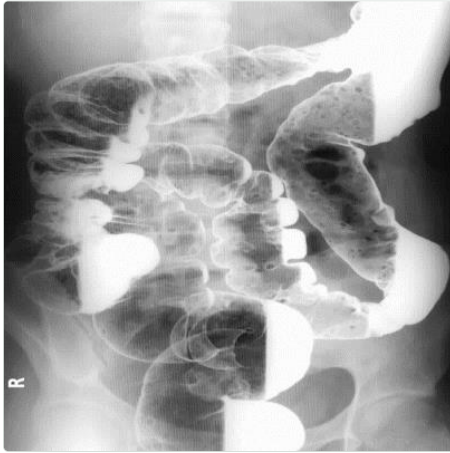


Figure 14.2 — Familial polyposis coli in contrast study

Management of familial adenomatous polyposis coli:

Surgery is mandatory for these patients as there is 100% chance of malignant transformation.

Options for surgical treatment:

1. **Total proctocolectomy with ileostomy.** Patient has a permanent ileostomy.
2. **Total colectomy with excision of rectal polyps and ileo-rectal anastomosis.** Needs regular follow up as the rectum is retained.
3. **Proctocolectomy with ileo-anal anastomosis with an ileal pouch reservoir.**

N.B. **Peutz Jegher's syndrome**

- Transmitted by autosomal dominant gene
- Gastrointestinal polyposis with pigmentation of oral mucous membrane or pigmentation in perianal region
- Polyps may be present in the whole gastrointestinal tract
- These are hamartomatous polyps
- Onset of symptoms is usually around the age of 10–30 years
- Patient may present with GI bleeding, intestinal obstruction due to intussusception



Colon and rectum

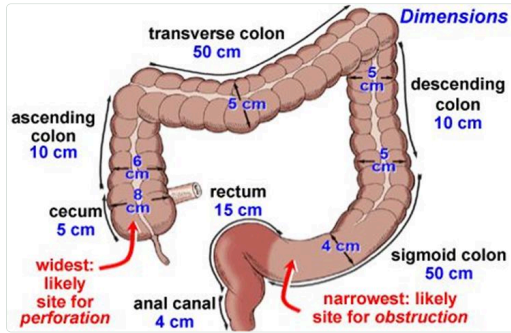


Figure 14.3 — Dimensions of the colon

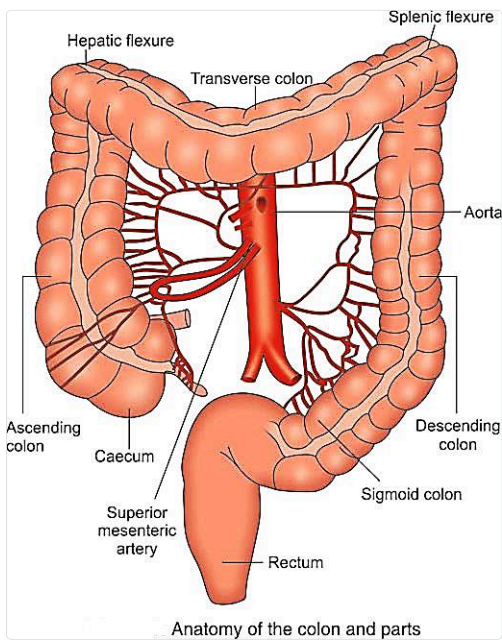


Figure 14.4 — Anatomy of the colon

Diameters of different parts

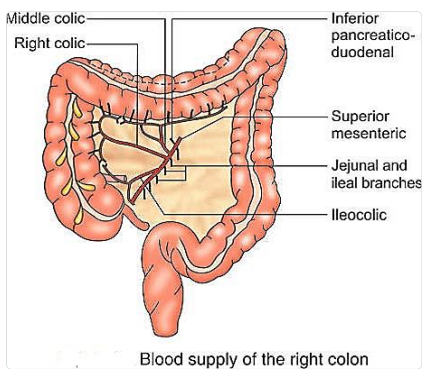


Figure 14.5 — Blood supply of the right colon

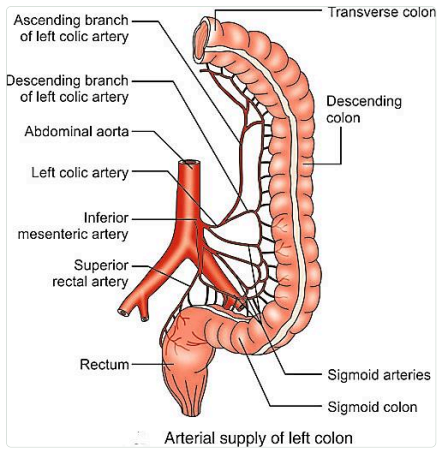


Figure 14.6 — Arterial supply of the left colon

Colorectal Cancer



15



Pathology of colorectal carcinoma

Age:

old age , however no age is immune

Sex :

more common in males than females

Aetiology:

1- Predisposing factors:

- Personal history of colorectal polyps or previous cancer
- Colo-rectal cancer family history
- Red meat, processed meat, fatty meals
- Obesity
- Smoking
- Alcohol

2- Precancerous lesions:

- FPC : (familial polyposis of colon)
 - Autosomal Dominant – 100 % turn malignant.
 - Inflammatory bowel disease (UC > Chron's)
 - Adenomatous polyp.

3- Microscopic picture:

Adenocarcinoma : More than 90% of colorectal carcinomas are adenocarcinomas originating from epithelial cells of the colorectal mucosa

- Mucinous adenocarcinoma (>50% mucinous material) is a variant of adenocarcinoma • According to the glandular contents :
 - a) Grade 1: Well differentiated carcinoma
 - b) Grade 2: Moderate differentiated (most common) c) Grade 3: Undifferentiated (poor prognosis)

Macroscopic picture: Naked eye:

- Cauliflower mass → with ulcerating bleeding surface
- Malignant ulcer → raised everted edge, hard base and necrotic floor
- Malignant stricture: hard and irregular



- Multicentricity is common

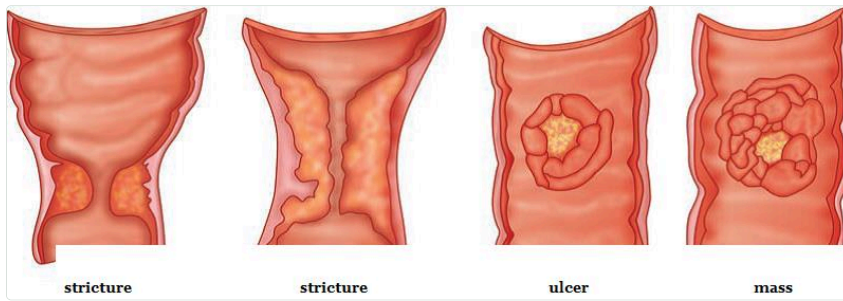


Figure 15.1 — Macroscopic appearances of colorectal cancer

Incidence of colorectal cancer in different parts:

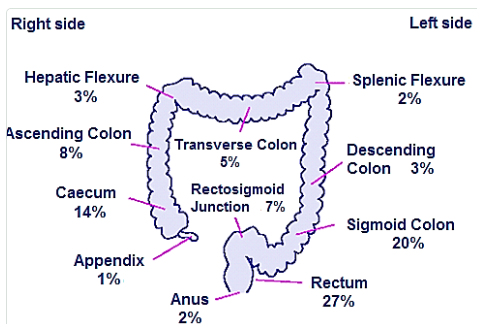


Figure 15.2 — Distribution of colorectal cancer

The most common site incidence is rectosigmoid part of the colon

Spread:

A- Direct: - mucosa → submucosa → muscle layer → serosa → nearby structures or Organs. It may lead to internal fistula as colo-vesical or recto-vaginal fistulae.

- **Right colon** → right ureter, abdominal wall, gonadal vessels, duodenum, small intestine, peritoneum, liver.
- **Transverse colon** → duodenum, stomach, liver, intestine, peritoneum.
- **Left colon** → spleen, abdominal wall, intestine, left ureter, gonadal vessels, peritoneum.

Rectum:

to urinary bladder, vagina → causing fistula → to sacrum, prostate, seminal vesicles, and peritoneum.

B- Lymphatic spread:

- 1- Epiploic lymph nodes
- 2- Para-colic lymph nodes